Economy

Inside the cost-of-living squeeze

04 November 2026

Key takeaways

- Not every essential has become less affordable. Despite the cumulative rise in prices, households are spending a slightly smaller share of income on gas and groceries than they did several years ago. That suggests the affordability squeeze is less broad-based than the inflation headlines alone might imply.
- Housing is the major exception, and the pressure extends beyond lower-income households. Both mortgages and rent have increased since 2019. Some lower-income renters look particularly stretched, but around one in four higher-income mortgage holders also spend more than half their income on housing.
- Wage gains and changing behavior have helped households absorb higher prices. But real wage growth has mostly softened, which may help explain why sentiment remains weak even as consumer spending continues to grow. In fact, declining sentiment likely reflects the sacrifices consumers have to make in order to keep spending.

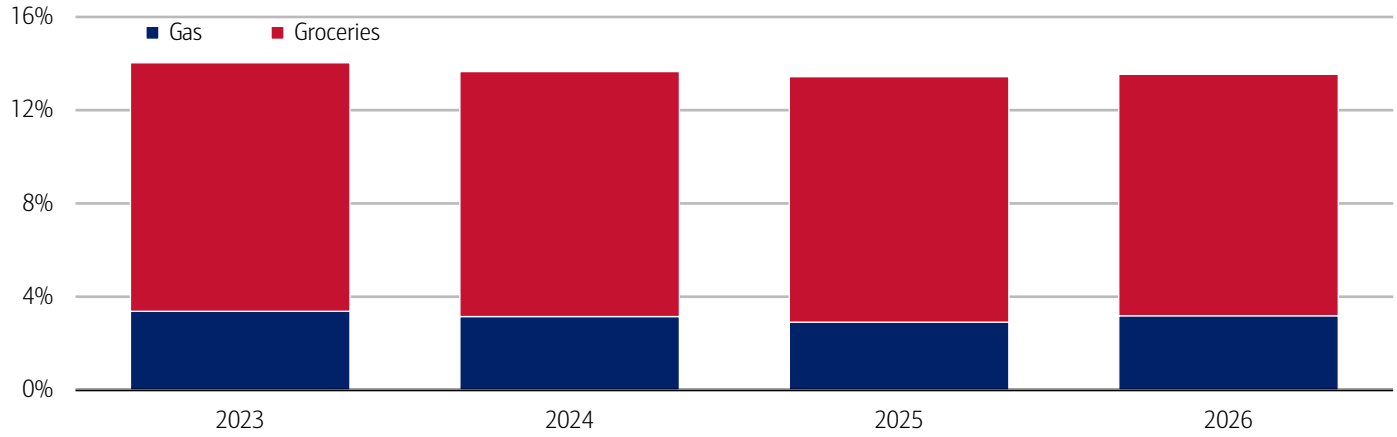
Everyday essentials take up less income than they did four years ago

Higher prices are still top of mind for consumers, especially when it comes to affordability. But just how stretched are households? Using Bank of America internal data, we look at how much of their income households are spending on everyday essentials like gas, groceries, mortgages and rent, and how that has changed over the past four years across income groups. We also explore how wage growth and shifts in spending habits have helped households cope with higher prices, and why consumers still feel downbeat even as they keep spending.

For example, while consumers overall spent slightly more of their income on gas and groceries compared to last year, outlays for these necessities are still below what they were four years ago, according to Bank of America payment and deposit data (Exhibit 1). Combined, gas and groceries accounted for nearly 13.5% of income in 2026, 10 basis points more than last year, but almost half a percentage point less than four years ago. And even with price increases over the past year, most consumers only spent 3.2% of their income on gasoline, up from 2.9% a year ago and below the 3.4% from four years ago.

Exhibit 1: Consumers overall spent slightly more of their income on gas and groceries compared to last year, though spending on these necessities is still below pre-pandemic levels

Median gas and grocery spending as a share of median income (annually, %)

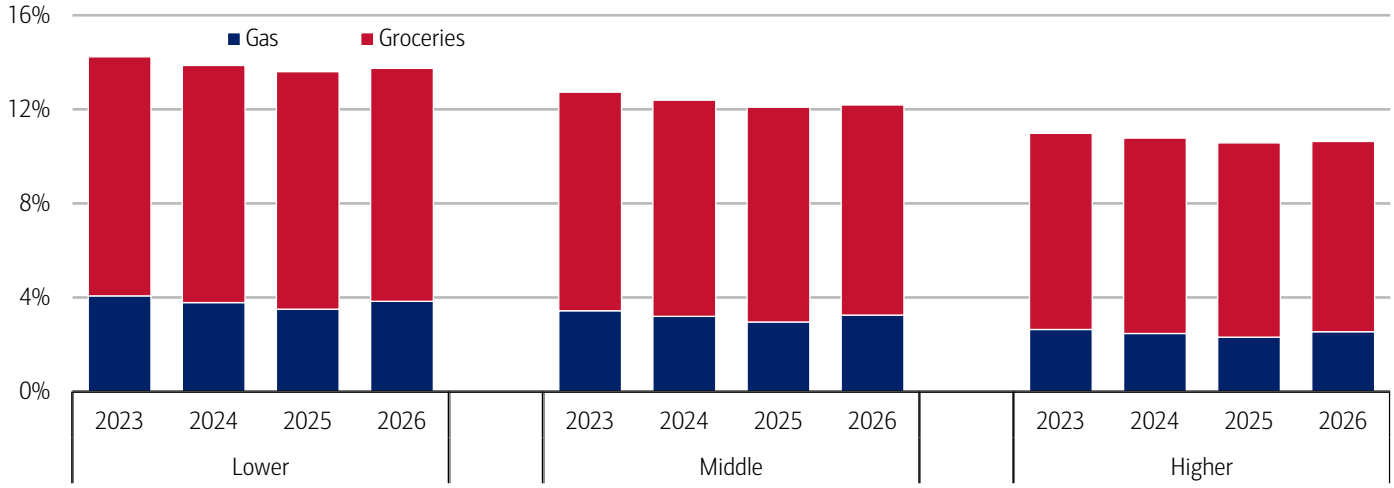


Source: Bank of America internal data
Note: 2026 data is through September.

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Across income groups, each cohort follows a similar pattern, though lower-income households do allocate the highest share of their income to gas and groceries (Exhibit 2).

Exhibit 2: All income cohorts have experienced an increase in gas spending as a share of income, offsetting a decrease in grocery spending in 2026
Median gas and grocery spending as a share of income by household income tercile (annual, %)



Source: Bank of America internal data

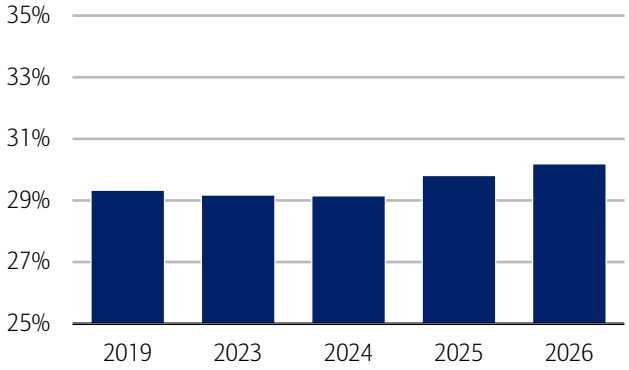
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Housing stress isn't just a low-income story

Housing is an altogether different picture. Whether you're a homeowner or a renter, housing is often the biggest expense of any consumer, accounting for around 30% of income. Median mortgage payments have steadily risen over the past two years and now account for more than 30% of a homeowner's income, according to Bank of America internal data (Exhibit 3).

While renters have experienced some easing, they tend to set aside more of their income to housing costs (Exhibit 4). Rents as a share of income peaked in 2025, with renters spending more than 32% of their income on shelter. Even so, a small improvement helps. Due to declines in housing payments, renters have seen a significant improvement in discretionary spending over the past year (read more in On the move: Renovation over relocation?). Also, some housing related costs have also eased in the past year (read more in Energy bills: Short-term relief, longer-term challenges and Insurance costs hit the brakes).

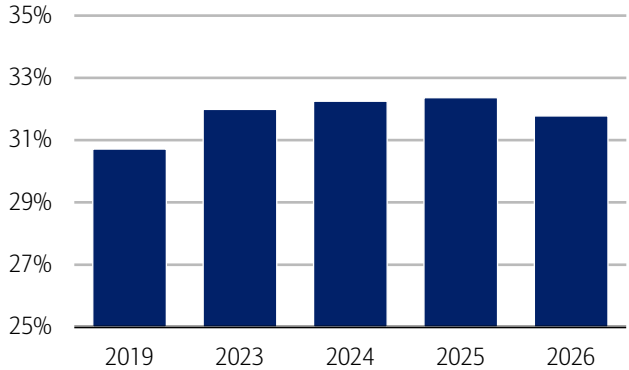
Exhibit 3: Mortgage payments as a share of income have risen over the past two years…
Median mortgage payments as a share of median income (annually, %)



Source: Bank of America internal data
Note: Includes both internal (BofA) and external (non-BofA) mortgage payments.

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Exhibit 4: …while rent payments have eased over the same time period
Median rent payments as a share of median income (annually, %)



Source: Bank of America internal data

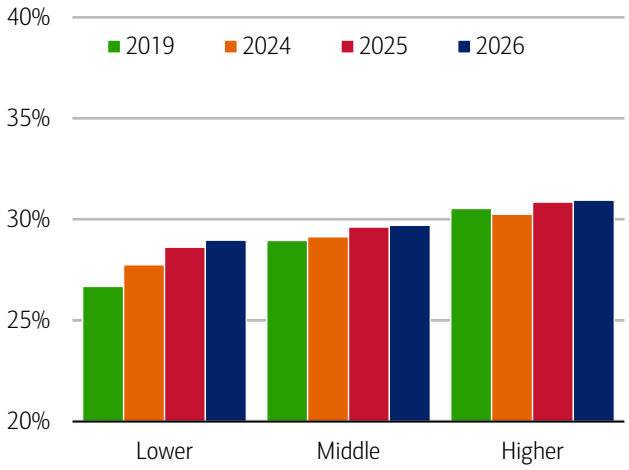
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Also, when it comes to housing, higher-income families typically spend more of their wages on mortgages than lower-income households (Exhibit 5). The converse is true for renters, with lower-income families spending around 35% of their paychecks on rent, five percentage points more than higher earners (Exhibit 6).

This suggests to us that despite higher mortgage rates, most homeowners are relatively insulated from increasing housing costs than renters, possibly because many purchased their house before prices and interest rates began rising sharply post-pandemic. Additionally, this buffer from housing inflation seems to skew more toward higher-income households.

Exhibit 5: Mortgage payments as a share of income have increased the most for lower-income households

Median mortgage payments as a share of income by household income tercile (annual, %)

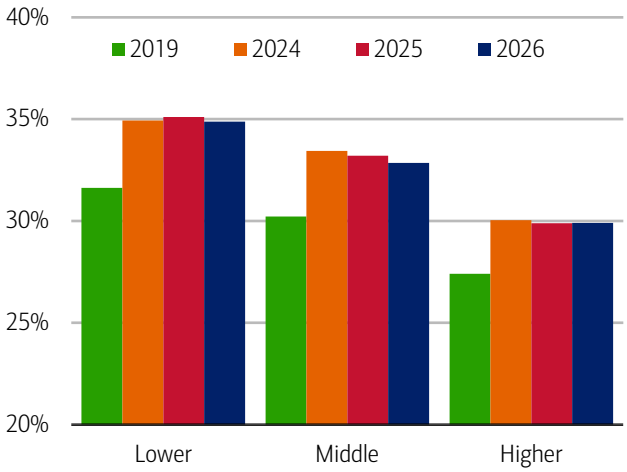


Source: Bank of America internal data
Note: 2026 data is through September. It includes both internal and external mortgage payments.

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Exhibit 6: Lower earners spend a much larger share of their income on rent

Median rent payments as a share of income by household income tercile (annual, %)



Source: Bank of America internal data
Note: 2026 data is through September.

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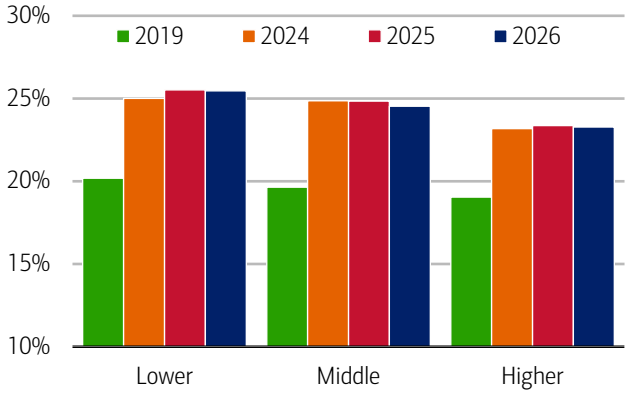
Some financially stretched households may be particularly affected by rising housing costs

Notably, in the first nine months of the year, roughly one in four lower-income renters spent more than half of their income on rent, up from about one in five in 2019 (Exhibit 7). One positive note, the share of renters who spend most of their income on rent does seem to have declined slightly in 2026.

Financial pressure from housing is not limited to lower-income households. Around one in four higher-income mortgage holders spend more than half of their income on home payments (Exhibit 8). And while only slightly more than one in five lower-income mortgage holders spent half of their income on housing, this is a nearly five percentage point increase compared to pre-pandemic levels.

Exhibit 7: Nearly one in four lower-income households spend at least half of their income on rent…

Share of households who spend more than 50% of their income on rent by household income tercile (annual, %)

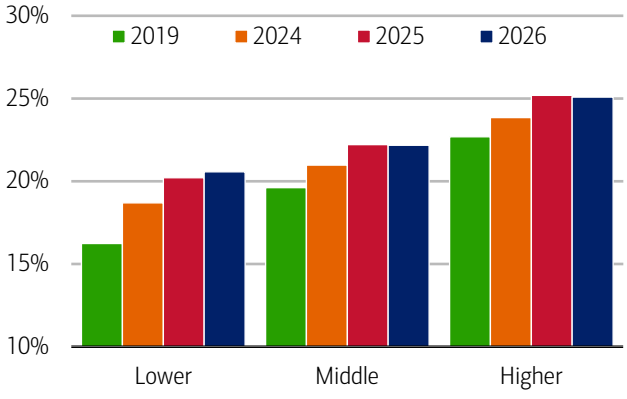


Source: Bank of America internal data

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Exhibit 8: …whereas one in four higher-income households spend at least half their income on their mortgage

Share of households who spend more than 50% of their income on mortgages by household income tercile (annual, %)



Source: Bank of America internal data
Note: Includes both internal and external mortgage payments.

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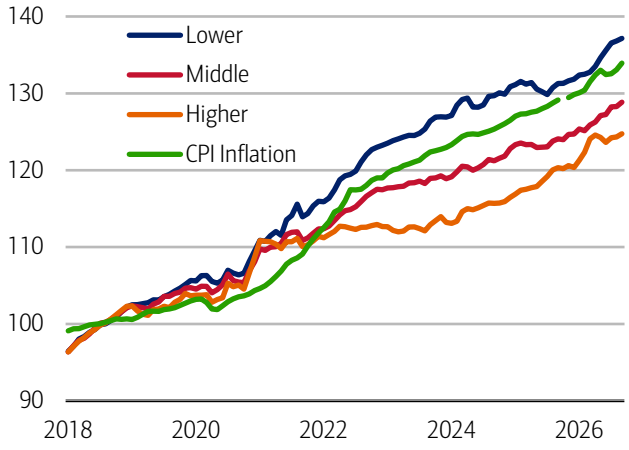
Wage growth and changing habits are insulating some households from inflation

Still, some families have not seen their bills, as a share of income, rise alongside inflation. Why is this? In our view, this is likely because after-tax wages have increased significantly for these people. This is especially true for some lower-income households that have landed steady employment since the pre-pandemic era, according to Bank of America internal data and the Consumer Price Index (CPI) (Exhibit 9).

Additionally, consumers have adjusted their behavior in response to higher prices. For one, we may be driving less than before the pandemic. Despite some increases in vehicle miles driven per capita over the past four years, we're still getting behind the wheel less on average compared to 2019 levels, according to the US Census Bureau and Bureau of Transportation (Exhibit 10). At the same time, some households have "traded down" to less expensive stores or items and services (read more in The rise of value).

Exhibit 9: Some lower-income households have seen their wages grow faster than inflation since 2019

After-tax wages, based on Bank of America deposit data (three-month moving average, index 2019 = 100) compared to CPI inflation, based on Bureau of Labor Statistics (BLS) data (monthly, index 2019 = 100)



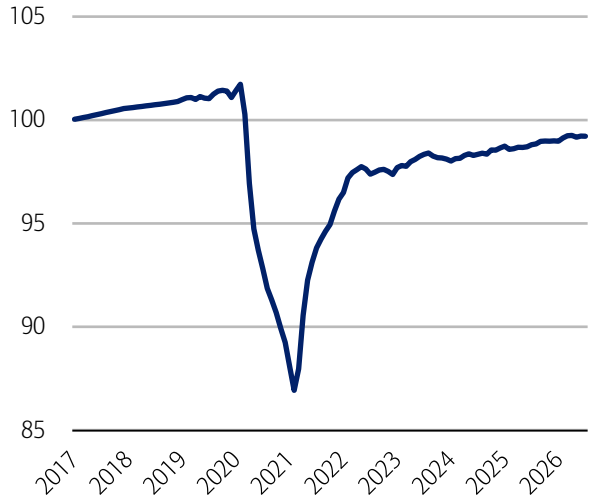
Source: Bank of America internal data, BLS

Note: Data not available for October 2025 due to federal government shutdown.

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Exhibit 10: Vehicle miles driven per capita have not returned to pre-pandemic levels

Total vehicle miles traveled per total resident and civilian population estimates (monthly, index 2017 average = 100)



Source: US Census Bureau, Bureau of Transportation

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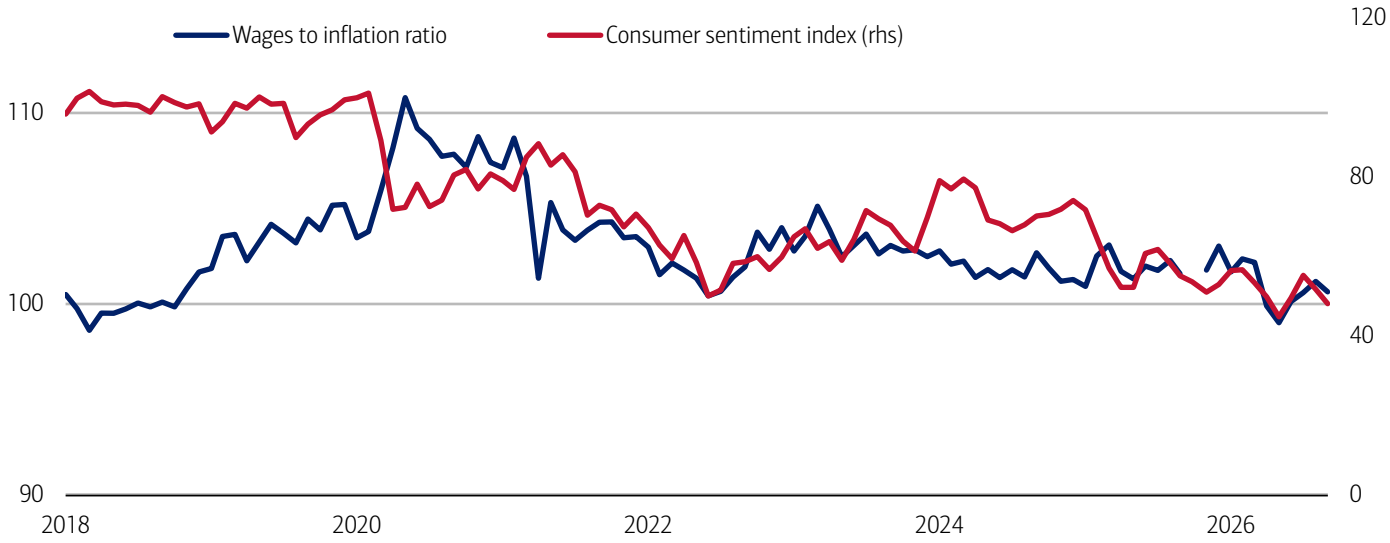
Inflation is driving households to make more sacrifices

There is one caveat to the affordability picture: wage growth among lower-income households slowed significantly throughout 2025, though they've recovered since then. Furthermore, some middle- and higher-income families have not seen their pay keep up with inflation, at least compared to 2019 levels.

In fact, using the ratio of after-tax wages to inflation as a proxy for "real wages," we can see they largely declined throughout 2022 and then again since mid-2023, according to Bank of America internal and BLS inflation data (Exhibit 11). Additionally, the drop in real wages is largely mirrored by a decline in the University of Michigan's Index of Consumer Sentiment. This suggests that people have been feeling worse about their finances and the broader economy as their purchasing power has declined. Consumers are spending more than they did last year (read more in Consumer Checkpoint: Strength in numbers), but sentiment likely reflects the sacrifices consumers have made in order to keep spending.

Looking ahead, in our view, the outlook for affordability is relatively encouraging: wage growth remains firm and has picked up for lower-income houses. That said, there are risks. BofA Global Research notes the potential for gasoline and grocery prices to increase. And higher mortgage rates make it difficult for those looking to make the jump from renting to homeownership.

Exhibit 11: The Consumer Sentiment Index has declined along with real wage growth since the post-pandemic period
The ratio of after-tax wages (year-over-year (YoY) %), based on Bank of America internal data, to CPI inflation, based on BLS data (monthly, index 2018 = 100) compared to Index of Consumer Sentiment (monthly, index Q1 1966 = 100, right-hand side (rhs))



Source: Bank of America internal data, BLS, University of Michigan
Note: October 2025 is missing due to the government shutdown.

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Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under $5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate (if discussed) can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.
2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.
3. Overall total card spending includes small business card spending while per household card spending does not.
4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).
5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

Grocery, as discussed in this analysis, includes spending at grocery stores, warehouse clubs and dollar/discount stores.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation, changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Generations, if discussed, are defined as follows: Gen Z, born after 1995; Younger Millennials: born between 1989-1995; Older Millennials: born between 1978-1988; Gen Xers: born between 1965-1977; Baby Boomer: 1946-1964; Traditionalists: pre-1946.

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Additional information about the methodology used to aggregate the data is available upon request.

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Disclosures

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Transformation

Quantum dictionary, part 2: From potential to progress

02 November 2026

Key takeaways

- In the second and final part of our quantum dictionary series, we move beyond the basics to explore where quantum could go next. Today's quantum computers remain limited by noise and errors, but their potential grows rapidly as they scale. Emerging applications like quantum AI and sensing could eventually unlock new capabilities and significant economic value.
- Quantum communication could change both how we connect quantum computers and how we protect information. A quantum internet could use entanglement to link quantum systems, creating additional computing power, while quantum networks could make eavesdropping detectable and offer new ways to securely transmit information.
- Meanwhile, the race for quantum leadership is already underway. Governments are pursuing greater quantum sovereignty as the technology's potential implications for competitiveness, security and defense grow.

Assessing quantum's performance and progress

Quantum advantage versus supremacy

Quantum advantage is reached when a quantum computer can solve a useful, real-world problem faster than a classical computer. The problem is, today's quantum computers are still too noisy, as we discussed in part one of this series, and qubit quality isn't yet high enough to demonstrate quantum advantage.

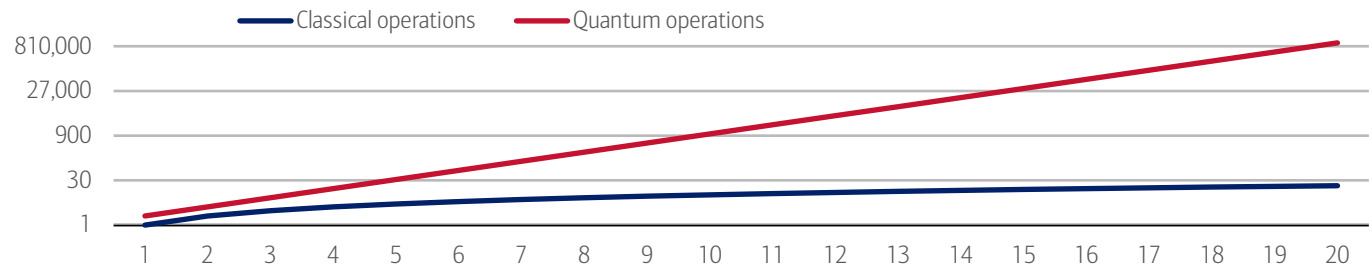
Quantum supremacy takes this one step further. It refers to the point when a quantum computer has computational advantage, meaning it can solve a problem that a traditional computer couldn't in any feasible amount of time.

Quantum scaling

Quantum scaling is about building bigger, more powerful quantum computers. It involves overcoming the challenges of increasing both the number and the quality of qubits. Today, we're in the noisy intermediate-scale quantum era. Quantum computers have tens to a few hundred qubits, but noise and errors still limit what they can do.[1]

However, when quantum computers do scale, they do so exponentially. Every additional qubit doubles the number of operations the system can perform. A quantum computer with just 10 qubits could perform more than 100 times the operations of a classical computer (Exhibit 1). More qubits could therefore mean more operations and, ultimately, more economic value in the industries where quantum computing can be put to work, according to BofA Global Research.

Exhibit 1: A quantum computer with just 10 qubits could perform more than 100 times the operations than classical
Number of classical versus quantum operations for a given (qu)bit number (log scale)



Source: BofA Global Research

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[1] Quandela. (n.d.). *NISQ*.

Quantum Benchmarking Initiative

The Defense Advanced Research Projects Agency's (DARPA's) Quantum Benchmarking Initiative (QBI) aims to answer a big question: Is it possible to build a commercially relevant quantum computer by 2033? To be clear, QBI is not a competition between companies. DARPA evaluates all viable approaches for which funding is available.

QBI has three stages:

- Stage A: Start with a concept. At this stage, there is a utility-scale quantum computer concept that has a plausible path to realization in the near term.
- Stage B: Put a plan behind it. This stage involves developing a research and development plan, identifying the associated risks and introducing prototypes to overcome them.
- Stage C: Put it to the test. In the final stage, companies work with the government to verify and validate that the utility-scale quantum computer concept can be built as intended.

Exploring emerging quantum applications

Quantum AI

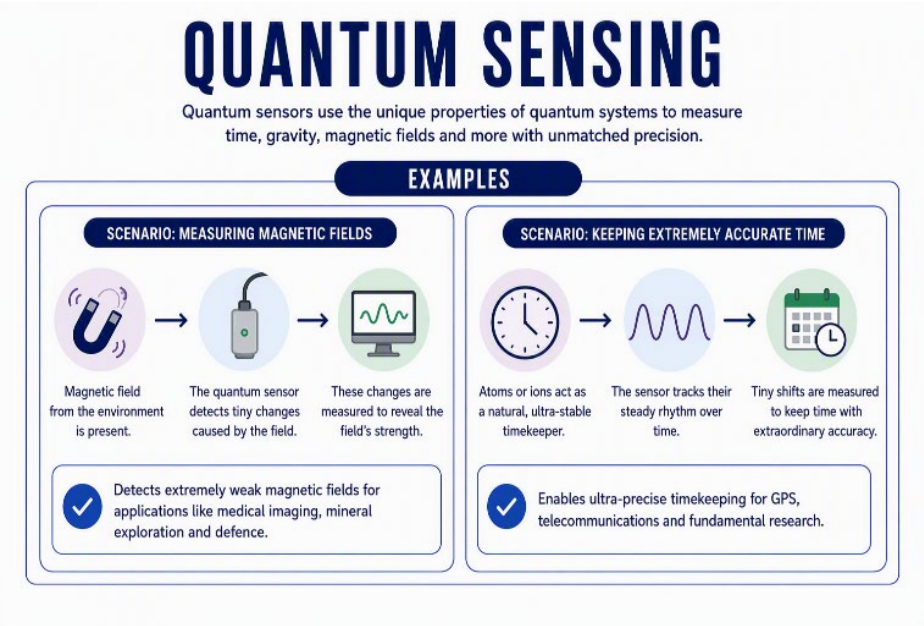
According to BofA Global Research, quantum AI brings together two powerful technologies: AI and quantum computing, with the potential to improve both the physical and the digital worlds. And the relationship could work both ways. Quantum computing could provide some of the compute power needed to train generative AI, while generative AI could help speed up the development and testing time of quantum systems.

AI tools like machine learning (ML) could also make quantum error correction (QEC) more efficient and scalable.[2] ML models trained on simulated datasets can identify and correct errors faster than traditional methods while using fewer computing resources. They can also more accurately predict errors before they happen.[3]

Quantum sensing

Quantum sensing uses the properties of quantum mechanics to measure things like magnetic fields, gravitational forces, time and rotation with greater precision than classical instruments (Exhibit 2).[4]

Exhibit 2: Quantum sensors use quantum properties to measure physical quantities with unmatched precision
An infographic of quantum sensing



Source: BofA Global Research

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[2] Wang, Z. & Tang, H. (2024, December 29). *Artificial Intelligence for Quantum Error Correction: A Comprehensive Review*. arXiv.
[3] Swayne, M. (2025, January 6). *AI for Quantum Error Correction: A Comprehensive Guide to Using Artificial Intelligence to Improve Quantum Error Correction.* Quantum Insider.
[4] Rehman, M.U. (2026, March 2). *Understanding Quantum Sensing and Its Industrial Potential.* Quantum Insider.

So, how does it work? In simple terms, there are three steps. First, a quantum system is prepared in a well-defined initial state, for example, by cooling an atom to near absolute zero. Next, the sensor is exposed to whatever is being measured, causing a subtle shift in its quantum state. In an atomic magnetometer, for example, a magnetic field changes the energy levels of an atom's electrons, which changes how the atom absorbs or emits light. Finally, the sensor measures how much the quantum state has shifted, revealing the strength of the external force.[5]

Benefits and challenges

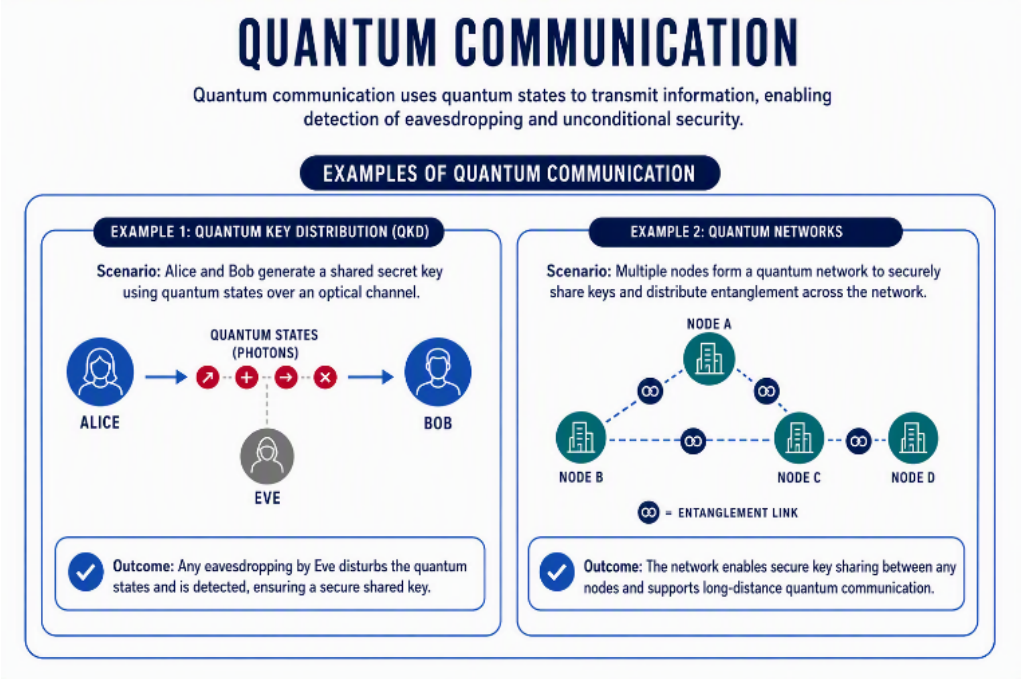
Compared with quantum computers, quantum sensors face some lower technical barriers. They don't need nearly as many qubits to be useful, entanglement can help but isn't required for every application, and even short coherence times can produce useful measurements.[6] The catch? They often require large vacuum chambers, laser systems and magnetic shielding to maintain coherence.

The quantum internet and communications

Quantum communication

Quantum communication is the study of how information can be encoded and transmitted between quantum systems over a distance (Exhibit 3). It uses properties like superposition and entanglement to enable the secure transmission of information. Why does that matter? One of the main goals is to keep information secure from potential eavesdroppers, particularly as more powerful quantum computers could pose new challenges for traditional communication methods. The ultimate goal is what some call the quantum internet, where quantum signals can move seamlessly across many quantum networks.[7]

Exhibit 3: Quantum communication uses quantum states to encode and transmit information
An infographic of quantum communication



Source: BofA Global Research

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Quantum internet

Think of the quantum internet as a scalable network that allows two or more quantum devices to exchange information (Exhibit 4). These connections could span vastly different distances, from within a silicon chip all the way to satellite links.[8]

But the quantum internet isn't meant to replace the internet we use today, nor is it simply a faster version of it. According to Cisco, the classical internet can move information between computers, but it is ultimately limited by what those computers can compute. Some problems become so complex so quickly that no classical machine can solve them in a useful amount of time.

[5] Ibid.
[6] Ibid.
[7] Quantinuum. (2024, July 30). *Coming Over the Horizon: Quantum Communication Enters the Mainstream.*
[8] Imperial. (n.d.). *Quantum Internet.*

When connection becomes computation

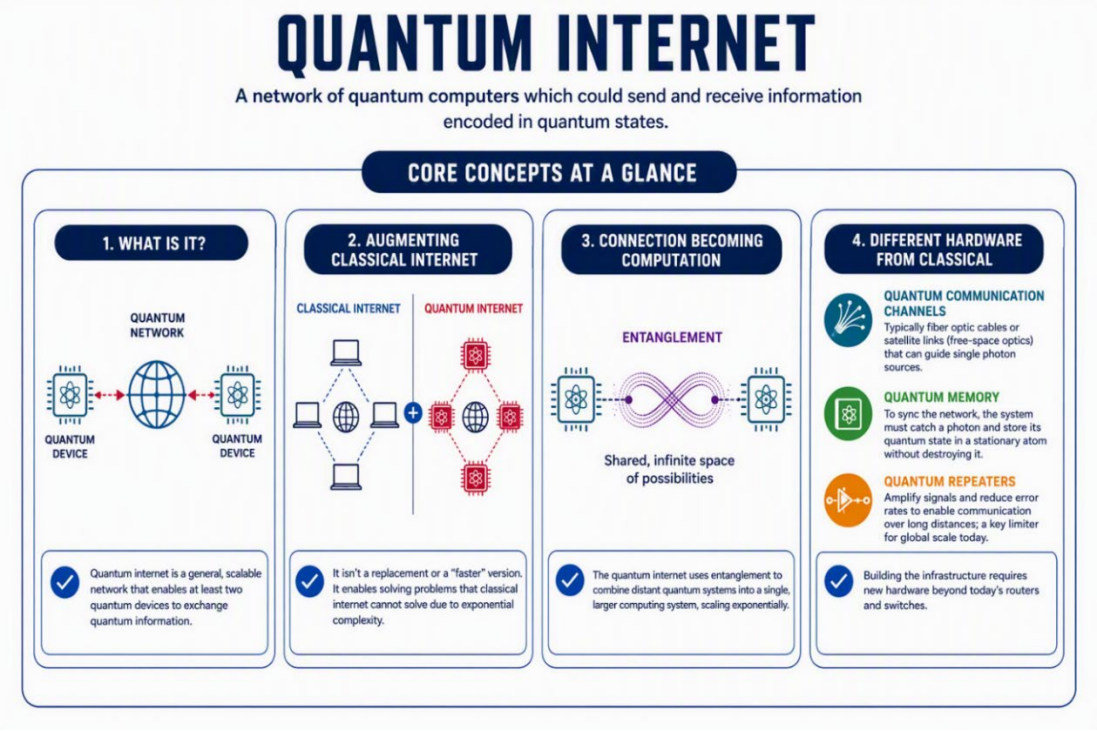
Here's what makes the quantum internet different: the connection itself can create new computing power. Using entanglement, a quantum network can connect quantum systems into a single, larger computing system that scales exponentially.[9] That's fundamentally different from classical computers, which communicate by passing copies of information back and forth. Quantum computers connected through a quantum network can instead create a feedback loop of entanglement and a shared space of possibilities that neither could produce on its own.

Different hardware from classical

Building a quantum internet requires a different set of hardware than the routers and switches that power today's internet. Key pieces include:[10]

- Quantum communication channels: These carry quantum signals, typically through fiber-optic cables or satellite links using free-space optics.
- Quantum memory: To keep the network in sync, the system must catch a photon and store its quantum state in a stationary atom without destroying it.
- Quantum repeaters: These help quantum signals travel longer distances by amplifying signals and reducing error rates. According to QuEra, the lack of viable quantum repeaters is currently a key limitation to building a global quantum network.

Exhibit 4: A network of quantum computers which could send and receive information encoded in quantum states
An infographic explaining the quantum internet



Source: BofA Global Research

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Quantum networking

Quantum networking is a way of sending information using quantum states. It generally includes three network types: quantum key distribution (QKD) for secure communications (more on this below), the quantum internet for distributed quantum computing, and quantum sensing networks that use entanglement to improve measurements.

Classical networks turn information into electrical or optical signals that can be copied or intercepted without the sender or receiver knowing. Quantum networks work differently. Because information is transmitted using quantum states, it cannot be

[9] Chan, M. (2026, June 1). *Quantum internet 101: What it is, and why it matters.* Cisco.
[10] QuEra. (n.d.) *Quantum Internet.*

intercepted and copied without disturbing the original information. That disturbance makes an eavesdropping attempt detectable.

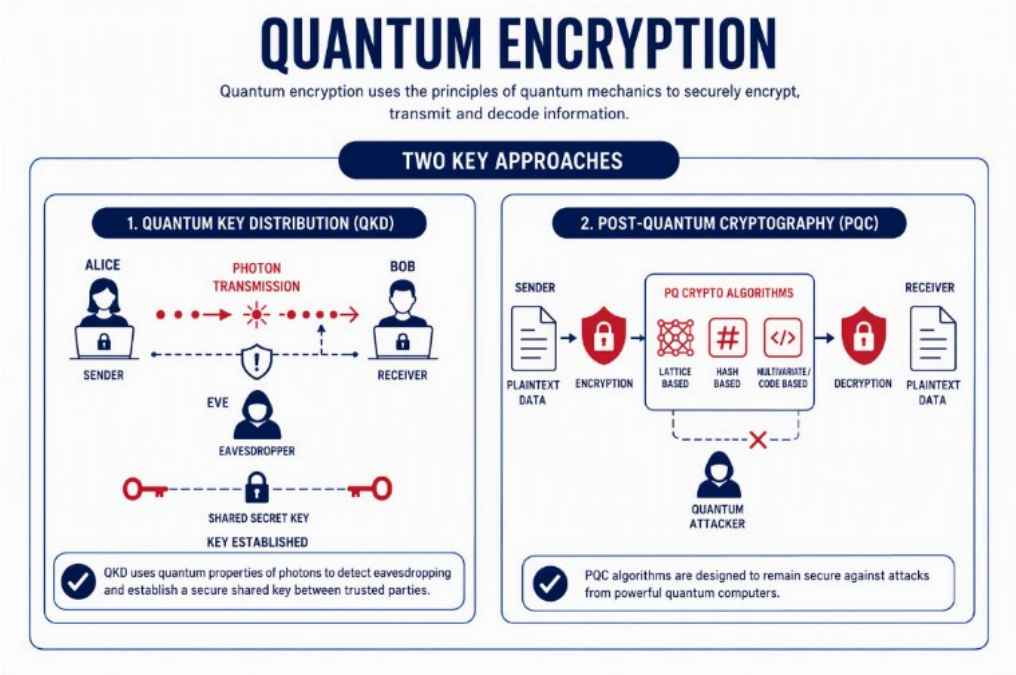
Evaluating security & strategic risks

Quantum encryption

Quantum encryption uses the principles of quantum mechanics, including superposition and entanglement, to encrypt and securely transmit data (Exhibit 5).

One potential risk of quantum computing is that it could eventually break widely used forms of encryption. And while large-scale quantum computers are still being developed, organizations may need to start preparing for that possibility today.

Exhibit 5: Examples include quantum key distribution and post-quantum cryptography
An infographic illustrating quantum encryption



Source: BofA Global Research

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Quantum key distribution

Quantum key distribution (QKD), a type of quantum encryption, uses the laws of quantum physics to create and distribute secure keys which prevent data decryption.[11] Think of a key as something that locks and unlocks a digital message. Two parties share a key to encrypt and decrypt information sent between them. If an eavesdropper tries to obtain the key, the attempt disturbs the quantum states, alerting the parties that the key may have been compromised and shouldn't be used.[12]

According to Quantum Insider, QKD generally involves three steps:[13]

- Create entangled photons: First, a quantum source, such as a laser and nonlinear crystal or a quantum dot, generates pairs of entangled photons. The photons are correlated in ways that classical light cannot replicate, so measurements of one reveal information about its partner.

- Send the photons: The photons travel to different locations through fiber-optic cables or free-space optical channels, such as satellite-to-ground or satellite-to-satellite links, while maintaining their quantum correlations. But those connections can be lost over long distances, with current systems reaching roughly 100 kilometers, according to Quantum Insider.[14] Classical networks get around distance limitations with repeaters that receive, amplify and retransmit signals.

[11] Toshiba. (n.d.). *Quantum Key Distribution (QKD).*
[12] QuEra. (n.d.). *Quantum Key Distribution.*
[13] Rehman, M.U. (2026, March 9). *Understanding Quantum Networking and Its Industrial Potential.* Quantum Insider.
[14] Ibid.

Quantum signals can't simply be amplified without destroying their quantum properties, so quantum repeaters take a different approach. They use entanglement swapping and quantum memory to extend the range of quantum networks.

- Measure the photons: Finally, the photons are measured at each location. This confirms whether the entanglement survived the journey and can reveal whether someone tried to intercept the photons along the way. The results can then be used to generate a shared secret key for encrypting classical messages.

Quantum sovereignty

Quantum sovereignty is an economic actor's ability to develop, produce and operate its own quantum technology.

Quantum is the first new computing platform technology since the arrival of digital computing. And the stakes are high. Stakeholders are interested in quantum because of its potential implications for economic competitiveness, security, supply chain resilience and defense. As a result, some may pursue "sovereign quantum," meaning quantum capabilities are owned and operated by a particular nation. That could involve investing in quantum technology, building domestic expertise or even export controls to limit other stakeholders' ability to develop their own sovereign capabilities.

The geopolitical race is on

Government quantum investment has reached at least about $58 billion cumulatively as of 2025, led by Europe at $15.4 billion and China at $15.3 billion.

The competition extends beyond investment. At least nine countries implemented export controls on quantum technology in 2024. Over the past 20 years, China and the US have led in patent requests filed, but Europe and China have led in the number of patents granted. The race for talent is global, too. China has trained the largest share of top quantum researchers at the undergraduate level, at roughly one-fourth, while the US is home to the largest share of top talent by employment, at one-third.

Quantum supply chain

Building quantum technology requires much more than qubits. A broader supply chain provides the equipment, components and materials needed to make quantum systems work, including:

- Cryogenics: Most types of qubits require some level of extreme cooling, with superconducting qubits among the most demanding at roughly 5–10 millikelvin. Keeping qubits this cold helps shield them from environmental noise that can cause decoherence.

- Qubit control electronics: Quantum gates direct qubits by sending them photons. That requires supporting electronics, including wiring, room-temperature electronics and amplifiers. For example, analog microwave signals used to read qubits must be amplified through several stages before they can be converted into digital signals and analyzed.

- Photonics: Lasers use light and atoms to help power quantum technology, while other devices can modify properties of photons, such as their path and polarization.

- Foundries: Quantum development also relies on several types of fabrication facilities, including research fabs at national research organizations and universities, pre-industry research fabs, independent foundries that manufacture semiconductors for third parties, and in-house fabs operated by quantum technology vendors.

- Raw materials: Quantum systems also depend on materials used in qubits themselves and in supporting components such as cables and cryostats. Helium-3 and helium-4 are used in cryostats to maintain very low temperatures, while rare earth metals used in quantum technology include yttrium, neodymium, europium, erbium and ytterbium.[15]

AI singularity

AI singularity refers to a hypothetical point when AI evolves to become self-improving and autonomous, surpassing human cognitive abilities.[16]

A complementary relationship between quantum and AI

Rather than developing in isolation, AI and quantum computing could ultimately help advance each other. Quantum computing may eventually provide some of the compute power needed to train generative AI, while generative AI could help speed up the development and testing of quantum systems.

[15] Swayne, M. (2025, October 15). *Rare-Earth Minerals And Quantum Computing: How Secure Are The Supply Chains?* Quantum Insider.
[16] ISB. (2025, March). *What is AI Singularity? Explore Its Future Impact.*

That relationship could become increasingly important as the computing power needed to train the largest generative AI models continues to grow exponentially. Classical computing may eventually struggle to keep pace, while quantum computing could help meet some of that demand because its computing power also scales exponentially.

By 2045? 2029? Now?

AI is becoming increasingly powerful and capable, raising a big question: when could it reach or even surpass human intelligence? Futurologist Ray Kurzweil expects AI to reach human levels of intelligence by 2029 and technological singularity by 2045. His view extends Moore's Law beyond processing power to other forms of technological progress.

Under that framework, computing power would continue growing exponentially. By 2045, Kurzweil projects that $1,000 worth of computing could perform as many calculations in one second as all human brains combined.

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Economy

Regional Roundup: Coast-to-coast growth

29 October 2026

Key takeaways

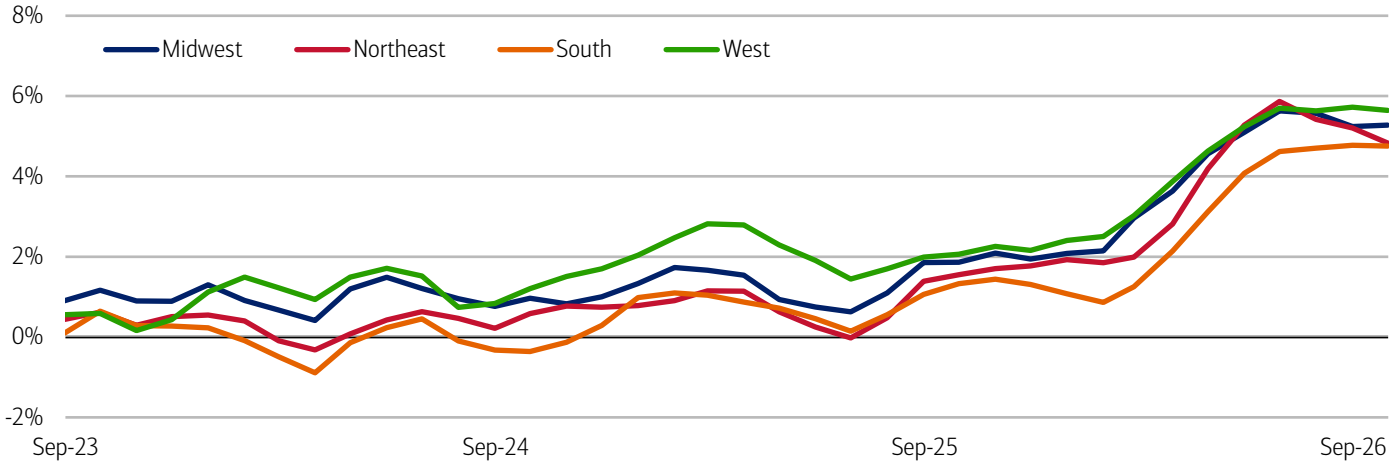
- Bank of America aggregated credit and debit card spending per household is growing at healthy rates across the country. Discretionary spending growth has improved markedly relative to the last two years. While the West has led recently and the South has lagged somewhat, no region appears to be experiencing a meaningful consumer slowdown.
- The South has seen softer discretionary retail spending, particularly in home-related durable goods categories. However, the region continues to post the strongest after-tax wage growth and has been the largest contributor to growth in new business applications nationwide.
- Business formation is occurring across a wide range of industries, with professional services, retail, healthcare, construction and hospitality all accounting for meaningful shares of new business applications. While entrepreneurial activity is broad-based across industries nationwide, industry revenue mixes vary considerably by region.

Spending growth is solid across all regions

Spending growth was strongest in the West in September, while spending growth lagged in the Northeast and South, though not by much. Total credit and debit card spending per household increased around 5.6% year-over-year (YoY) in the West, outpacing other regions by less than a percentage point, according to Bank of America aggregated card data (Exhibit 1). All regions also experienced major improvements in spending growth compared to last year, increasing by over three percentage points in every region.

Spending growth in the Northeast peaked at around 6% YoY in June but remained a solid 4.8% YoY in September. The slowest growth was in the South, though spending was still up nearly 4.8% YoY.

Exhibit 1: Spending growth has mostly improved since the start of the year across all regions
Total card spending per household by US Census region (three-month moving average, YoY%)

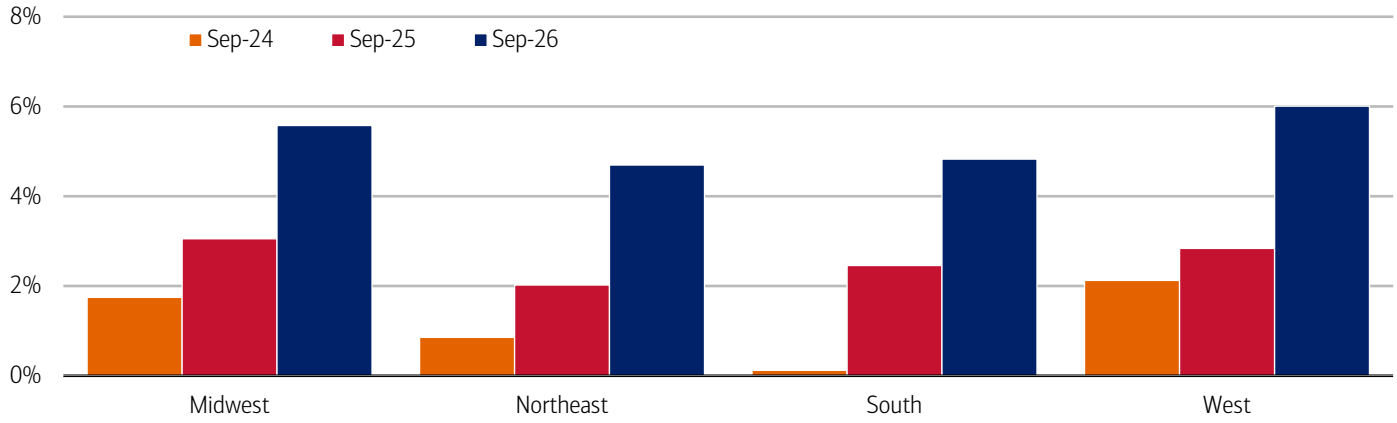


Source: Bank of America internal data

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Discretionary spending growth was also somewhat weaker in the Northeast and South than in the Midwest and West (Exhibit 2). However, these two regions have also seen the most improvement in “nice-to-have” spending over the past two years. And here spending was also up around 4.8% YoY in both regions, still a very healthy rate of growth.

Exhibit 2: Discretionary spending growth has improved across all regions compared to the past two years
Discretionary card spending per household by US Census region (three-month moving average, YoY%)



Source: Bank of America internal data
Note: Discretionary spending is all card spending minus gas, groceries and utilities.

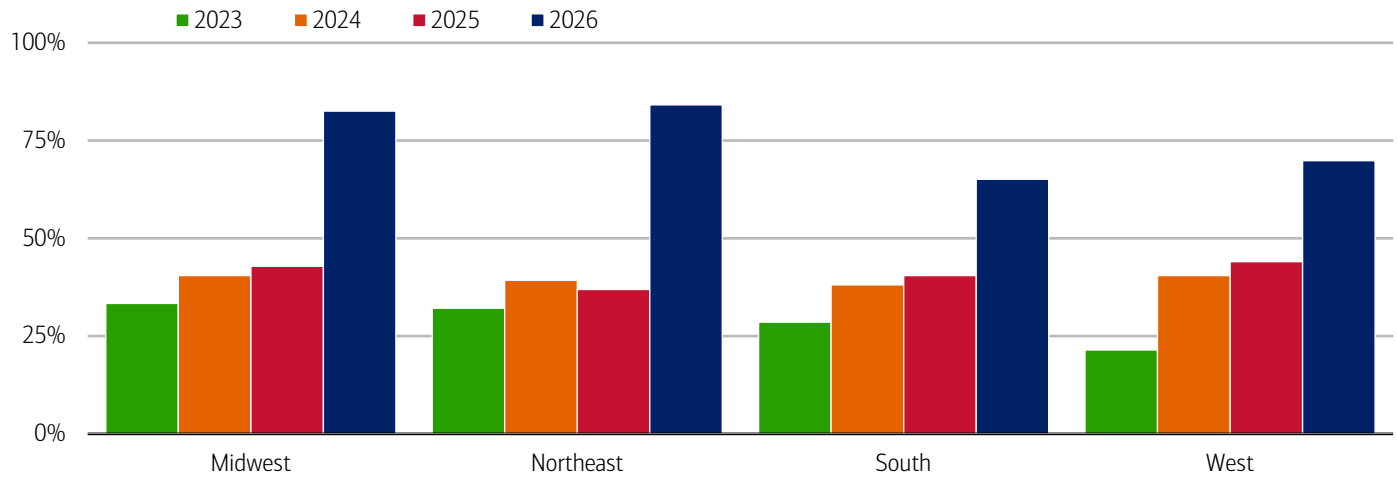
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Durables spending is weighing on discretionary retail spending in the South

One of the key themes from September in the overall spending data was a broadening in spending growth, particularly in "more discretionary" retail categories like clothing, electronics, building materials and non-store/miscellaneous retailers (read more in Consumer Checkpoint: Strength in numbers). Bank of America card data shows that this improvement is occurring across all regions.

However, the South lags, though this is mostly due to declines in spending on home-related durables in, for instance, home improvement stores and at furniture retailers. In our view, this could reflect some slowdown in domestic migration to the Sun Belt and a corresponding easing in home/moving-related spending (read more in On the move: Renovation over relocation?)

Exhibit 3: The Midwest and Northeast have seen a higher share of discretionary retail categories with spending growth when compared to the South and West
Diffusion index across "more discretionary" retail categories by spending across US Census regions (monthly, %)



Source: Bank of America internal data

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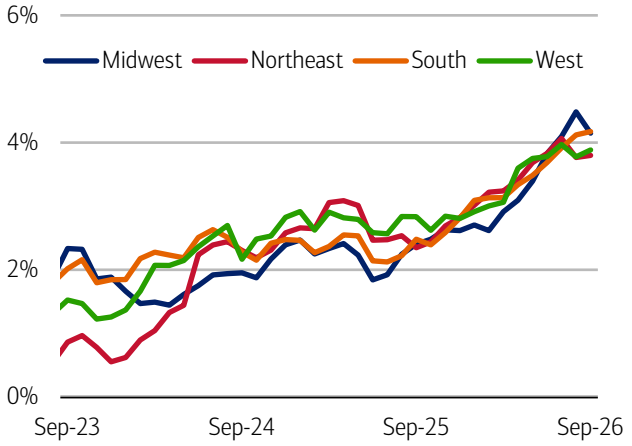
The South leads in after-tax wage and business application growth

One reason we are not overly concerned about consumer spending in the South is the strength of after-tax wage growth in the region. Bank of America deposit data suggests that after-tax wage growth was strongest there, though only by a slim margin, up nearly 4.2% YoY, just ahead of households in the Midwest, and outpacing the West and Northeast by less than half a percentage point (Exhibit 4).

Beyond wage income, business ownership offers another potential channel for wealth accumulation. The South has been the primary contributor to overall US growth in new business applications so far in 2026, according to US Census Bureau data (Exhibit 5).

While historically only a small share of these applications is likely to become a business employing workers, this surge in applications points to a potential pipeline of entrepreneurial activity. For businesses that become established and profitable, ownership can provide a pathway to wealth through retained earnings and the accumulation of business equity. Still, the extent of that opportunity depends on businesses' survival, profitability and financing needs (read more in Business Checkpoint: Looking into logistics).

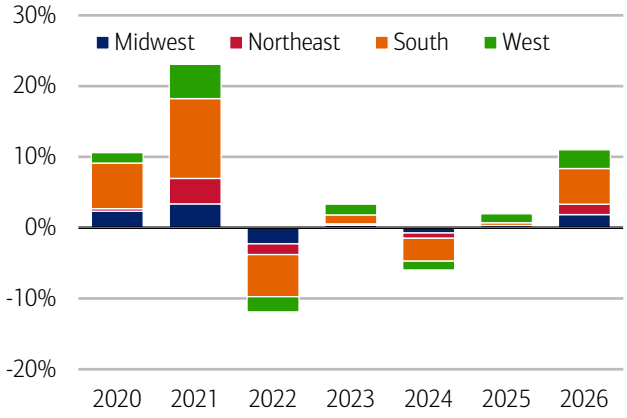
Exhibit 4: After-tax wages are accelerating across all regions, though growth was strongest in the South and Midwest
After-tax wage and salary growth by US Census region (three-month moving average, YoY%, seasonally adjusted (SA))



Source: Bank of America internal data

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Exhibit 5: Nationwide business formation growth was driven primarily by new business applications from the South in 2026
Percentage point contribution to overall new business application YoY growth by US Census region (yearly, percentage points, SA)



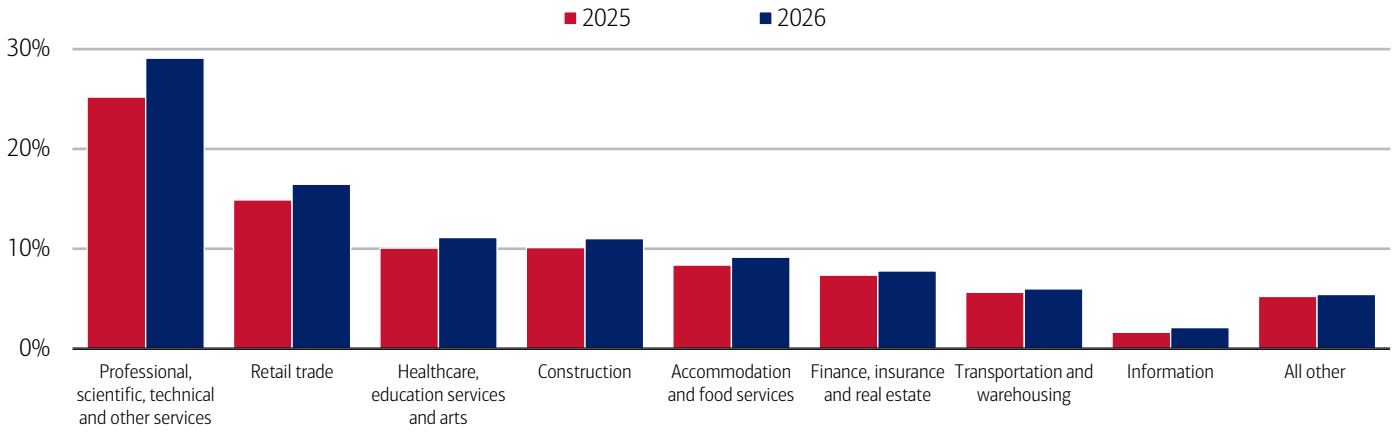
Source: US Census Bureau
Note: 2026 is YTD through September

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Business application growth is widespread across regions and industries

Looking at 2026 business applications by industry, entrepreneurial activity spans multiple industry groupings (Exhibit 6). Professional, scientific, technical and other services account for the largest share, followed by retail trade and the grouping comprising healthcare, education services and the arts. Additionally, businesses provide a significant source of income for US households. In fact, business revenues accounted for nearly 14% of total pre-tax income, according to the US Census' 2025 Survey of Consumer Finances.

Exhibit 6: Professional, scientific, technical and other services industries comprise a significant share of business formation
Share of new business applications by industry groupings (annual, 2026 year-to-date, %)



Source: US Census Bureau
Note: All other includes manufacturing, mining, utilities, wholesale trade, management of companies and enterprises as well as agriculture.

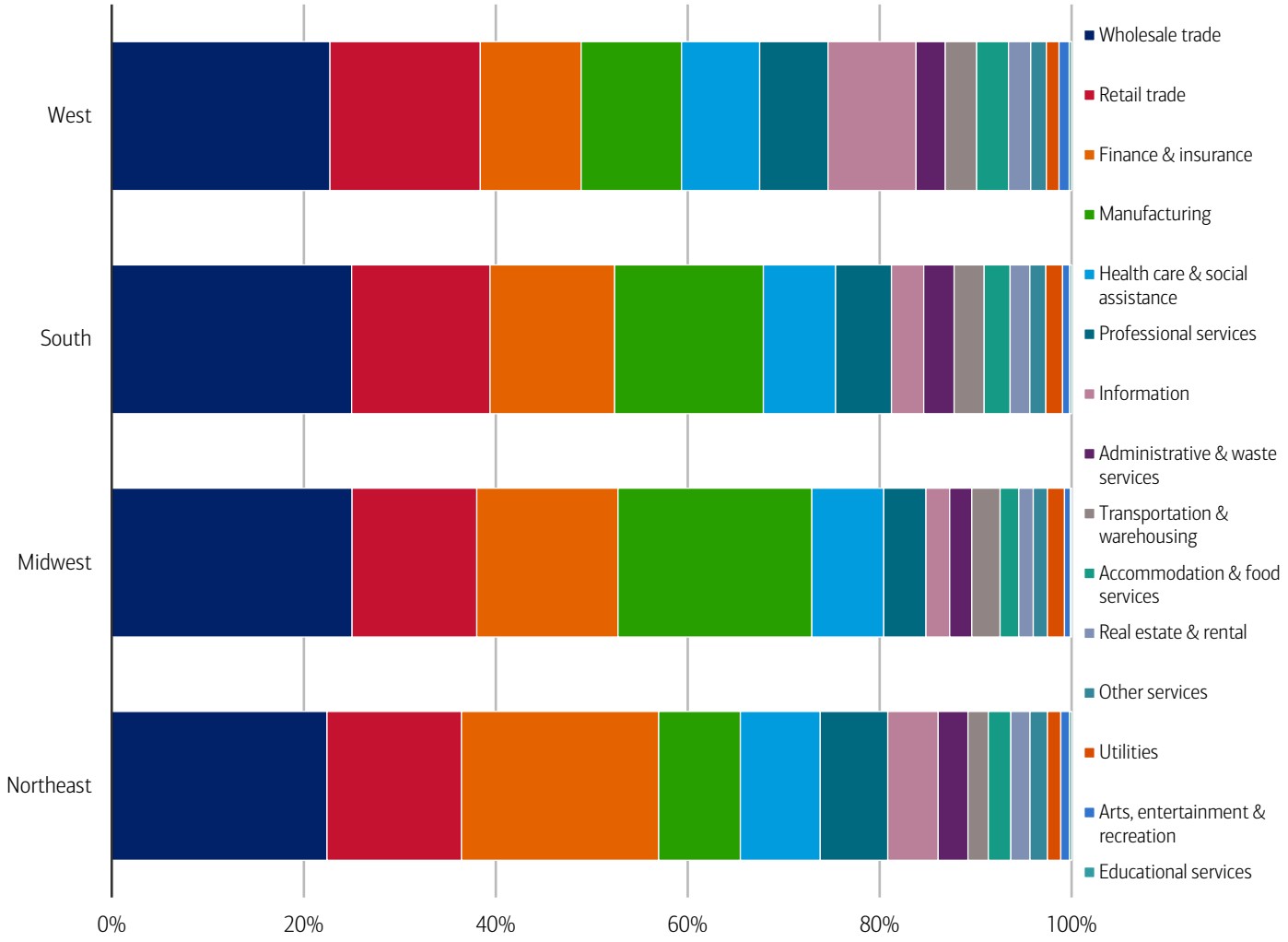
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Combined, this suggests that potential pathways to business ownership extend not only across various households (read more on generational shifts in entrepreneurship in July's A new generation of business formation) but also across many industries, rather than a single sector.

While business applications show the breadth of prospective entrepreneurial activity nationwide, the industry composition of existing business revenue offers a regional lens on business activity. Exhibit 7 shows how revenue was distributed across industries within regions according to the 2024 Annual Integrated Economic Survey from the US Census Bureau, highlighting the extent to which regional economies are concentrated in particular sectors.

Although wholesale trade was the sector with the largest share of revenue in each region in the US, there is notable variance in economic activity from business revenues across the country. For example, finance and insurance accounted for more than 20% of revenue in the Northeast, whereas it's about half that in the West. Also, manufacturing was only 8.5% of revenue in the Northeast, and more than 20% in the Midwest. Overall, the data shows a dynamic national business landscape, with broad entrepreneurial activity but distinct regional differences in the industries generating revenue.

Exhibit 7: Wholesale trade was the sector with the largest share of revenue in each region in the US
Share of revenue by industry by region in 2024 (%)



Source: US Census Bureau
Note: "Revenue" represents "Sales, Value of Shipments, or Revenue" in the source data

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Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend includes total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under $5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards is excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs. Value grocers is defined using a combination of MCC code and proprietary methods.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation, changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Generations, if discussed, are defined as follows: Gen Z, born after 1995; Younger Millennials: born between 1989-1995; Older Millennials: born between 1978-1988; Gen Xers: born between 1965-1977; Baby Boomer: 1946-1964; Traditionalists: pre-1946.

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Additional information about the methodology used to aggregate the data is available upon request.

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Economy

COLA comfort for older consumers?

27 October 2026

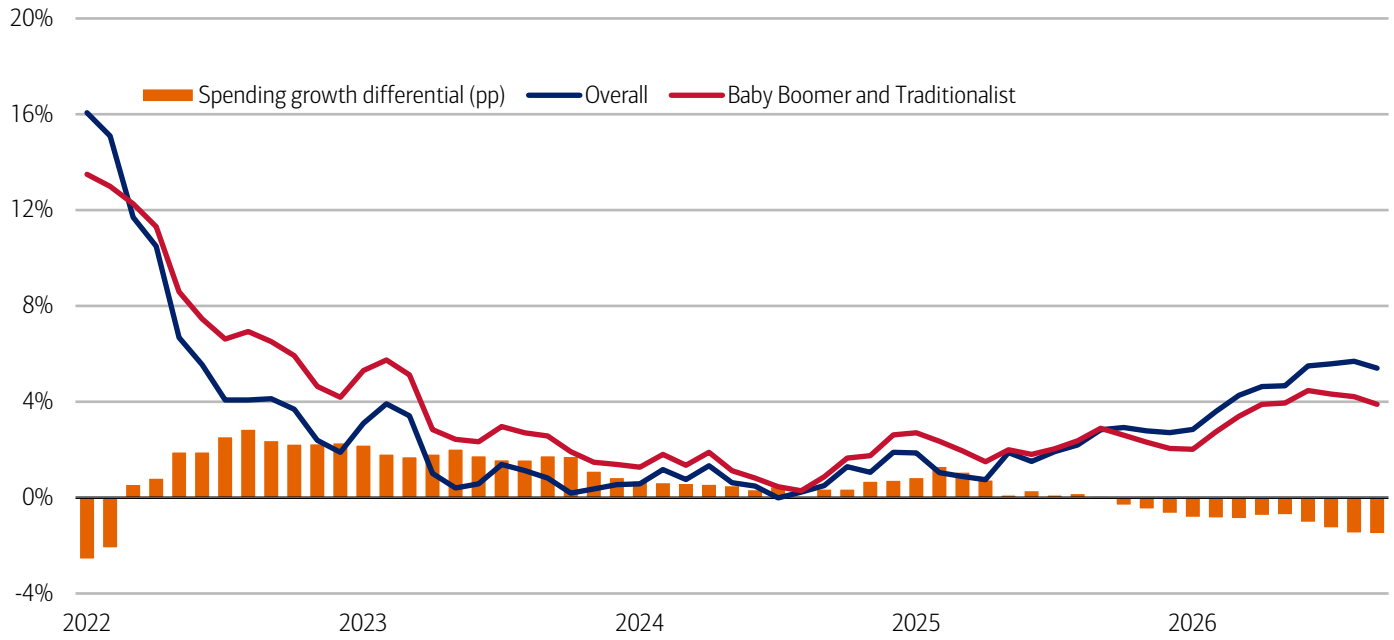
Key takeaways

- Older generations' spending growth has lagged other cohorts over the last year in Bank of America aggregated credit and debit card data, despite sizeable wealth gains. This likely partly reflects the highly uneven distribution of wealth among older households, limiting the extent to which rising wealth supports spending across the cohort.
- More recently, stronger consumer spending appears to be primarily a labor-income story. However, older households are less exposed to accelerating wage growth. In addition, large differences in their income sources help explain the considerable variation in spending among this cohort.
- The recently announced 3.6% Social Security cost-of-living adjustment could provide a meaningful boost in 2027. If inflation moderates as expected, benefit increases may outpace price growth, supporting older households' purchasing power and potentially narrowing their spending gap with other generations.

Older generations' spending growth has lagged

While total credit and debit card spending per household has picked up in 2026 in Bank of America internal data, a notable laggard has been the spending growth of older generations (Baby Boomers and Traditionalists). For around a year, discretionary spending growth for these older generations has been lower than that of overall households. This is in stark contrast to 2022-2023, when they were significantly outstripping overall household spending growth (Exhibit 1).

Exhibit 1: Discretionary spending growth has been slower among older (Baby Boomer and Traditionalist) households despite wealth effects
Total credit and debit card spending per household on discretionary categories by older generations and overall (three-month average, % YoY)

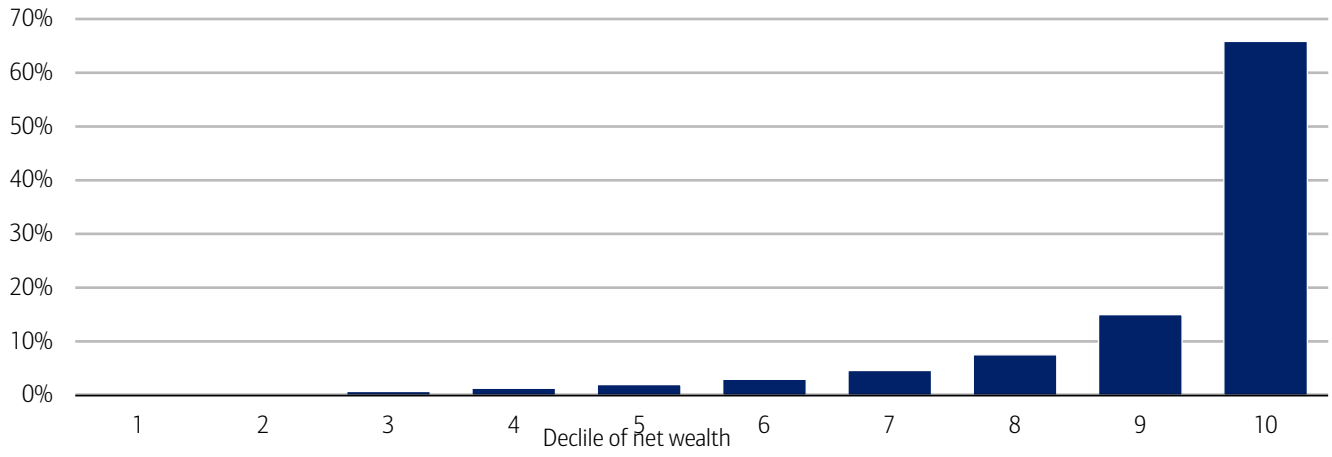


Source: Bank of America internal data

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On one level, this observation seems surprising given that in our piece on the older consumer we noted that a large and growing proportion of net worth is held by older cohorts (read: Entering the age of the older consumer). So what's going on? One explanation is that wealth is very unevenly distributed even within the older cohort. Exhibit 2 uses the latest Federal Reserve (2025) Survey of Consumer Finances to illustrate that, across households older than 65 years of age, around two-thirds of net wealth is held by the wealthiest 10% of households. As a result, wealth gains from rising stock markets may not trickle too far down the wealth distribution of older consumers.

Exhibit 2: The wealth distribution for older households is very skewed
Distribution of net wealth for households over 65 years old (%)



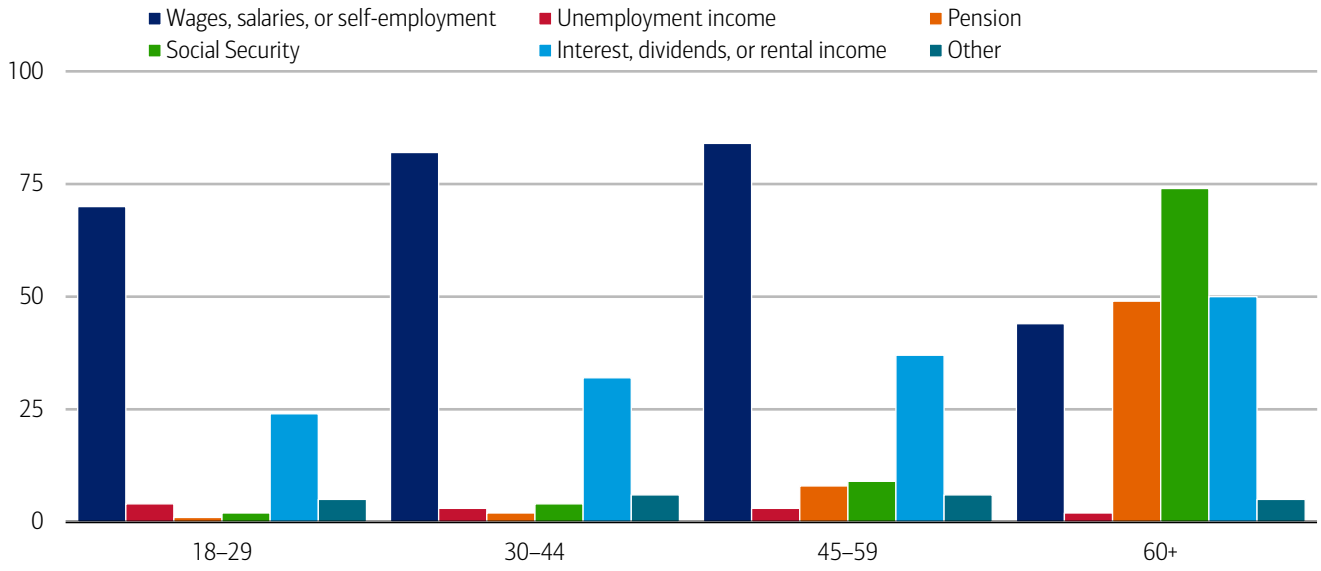
Source: Federal Reserve Board of Governors (2025 Survey of Consumer Finances)

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Income and spending become more diverse and unequal with age

Another important part of the story is that the rise in overall spending growth over the past year appears to be supported by labor market strength as well as rising wealth. The October Consumer Checkpoint described how a pickup in after-tax wage growth among lower- and middle-income households was likely a key factor driving a recovery in these cohorts' spending growth (read: Consumer Checkpoint: Strength in numbers). As households age, their income sources become more diverse (Exhibit 3), which means older generations will likely benefit less from this pickup in wage growth.

Exhibit 3: As households age their sources of income become more diverse
Sources of income by age (%)



Source: Federal Reserve Board of Governors (2025 Survey of Household Economics and Decision making)

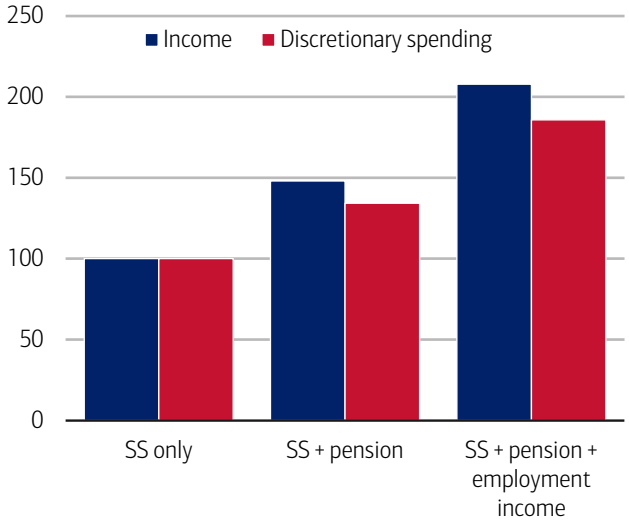
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The greater diversity in income sources among older households helps explain the wide dispersion in their total incomes and spending. For example, using Bank of America internal data, Exhibit 4 shows that households over age 62 that rely solely on

Social Security have average income and discretionary spending levels close to half those of households that also draw income from both pensions and employment. Consistent with this, discretionary spending is distributed more unevenly among older generations by income than it is across all households (Exhibit 5).

Exhibit 4: There is considerable diversity in income and spending across older households…

Income and discretionary card spending in Bank of America internal data for households over age 62 by form of income (year to September 2026, index social security income only=100)

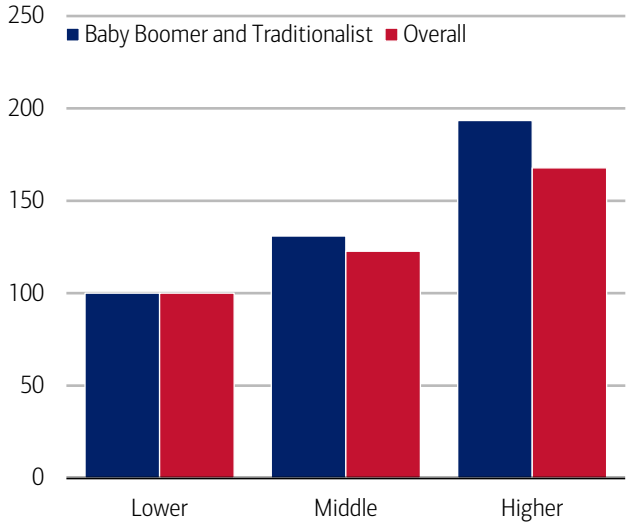


Source: Bank of America internal data. (SS = Social Security payments)

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Exhibit 5: …and the divergence across older generation income cohorts is larger than the overall population

Discretionary card spending per household by income relative to lower-income cohort for older generations and overall households (year to September 2026, index lower-income =100)



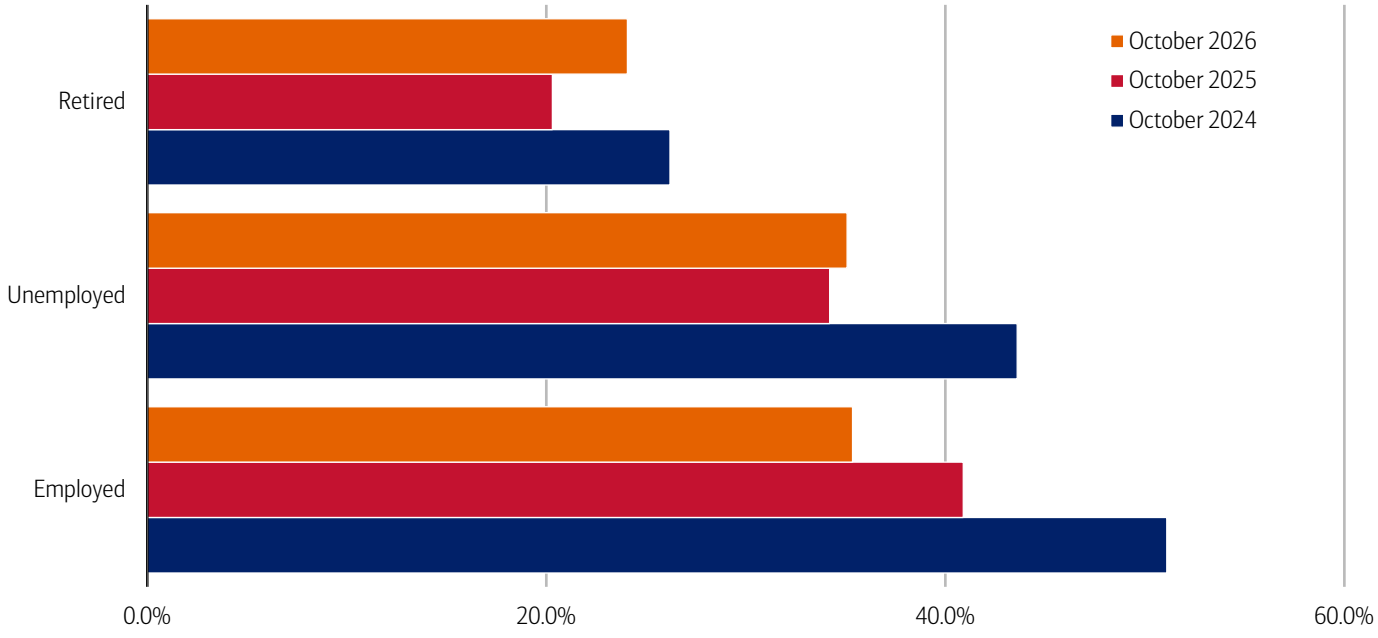
Source: Bank of America internal data

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And, interestingly, the latest Bank of America Market Landscape Insights Study shows that retirees are least confident in an improvement in their finances compared to those in work or unemployed over the next six months (Exhibit 6). Again, this may reflect the lower exposure of many retirees to the labor market, with many on relatively fixed income sources.

Exhibit 6: Retirees expect their finances to be little changed over the next six months

Response to a question on the state of respondent's expectation on their finances over the next six months (% responding "much better" or "somewhat better")



Source: Bank of America Market Landscape Insights Study (October 2026)

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The 2027 COLA may offer some respite

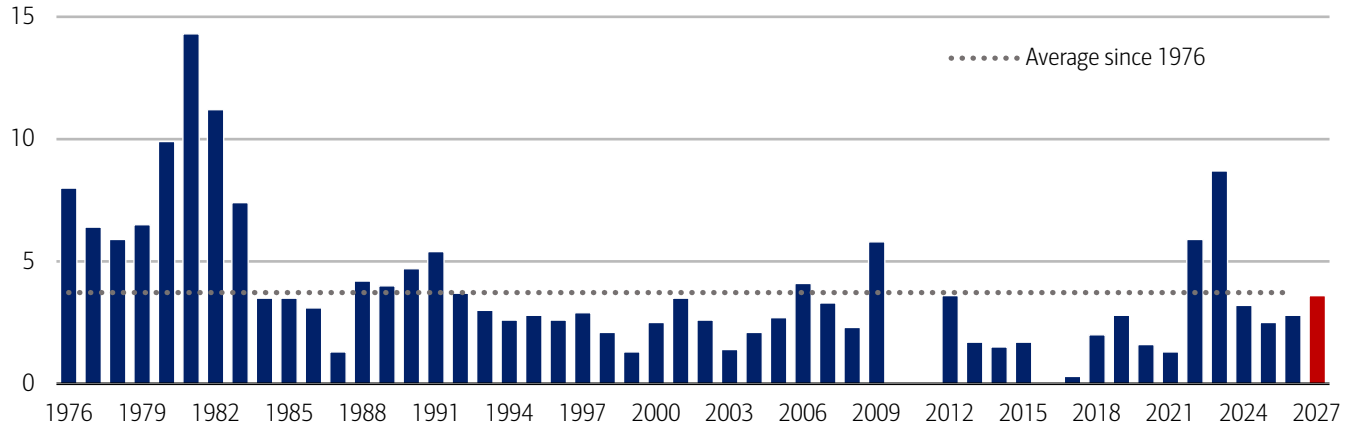
Still in our view there are some reasons for optimism around older generations spending outcomes.

For some older households in the middle of the income distribution, tax policy is likely providing some support to their spending growth. The One Big Beautiful Bill Act introduced a temporary deduction of up to $6,000 for each taxpayer aged 65 or older for 2025-2028, with the benefit phasing out above $75,000 of modified adjusted gross income for single filers and $150,000 for joint filers. This measure is likely lifting after-tax income and spending capacity for older households – particularly those in the middle of the income distribution.

However, households on lower incomes may see less benefit from these policies as they likely already owe limited federal income tax. For these households relying on Social Security either as their sole or as a large component of their income, the forthcoming annual cost-of-living adjustment (COLA) to social security benefits will likely be a more important determinant of their budgets for 2027. The announced COLA for 2027 is a rise of 3.6% (Exhibit 7) – the largest since 2023, though lower than the 8.7% rise in 2022.

Exhibit 7: The 2027 COLA increase is the largest since 2023 and close to the historical average

Social Security cost of living adjustments (% increase as of January effective date)



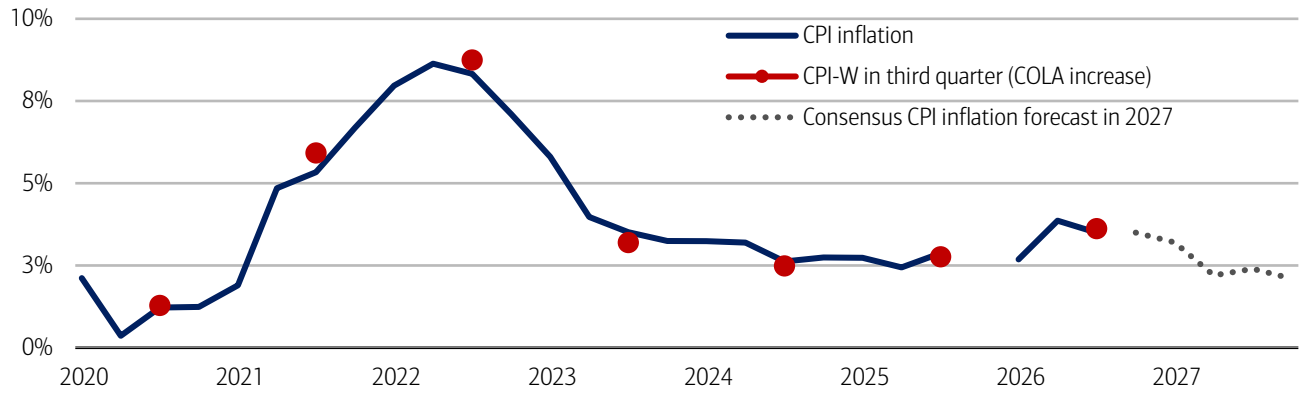
Source: US Social Security Administration (SSA)

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The COLA increases are based on the year-over-year (YoY) % change between the third quarters of successive years in the Consumer Price Index for Urban Wage Earners and Clerical Workers (CPI-W) and are typically announced in October and effective the following January. If the COLA increase simply keeps benefits in line with inflation, why is it particularly interesting? For sure, many times it is not. But when inflation is close to a turning point, the increase can end up looking relatively high or low compared with inflation over the course of the following year. The 2022 COLA increase was a case in point, as inflation in the subsequent year slowed sharply. In our view, this was one reason older generations' spending growth was stronger than that of other age cohorts over 2022-2023.

Exhibit 8: Inflation is expected to decline next year

Consumer Price Index (CPI) (% YoY), third quarter CPI-W increases (red dots, % YoY) and latest consensus CPI forecast for 2027 (dotted line, % YoY)



Source: Haver Analytics, Bloomberg. Some fall 2025 months might be missing due to the government shutdown.

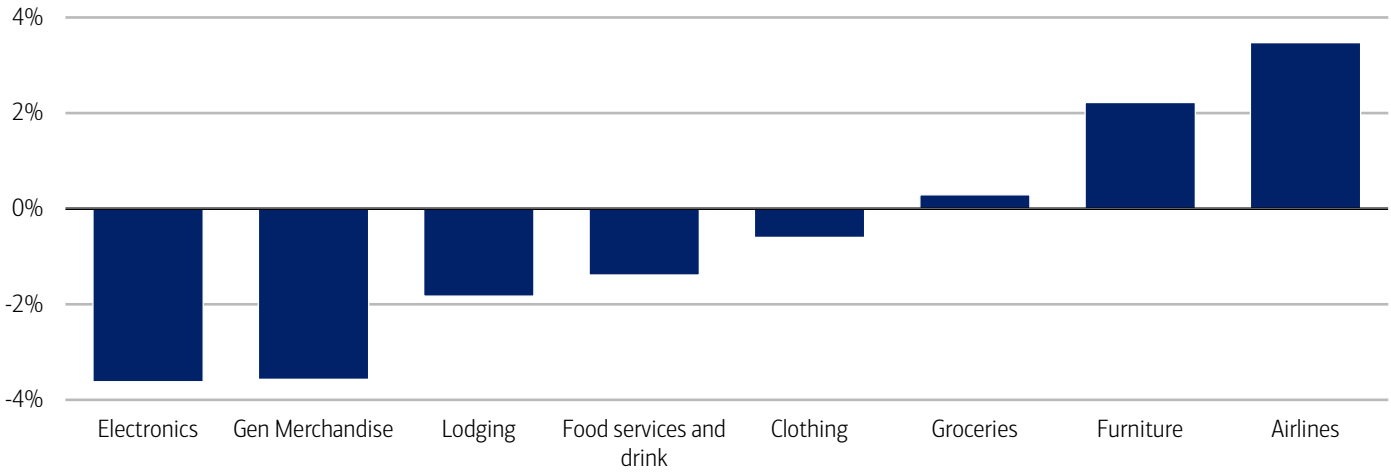
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Regarding the 2027 COLA announcement, there are some parallels with 2022, albeit on a smaller scale. In particular, the consensus expectation is that CPI inflation is likely to cool over 2027, largely reflecting the lessening impact of higher gasoline prices (Exhibit 8).

So if these expectations prove correct, households that rely on Social Security may receive a more impactful boost to their budgets next year than they have for the last few, potentially narrowing the gap between older generations' spending and other households.

What categories of spending growth might benefit from this? It is hard to be sure, but notable growth deficits for older generations currently reside in discretionary areas like electronics, lodging and restaurants (Exhibit 9). Some of these may reflect the preferences of older households, but it could be that there is room for some narrowing in these areas.

Exhibit 9: Large spending growth gaps for older households exist in a range of discretionary goods and services
Difference between older generations' (baby boomer and traditionalist) spending and overall households' YoY spending growth by category (three-month moving average, percentage points)



Source: Bank of America internal data

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Methodology

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Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under $5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate (if discussed) can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.
2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.
3. Overall total card spending includes small business card spending while per household card spending does not.
4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).
5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

Store category aggregation for "more discretionary", "partly discretionary", and "mostly necessity" is defined as follows:

More discretionary: furniture, electronics, building materials, hobbies, miscellaneous stores, nonstore retail, clothing.

Partly discretionary: auto, health and personal care, general merchandise.

Mostly necessity: Gasoline, food and beverage.

We consider a measure of services necessity spending that includes but is not limited to groceries, utilities, and gasoline. Discretionary services includes but is not limited to charitable donations, leisure travel, entertainment, and professional/consumer services. Discretionary retail includes but is not limited to general merchandise, miscellaneous, clothing, electronics and furniture. It excludes categories like groceries and gasoline. Holiday spending is defined as items in which spending in the November-December period is usually at least 20% of total annual spending on the category. Value and premium grocers are determined judgmentally on a proprietary analysis of merchants by equity analysts. The "more discretionary" retail category includes furniture, electronics, building materials, online marketplaces, clothing retailers and miscellaneous stores; while "partly discretionary" retail contains auto parts and services, health and personal care retailers, general merchandise and "mostly necessity" retail contains gas and groceries.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation, changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Generations, if discussed, are defined as follows: Gen Z, born after 1995; Younger Millennials: born between 1989-1995; Older Millennials: born between 1978-1988; Gen Xers: born between 1965-1977; Baby Boomer: 1946-1964; Traditionalists: pre-1946.

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Bank of America Proprietary Market Landscape Insights Study is an online quantitative survey among Bank of America customers and noncustomers sampled and balanced to provide a representative view of the U.S. adult population. Insights are based on aggregated and anonymized responses to surveys. Significance testing is done at the 90 percent confidence interval.

Additional information about the methodology used to aggregate the data is available upon request.

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Economy

Kicking and screaming: Child care costs persist

20 October 2026

Key takeaways

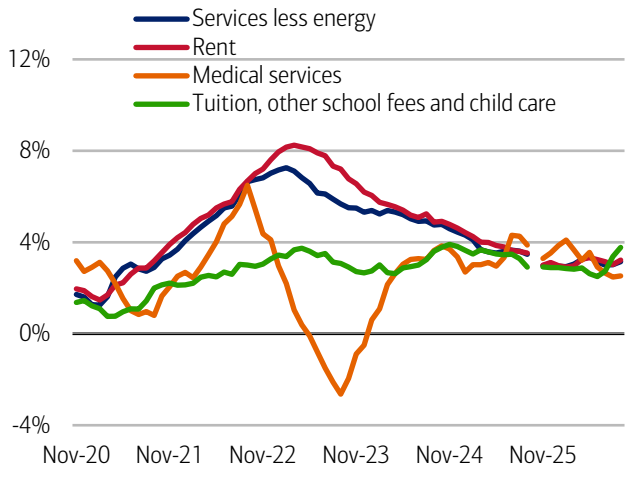
- Child care remains a growing financial burden for many households, even as growth in the median payment has moderated, according to Bank of America payments data. More families are paying more than $2,000 per month for care, suggesting affordability pressures are becoming increasingly concentrated among households facing the highest expenses.
- Child care cost pressures vary significantly across the country. Nashville and Jacksonville, FL posted the fastest year-over-year growth among major metro areas, while New York and Washington DC saw the biggest decrease in median payment growth, according to Bank of America payments data.
- Limited supply may continue to constrain access to child care for some households. Child care employment fell 0.7% year-over-year in September, suggesting provider capacity remains under pressure and could continue to affect household finances and labor force participation.

Child care costs outpace rent and medical services

Child care remains a significant expense for many families. In September, the growth in child care and other school-related fees surpassed other major household expenses such as rent and medical services, according to the Bureau of Labor Statistics (BLS) (Exhibit 1).

Yet, while child care prices continue to rise in official inflation data, median household payments have largely moderated since February, according to Bank of America data (Exhibit 2). More importantly, the growth in the number of households paying for child care has fallen since the start of 2025.

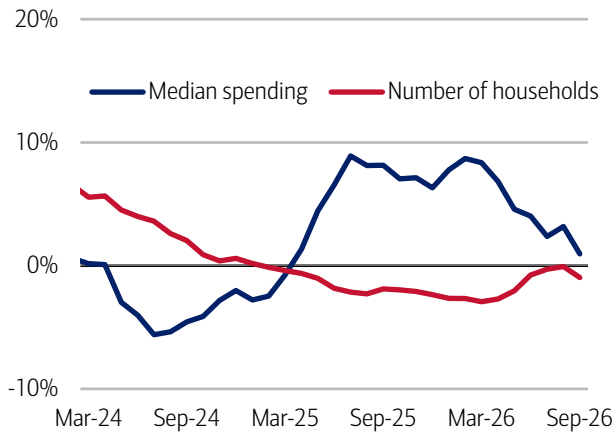
Exhibit 1: The cost of child care has surpassed other services such as rent in the consumer price index
Consumer price index by category (monthly, seasonally adjusted, YoY%)



Source: Haver Analytics

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Exhibit 2: The median child care payment per household was up 1.0% year-over-year (YoY) in September but the number of households paying for child care fell 1.0% YoY
Median child care payment per household and number of households (monthly, three-month moving average, YoY%)



Source: Bank of America internal data

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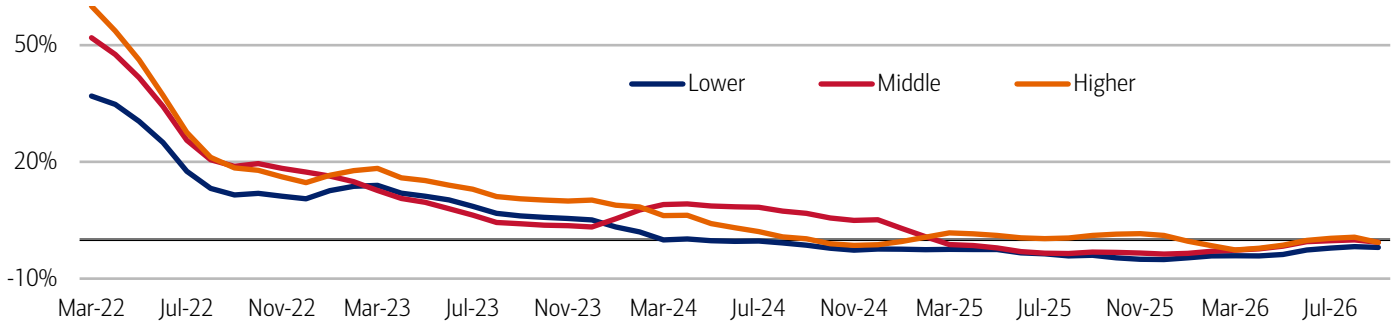
Fewer lower-income households are paying for child care

Why is Bank of America payment growth trending in the opposite direction of the Consumer Price Index? Households are likely changing the amount of care they use, which is reflected in slower growth in the number paying for child care. As more

households shift into lower-cost care (school-age, public pre-K, subsidized or free after-school programs), median payments can fall even as prices continue rising.

The decline in the number of households paying for child care may also reflect families seeking alternatives to formal care arrangements. Some may be relying more heavily on relatives, shifting parents' work schedules or dropping out of the workforce altogether to care for their children in response to elevated costs. This could explain why, among all income groups paying for care, the number of lower-income households has seen the biggest decline, falling 2.0% year-over-year (YoY) in September (Exhibit 3).

Exhibit 3: The number of lower-income households paying for child care has fallen the most, down 2.0% YoY in September
Number of households with child care payments (monthly, three-month moving average, YoY%)



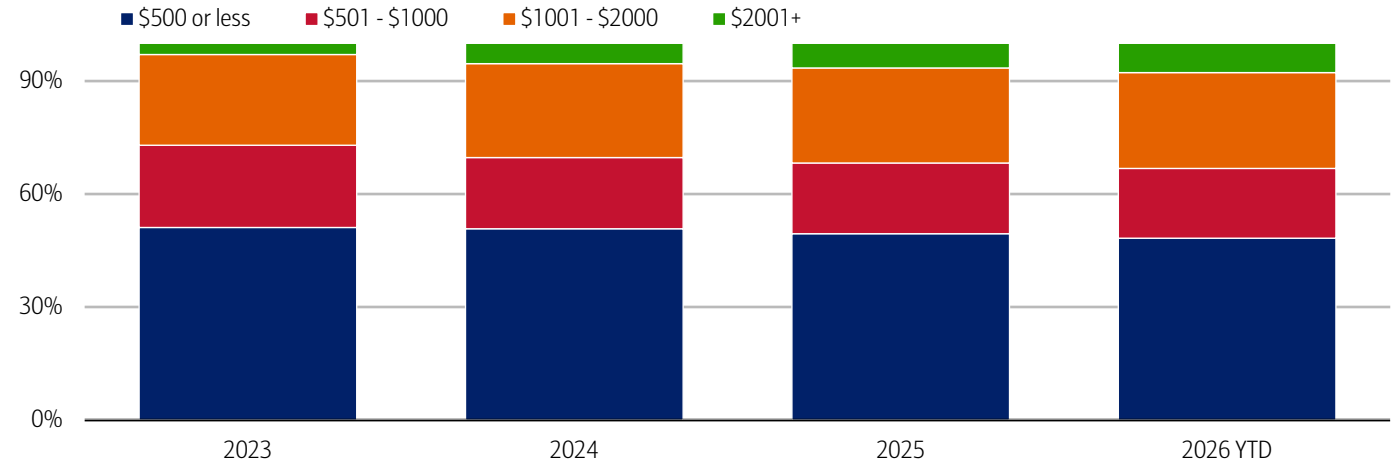
Source: Bank of America internal data

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More families face larger monthly bills

Yet even as fewer households pay for child care, the share of those shelling out more than $2,000 per month has increased substantially (Exhibit 4). In fact, it has more than doubled since 2023, reaching 7.7% in 2026 year-to-date. Meanwhile, the share of those paying $1,000 or less has declined, suggesting higher monthly payments are becoming more common even as the national median grows more slowly.

Exhibit 4: The share of households paying >$2000 a month for child care has more than doubled since 2023
Monthly child care payments by amount (annual average, share of households, %)



Source: Bank of America internal data

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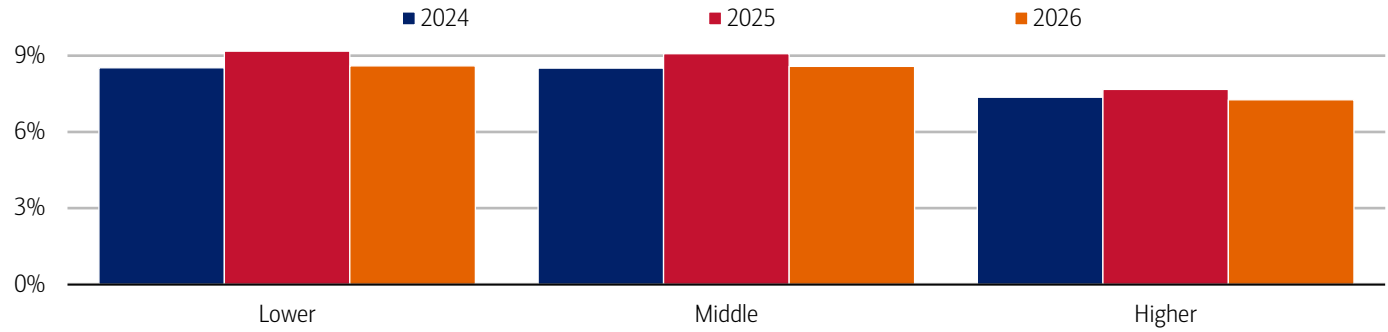
Rising costs take a bigger bite out of households' paychecks

What makes child care "affordable"? According to the U.S. Department of Health and Human Services, child care is considered affordable if it accounts for no more than 7% of family income. Yet, in Bank of America data, the average child care cost was 8% in September.

While an 8% share may not appear dramatically above the federal affordability benchmark of 7%, it represents an average across households' income levels. For families in high-cost metro areas or those paying more than $2,000 per month, the burden is likely considerably higher.

Notably, lower- and middle-income households see outsized shares of their pay dedicated to child care in Bank of America payments data, though this has moderated slightly from 2025 (Exhibit 5).

Exhibit 5: In September, the average child care payment was 8.6% of lower- and middle-income total households' pay, whereas for higher-income families it accounted for 7.3%
Average child care payment as a share of pay in September by income (annual, %)



Source: Bank of America internal data

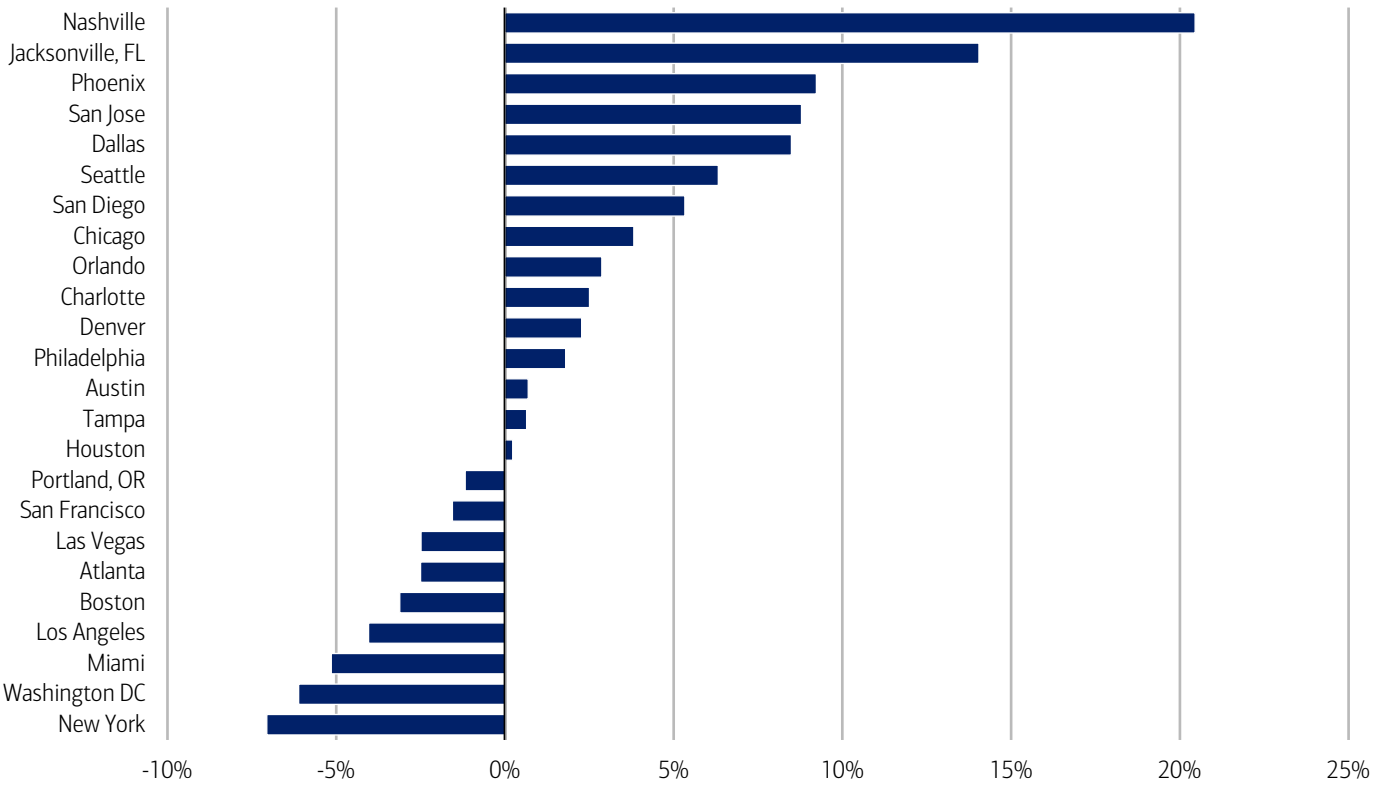
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Where child care costs are climbing

Child care pressures also differ substantially by location. In Bank of America payments data, Nashville posted the fastest YoY median payment growth in September, followed by Jacksonville, FL and Phoenix (Exhibit 6). This dispersion of growth across the country suggests child care cost pressures are no longer confined to historically expensive coastal markets.

Interestingly, New York and Washington DC saw the sharpest decline in growth this year. Why might this be? The supply side of the market may help explain why – for example, New York City recently introduced universal Pre-K.[1]

Exhibit 6: Nashville and Jacksonville, FL saw the greatest increase in median child care payment growth among select cities in September
Median child care payments by core-based statistical area in September (YoY%)



Source: Bank of America internal data

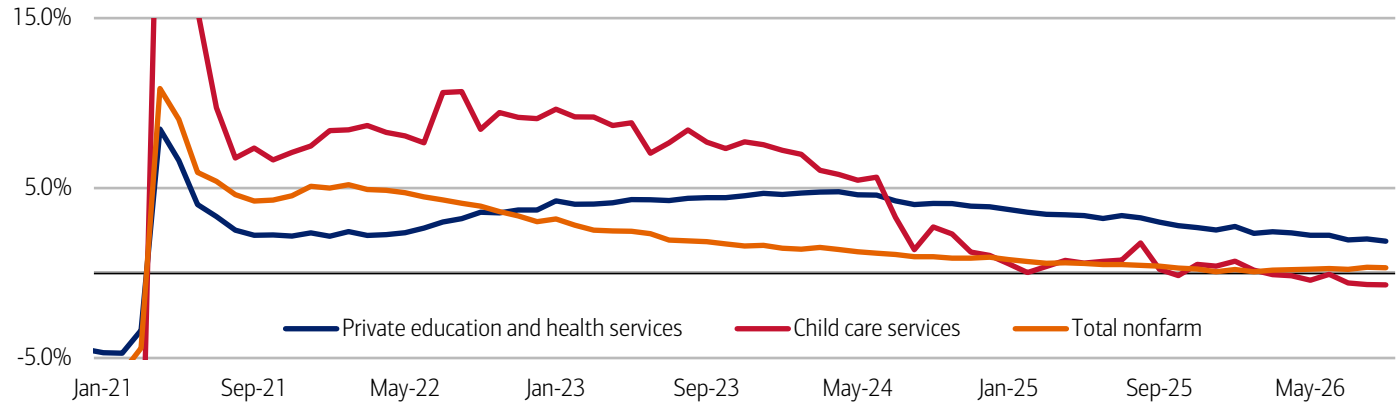
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[1] NYC Public Schools. 2026. Pre-K. New York City Department of Education.

Fewer providers, higher costs

As of September 2026, employment in child care services was down 0.7% from a year earlier, suggesting the sector's workforce recovery has stalled. Fewer workers can constrain the number of available places or the hours providers offer. That may leave families facing longer searches, fewer workable arrangements or continued price pressure. In turn, access to care can influence parents' ability to enter, remain in or expand their participation in the workforce.

Exhibit 7: Employment within child care services fell 0.7% YoY in September
Employees by sector (monthly, seasonally adjusted, YoY%)



Source: Haver Analytics

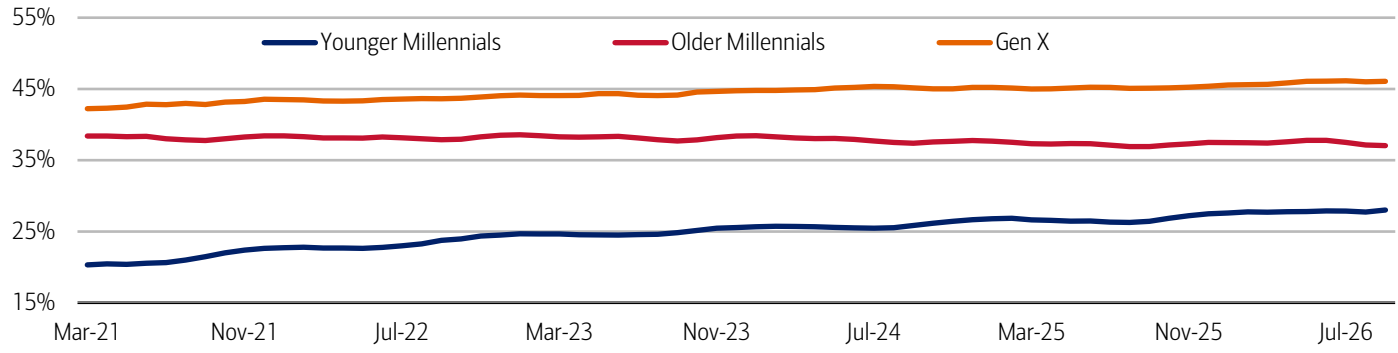
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What are the economic implications of rising child care costs?

Child care costs may have important labor market implications for families with young children. Of those households paying for child care in 2026, more than half of them were Older Millennials (52%), followed by Younger Millennials (20%) and Gen X (16%), according to Bank of America payments data. This means child care costs weigh more heavily on Older Millennials.

The share of Older Millennial households receiving more than one paycheck has continued to grow from a year earlier, but the pace of that improvement has moderated over the past three months (Exhibit 8). Given that this group accounts for the majority of households paying for child care in Bank of America payments data, they may offer early signs of how affordability affects their employment decisions.

Exhibit 8: The share of Older Millennial households paying for child care that have more than one person with a paycheck has declined since 2021
Share of households with a child care payment with more than one person with a paycheck (%, monthly, three-month moving average)



Source: Bank of America internal data

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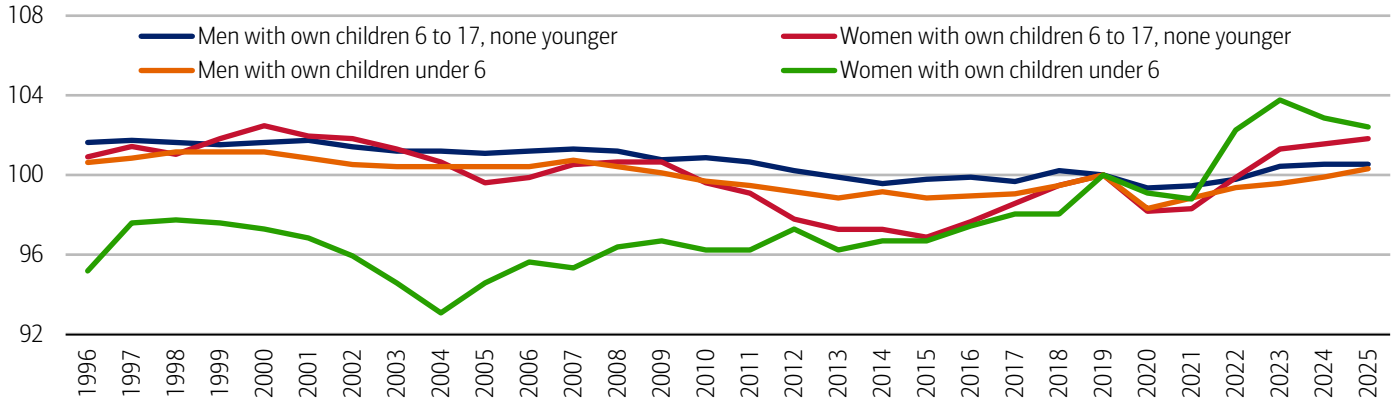
Child care challenges can weigh on women's employment

According to data from the American Time Use Survey, women continue to perform significantly more child care and household work than men, even as their participation in paid work has increased. This suggests that persistent gaps in access and affordability may have disproportionate consequences for women's employment. Indeed, the percentage of women who cited child care as a reason for temporary unemployment was nearly three times higher than men's in 2026, though the average fell from 2025, according to data from the BLS.

Yet labor force participation data indicates a more nuanced picture. Despite persistent child care pressures, mothers are participating in the workforce at rates that remain above their 2019 averages.

The strongest gains have occurred among women with children under age six, whose labor force participation rate remains well above pre-pandemic levels (Exhibit 9). Women with school-age children have also seen participation rise relative to 2019, though by a smaller margin.

Exhibit 9: In 2025, women with children under age six saw the greatest increase in their participation rate compared to the 2019 average
Annual estimates for labor force participation rates by sex and age of children (annual averages, indexed, 2019 average = 100)



Source: Bureau of Labor Statistics

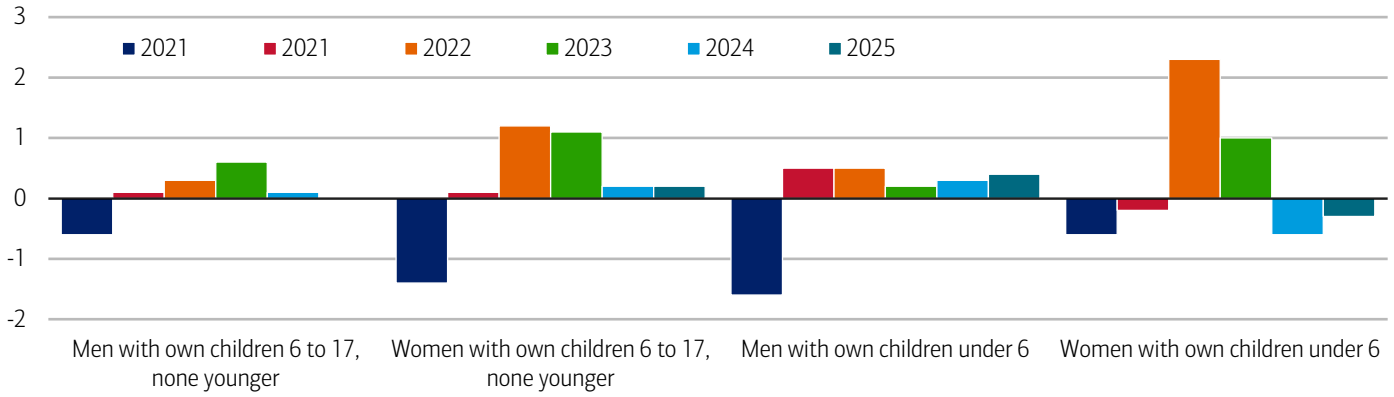
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Mothers' workforce participation remains elevated, but recent gains have slowed

However, recent momentum has slowed. While mothers of young children experienced particularly large participation gains in 2022 and 2023, those have partially reversed in 2024 and 2025 (Exhibit 10).

In contrast, participation among women with children ages 6 to 17 has continued to edge higher. This signals that the post-pandemic increase in working has largely been sustained in this group, but the pace of improvement has moderated, especially for women caring for younger children.

Exhibit 10: The labor force participation rate for women with children under age six fell in 2025 from the prior year
Annual estimates for labor force participation rates by sex and age of children (annual averages, percentage point difference from prior year)



Source: Bureau of Labor Statistics

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Economy

Insurance costs hit the brakes

16 October 2026

Key takeaways

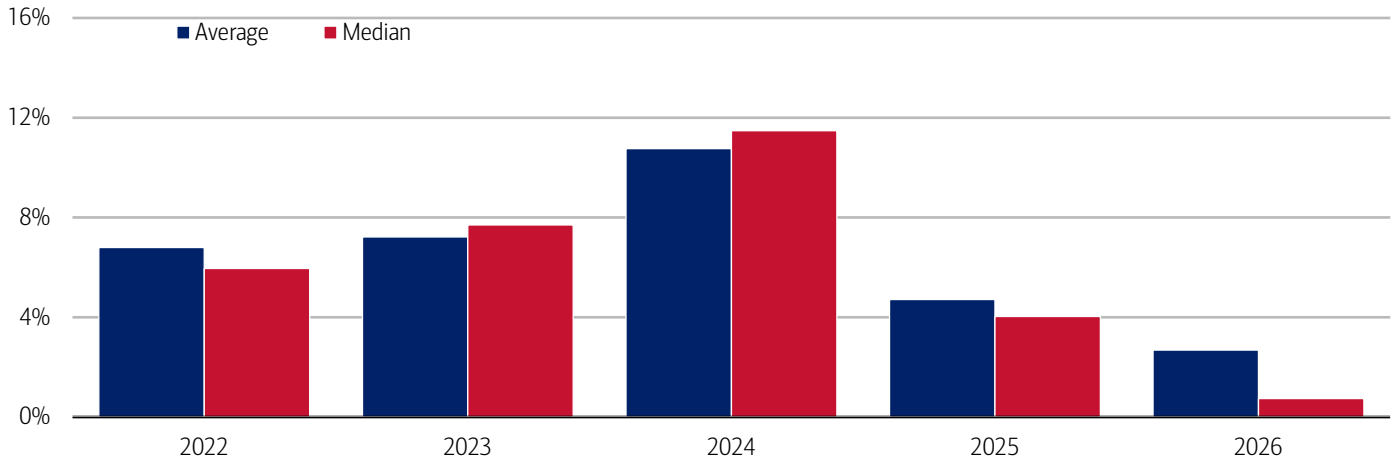
- Property insurance (e.g., auto, homeowners, renters) payment growth has slowed considerably, according to Bank of America payments data. Median payments rose just 1% year-over-year (YoY) in 2026, easing further from the 6% growth seen in 2025, after a roughly 12% increase in 2024, suggesting that many consumers are getting some relief from rising insurance payments.
- However, the moderation is uneven across consumers. Bank of America payments data suggests that Gen Z continues to experience insurance payment growth of nearly 6%, while increases for other generations are closer to 1% to 2%. Payments also rose nearly 1.5% for higher-income households but edged lower for those with lower incomes.
- Lower accident rates, easing vehicle prices, and softer major-purchase growth may be reducing cost pressures, but geography remains a major dividing line. Florida metros recorded declines of roughly 7% to 8%, while California and several Northeastern metros continued to see relatively strong increases.

Insurance payments stabilize after years of painful growth

The burden of higher property (e.g., auto, renters, homeowners) insurance payments may be easing for some. Notably, while the average payment increased 3% year-over-year (YoY), the median payment was only up 1%, according to Bank of America payments data. This suggests that most consumers have only seen a small increase in their premiums, though there may be a few outliers (Exhibit 1).

The latest data stands in contrast to just two years ago in 2024, when most consumers saw their insurance payments surge about 12% YoY, after a nearly 8% YoY increase in 2023.

Exhibit 1: Insurance payments have increased only modestly through August of this year, with growth peaking in 2024
Median and average insurance payments (September to August yearly, YoY%)

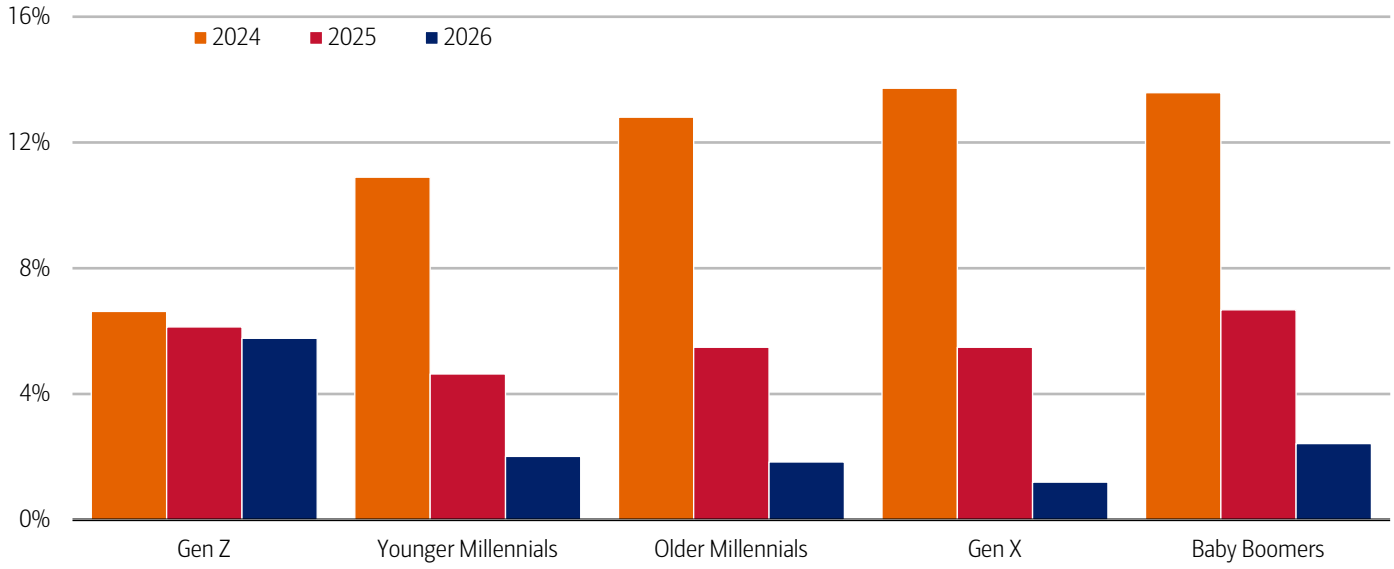


Source: Bank of America internal data

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Insurance payments by age show a similar pattern, with much stronger growth in 2024 across most generations followed by a cooling since then (Exhibit 2). Gen Z is the outlier here, with their payments still growing at a healthy rate, increasing almost 6% YoY for both 2026 and 2025, after a 7% increase in 2024. Meanwhile, Millennials and Baby Boomers experienced a 2% YoY increase so far this year, with Gen X only seeing around a 1% rise.

Exhibit 2: Insurance payment growth is easing significantly for every generation except Gen Z
Median insurance payments by age generation (September to August yearly, YoY%)

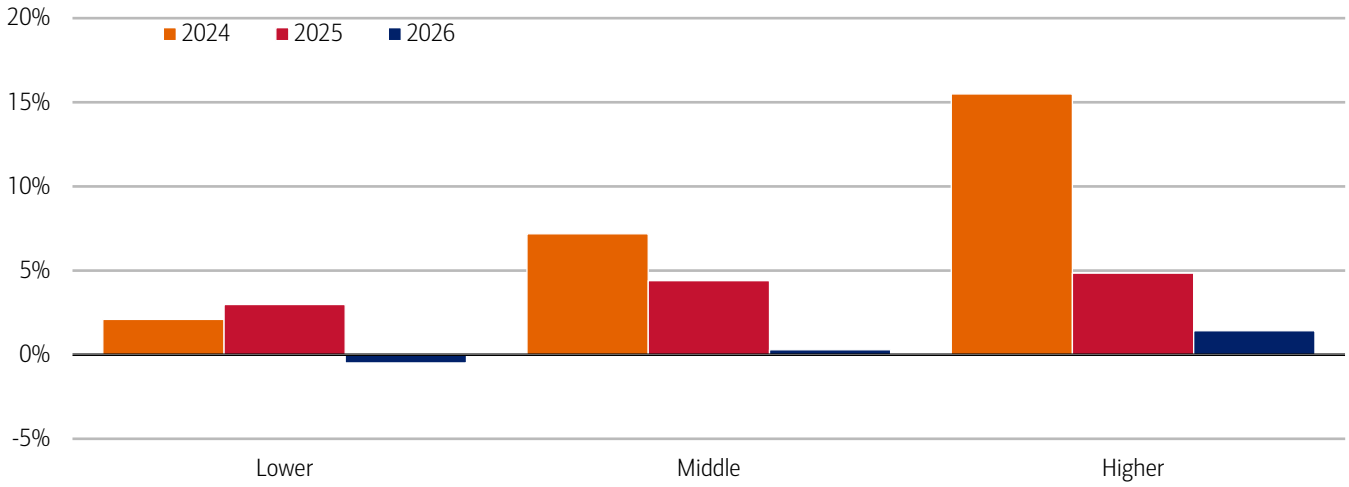


Source: Bank of America internal data

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Bank of America payments data by income provides even more insight. Higher-income households paid nearly 1.5% more in 2026 than they did a year ago, while middle-income households saw only modest increases or none at all (Exhibit 3). Conversely, lower-income households actually paid slightly less so far this year than they did last year.

Exhibit 3: Higher-income households are still paying more for insurance compared to last year, while lower earners are paying less
Median insurance payments by income cuts (September to August yearly, YoY%)



Source: Bank of America internal data

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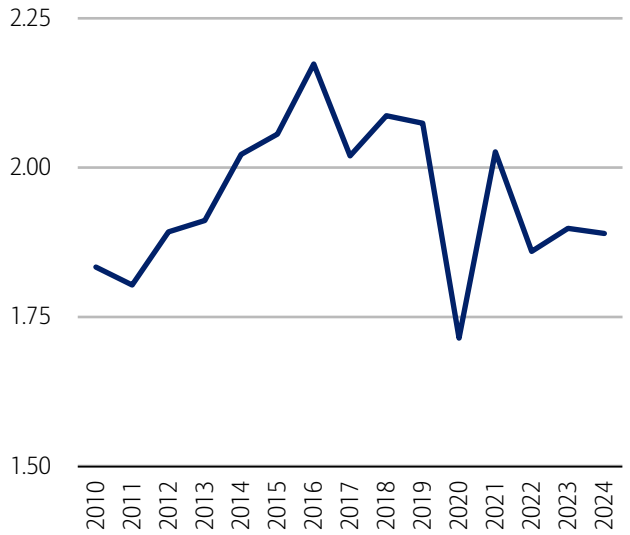
Fewer accidents and lower vehicles prices may be driving insurance payments down

But why might insurance payments be declining at all? One thing to keep in mind is that insurance companies often raise rates to cover events that occurred in previous years to recover costs, while adjusting for changing risks.

For example, the crash rate – the number of vehicle crashes per one million miles driven – suggests that there was a sharp rise in accidents in 2021, according to the Department of Transportation (Exhibit 4). Additionally, new and used vehicle prices increased significantly after the pandemic, according to Consumer Price Index (CPI) data from the Bureau of Labor Statistics (BLS) (Exhibit 5). So, insurers were not only facing more vehicle replacements due to a higher crash rate, but a higher cost to replace vehicles as well.

However, the crash rate mostly declined from 2021 to 2024, and the Department of Transportation reported that the number of accidents likely decreased again in 2025. Also, while the cost to maintain and repair vehicles has consistently grown since 2021, the price to replace vehicles peaked in 2022 and has eased since then – the latter likely a more important factor in insurance costs than the former. Additionally, while the CPI index for motor vehicle insurance has fallen 5% YoY, it's possible that this may partially be due to rebates, according to BofA Global Research. So, the price figured into CPI may not be fully realized by the consumer.

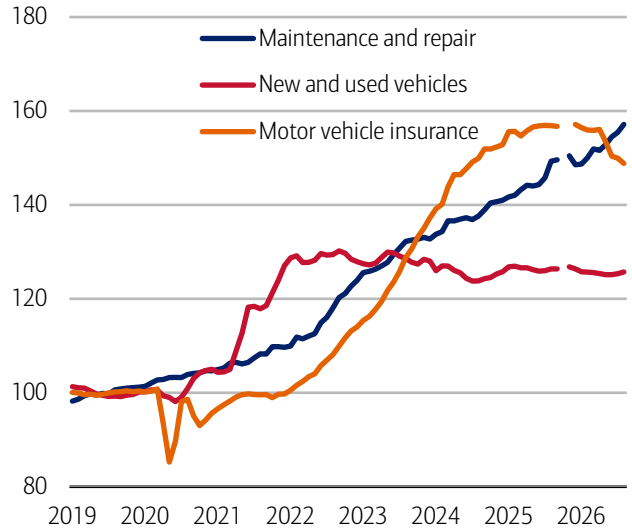
Exhibit 4: The rate of crashes is below pre- and post-pandemic levels
Number of crashes per one million miles driven (yearly, %)



Source: Department of Transportation

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Exhibit 5: New and used vehicle inflation has eased since 2022
CPI for vehicle maintenance and repair, new and used vehicles and motor vehicle insurance (monthly, 2019 average = 100)

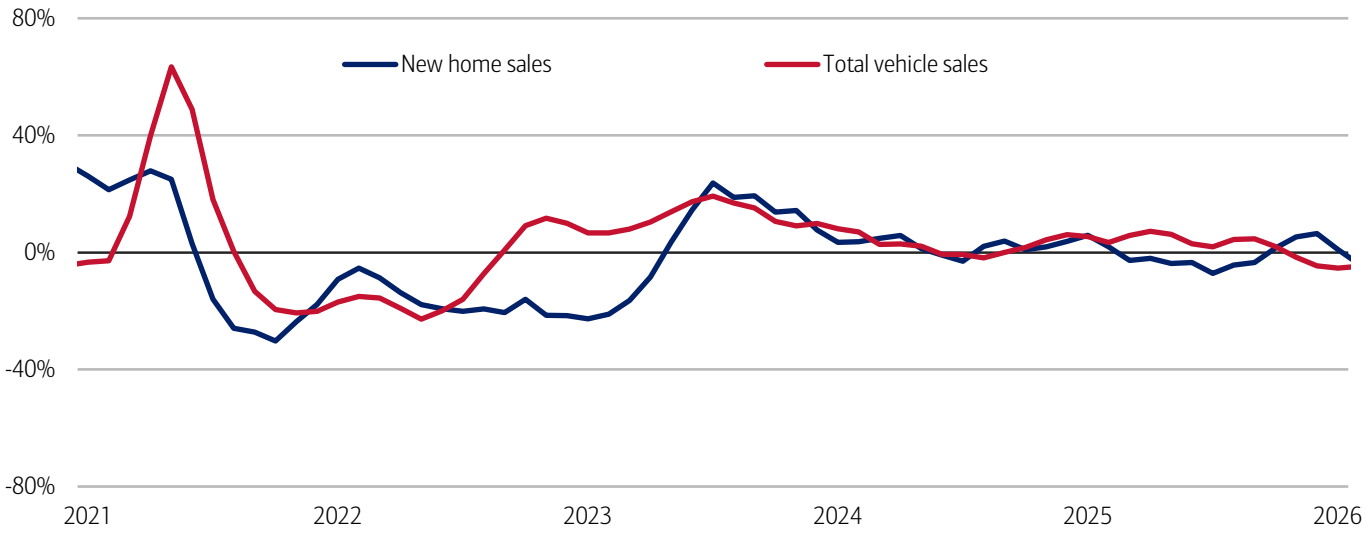


Source: BLS. Some fall 2025 months might be missing due to the government shutdown.

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Another factor in the divergence between a declining CPI and rising insurance payments in our data may be the growth in the amount of property people are insuring now. Notably though, this growth has eased in recent years. YoY growth in both new home and total vehicle sales peaked in mid-2023 and has grown modestly, or even declined, since (Exhibit 6). And given that these major purchases are still K-shaped (read more in Autos: In gear, but the "K" is still in the headlights and On the move: Renovation over relocation?), it could help to explain the divergence in insurance payment growth across income cohorts.

Exhibit 6: Growth in new home and total vehicle sales peaked in mid-2023 due to a post-pandemic boom but has eased since
New home and total vehicles sales (three-month moving average, YoY%)



Source: Haver analytics

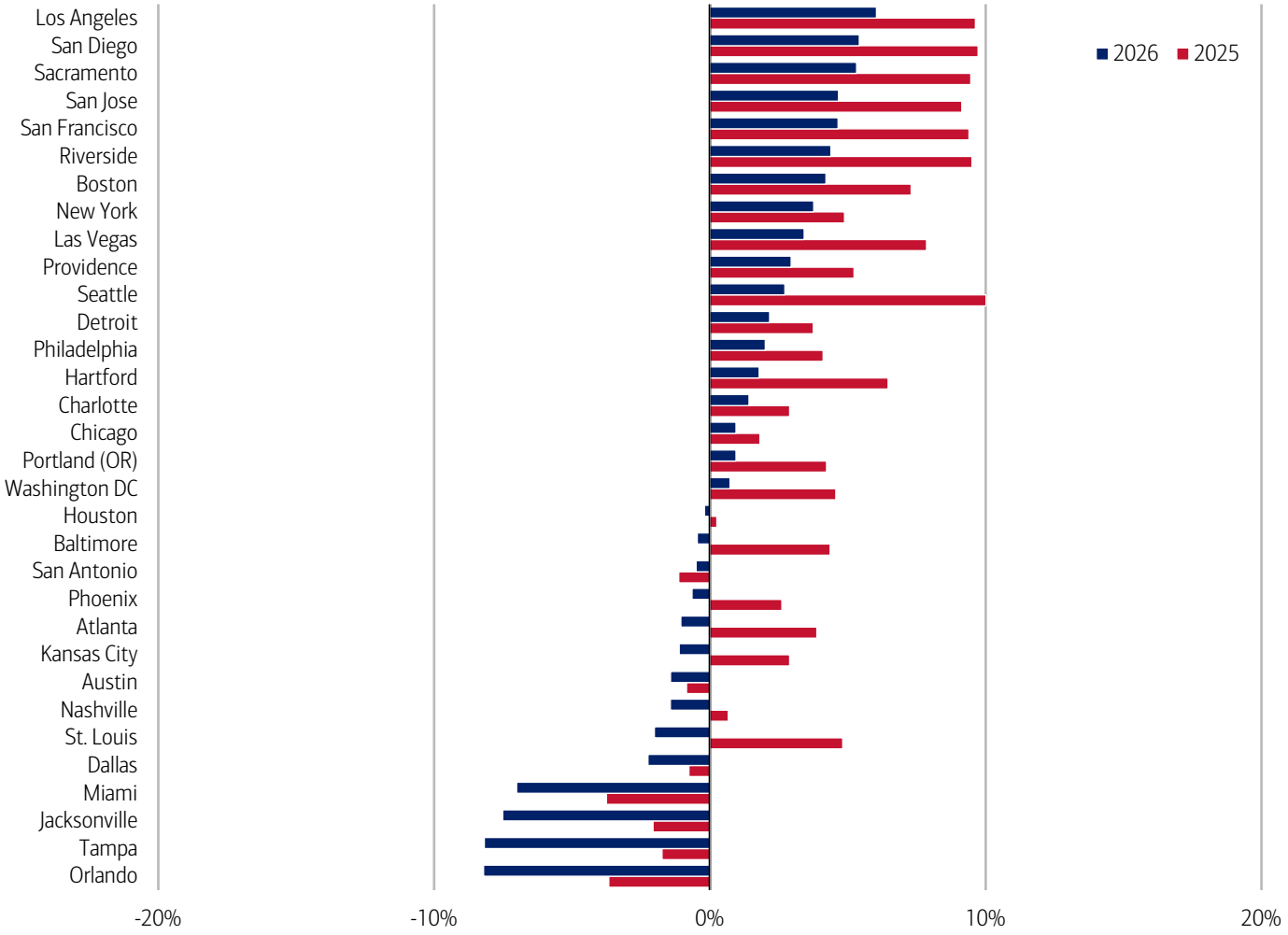
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Florida's metros have seen the largest decline when compared to other major cities

Examining different cities across the US also sheds light on insurance trends. Consumers in major California metros saw the largest increases in 2026, following near double digit growth in 2025, likely driven by widespread property damage from wildfires, according to BofA Global Research (Exhibit 7). Major coastal metros in the Northeast like Boston and New York also experienced sizable increases as well.

Conversely, many cities throughout the Sun Belt like Houston, Nashville and Dallas saw flat or even declining insurance payments. BofA Global Research points to several reasons for these declines: part of the equation in this region is that the massive runup in the cost of property insurance and domestic migration to the region through 2024 has since eased or reversed course.

Exhibit 7: California major metros have seen steady insurance payment growth, while Miami metros have seen declines
Median insurance payments by core based statistical areas (September to August yearly, YoY%)



Source: Bank of America internal data

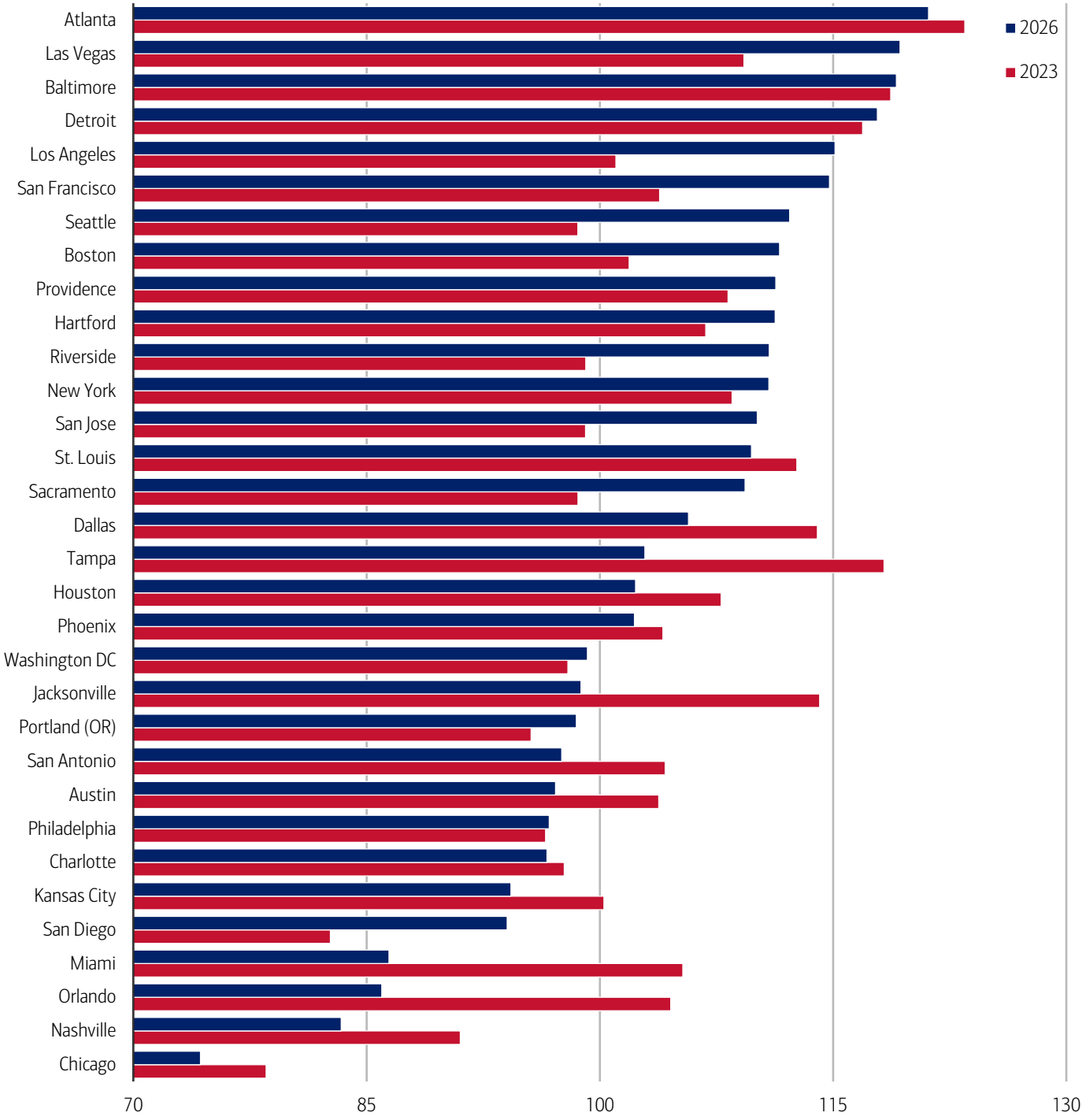
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In Florida, legislation that went into effect in 2023, eventually made its way into insurance prices in 2025, leading to widespread declines in premiums. The state also has a "lookback period" of three years, where insurance profits can be deemed too high and a rebate is issued to consumers. It's possible this rebate is factoring into declines this year, as some carriers only allow rebates upon renewals.

The Sunshine State was a notable standout for declines, with Orlando, Tampa and Jacksonville each down around 8% YoY in 2026, followed closely by Miami with a 7% decrease. Additionally, Bank of America payments data suggests that Jacksonville, Miami and Orlando all had median insurance payments above the national level in 2023, though their payments were fell below that in 2026 (Exhibit 8).

Major metros in Texas also saw easing insurance payments relative to 2023, while Seattle, San Jose, and Riverside, CA are among the major metros that went from below the national median to above in the three-year period ending in 2026.

Exhibit 8: Chicago, Nashville, Orlando and Miami residents have some of the lowest insurance payments in the nation among major metros
Median insurance payments by core based statistical areas (September to August yearly, index to national median = 100)



Source: Bank of America internal data

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Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under $5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate (if discussed) can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.
2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.
3. Overall total card spending includes small business card spending while per household card spending does not.
4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).
5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

Insurance payments include renters, auto or home, but do not include insurance escrowed into mortgage payments or deducted from a paycheck.

We consider a measure of services necessity spending that includes but is not limited to groceries, utilities, and gasoline. Discretionary services includes but is not limited to charitable donations, leisure travel, entertainment, and professional/consumer services. Discretionary retail includes but is not limited to general merchandise, miscellaneous, clothing, electronics and furniture. It excludes categories like groceries and gasoline. Holiday spending is defined as items in which spending in the November-December period is usually at least 20% of total annual spending on the category. Value and premium grocers are determined judgmentally on a proprietary analysis of merchants by equity analysts. The "more discretionary" retail category includes furniture, electronics, building materials, online marketplaces, clothing retailers and miscellaneous stores; while "partly discretionary" retail contains auto parts and services, health and personal care retailers, general merchandise and "mostly necessity" retail contains gas and groceries.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation, changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Generations, if discussed, are defined as follows: Gen Z, born after 1995; Younger Millennials: born between 1989-1995; Older Millennials: born between 1978-1988; Gen Xers: born between 1965-1977; Baby Boomer: 1946-1964; Traditionalists: pre-1946.

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Additional information about the methodology used to aggregate the data is available upon request.

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Disclosures

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Economy

Business Checkpoint: Looking into the logistics

15 October 2026

Key takeaways

- Small and mid-sized businesses entered Q3 in relatively solid financial condition, with profitability growth up across both groups, according to Bank of America business account data. Nearly half of estimated tariff payments had been refunded by September, though the share of these refunds relative to revenue has peaked.
- In Bank of America business payments data, logistics spending growth per client was up four times as much for small than mid-sized firms. Additionally, distribution and inventory transactions growth per client has outpaced spending since the spring of 2025.
- Growing transportation bottlenecks are re-emerging ahead of the holiday season. Trucking capacity has tightened as driver availability remains constrained, according to BofA Global Research. This has pushed freight rates higher and driven strong transportation sector payroll growth, particularly among mid-sized businesses, according to Bank of America payments data.

Business Checkpoint is a new, regular publication from Bank of America Institute. It provides a real-time view of the financial health, opportunities and challenges facing small and mid-sized businesses across the US economy, leveraging the depth and breadth of Bank of America's proprietary data. By analyzing activity among both small and mid-sized firms, Business Checkpoint offers timely insights into business profitability, hiring, investment and spending trends. Such data is not intended to be reflective or indicative of, and should not be relied upon as, the results of operations, financial condition or performance of Bank of America.

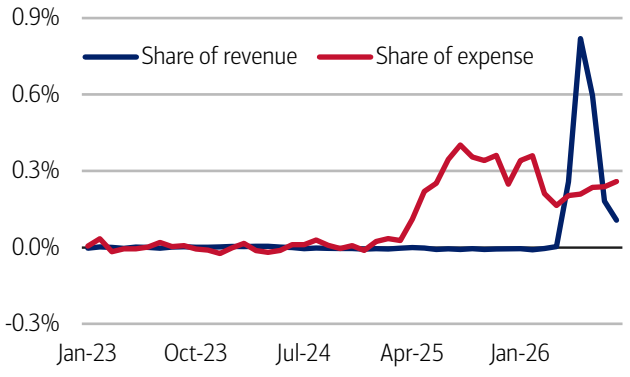
Businesses end Q3 in fair shape

While profitability growth remained positive for both small and mid-sized firms in the third quarter, small firms outpaced their mid-sized counterparts for the second consecutive quarter (Exhibit 2). Still, there's been a notable reversal in the profitability of mid-sized businesses, growing this quarter after declining slightly for the past two.

The continued rollout of tariff refunds is serving as a tailwind to some firms' cash positions. That is particularly true for small firms. According to Bank of America account data, by September approximately 47% of estimated International Emergency Economic Powers Act-related tariff payments had been refunded to companies with less than $50 million in annual revenue.

Exhibit 1: Tariff refunds as a share of income for businesses <$50M in annual revenue peaked in June

Share of tariff payments of income and expense (percentage point difference from 2023 average share)

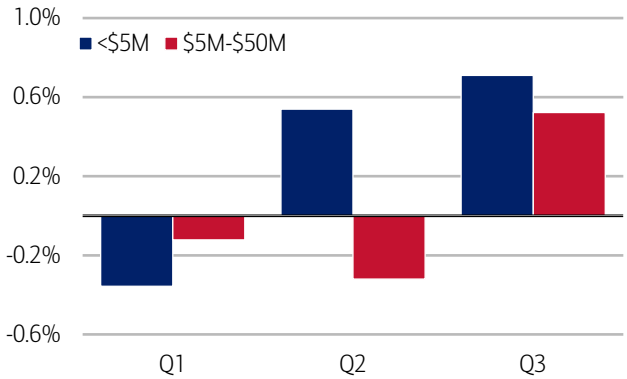


Source: Bank of America internal data

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Exhibit 2: Small firms' profitability growth outpaced mid-sized firms in Q3 2026

Profitability growth by firm size in 2026 (quarterly average, year-over-year (YoY)%)



Source: Bank of America internal data

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According to Bank of America business payments data, the percentage point difference in tariff refunds as a share of revenue for firms with less than $50 million in annual revenue spiked in June but dropped to nearly zero in September. At the same time, the percentage point difference in tariff payments as a share of expenses began rising again relative to the 2023 average (Exhibit 1). This suggests that while tariffs continue to affect companies, the refund program has provided temporary relief for some businesses.

Wholesale trade along with transportation and warehousing posted solid profitability growth across business size

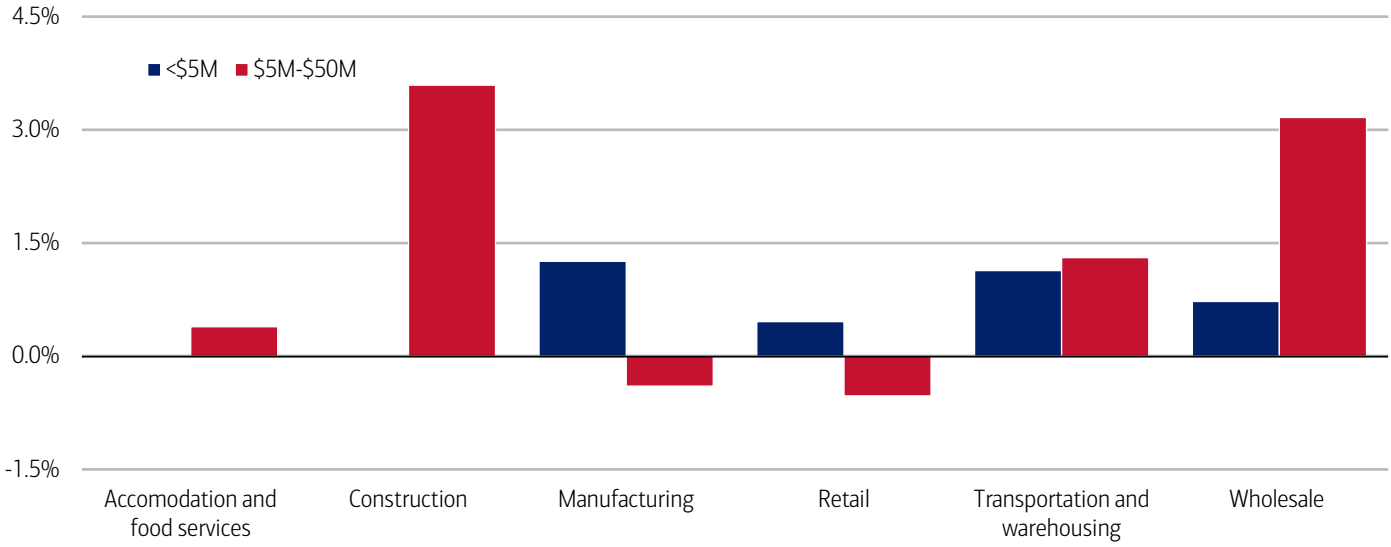
However, the benefits appear uneven. Such refunds have boosted cash flow, but they have not fully insulated firms from broader cost pressures.

Looking across industries, mid-sized construction and wholesale firms had the strongest profitability growth in September (Exhibit 3). Notably, transportation and warehousing, as well as wholesale trade were among the few sectors showing positive profitability growth for both small and mid-sized firms last month.

The persistence of logistics, freight and inventory-related cost pressures may help explain why profitability gains have remained relatively concentrated in transportation and wholesale trade. However, these sectors also have a higher exposure to rising gas prices within their margins (read more on this in the August Small Business Checkpoint).

Exhibit 3: For both small and mid-sized businesses, profitability growth was positive in transportation and warehousing as well as wholesale trade in September

Profitability growth by sector and business size in September (YoY%, six-month moving average)



Source: Bank of America internal data

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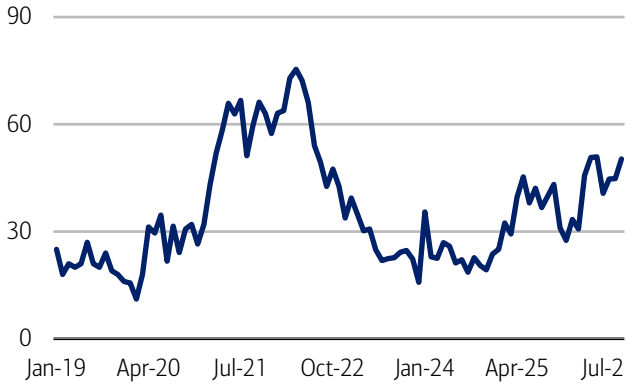
The cost of maintaining supply chains is rising

In fact, the prices paid component of the September services reading from the Institute for Supply Management indicates still-building price pressure for firms (Exhibit 4). This marked the highest reading since July 2022.

Using Bank of America internal data, we identified payments to logistics and freight companies to assess a key part of supply chain costs. In June, logistics and freight payments' growth per client from small businesses grew nearly 30%, though has moderated to 12.7% YoY in September (Exhibit 5). Meanwhile, mid-sized firms' payments growth in this category was up just 3.2% YoY in September.

The divergence suggests logistics costs may be disproportionately affecting smaller firms, who are typically less capable of absorbing rising costs than larger firms. Larger companies also often benefit from dedicated logistics contracts and greater purchasing power, while smaller ones may be more exposed to spot-market transportation costs and fluctuations in freight pricing.

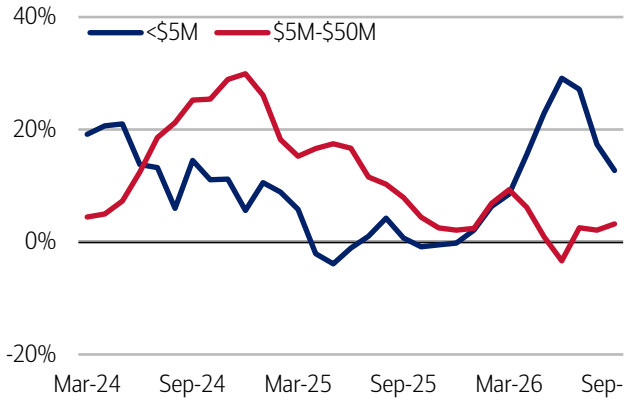
Exhibit 4: The September services Institute for Supply Management higher prices component was the highest reading since July 2022
Institute for Supply Management: Services: Prices, Higher (% of respondents)



Source: Haver Analytics

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Exhibit 5: Logistics and freight payments' growth by small businesses peaked in June
Logistics and freight payments by business size (YoY%, monthly, three-month moving average)



Source: Bank of America internal data

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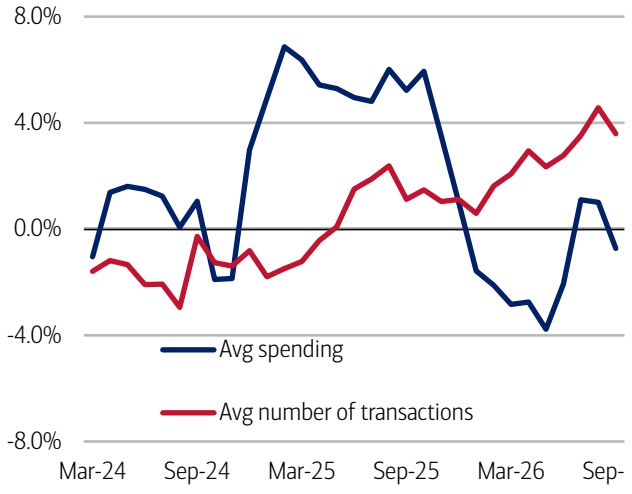
Evidence of more moving parts in companies' supply chains

Yet there is increasing evidence that tariffs and geopolitics are adding complexity to small and mid-sized companies' supply chains.

In Bank of America payments data, average distribution and inventory transaction growth per business client has increased steadily since the spring of 2025 despite spending growth in this category falling in September (Exhibit 6). Rising transaction growth alongside softer spending growth suggests businesses may be engaging with more suppliers, shipments or inventory movements even as overall procurement outlays moderates.

Furthermore, in May 2026, the Global Supply Chain Pressure Index reached its highest reading since the summer of 2022 and has continued to remain elevated into October (Exhibit 7). This further emphasizes that transportation bottlenecks, delays and procurement challenges are becoming more common globally. While current readings remain below the extreme levels observed during the pandemic, the recent move higher indicates supply chain normalization may be reversing.

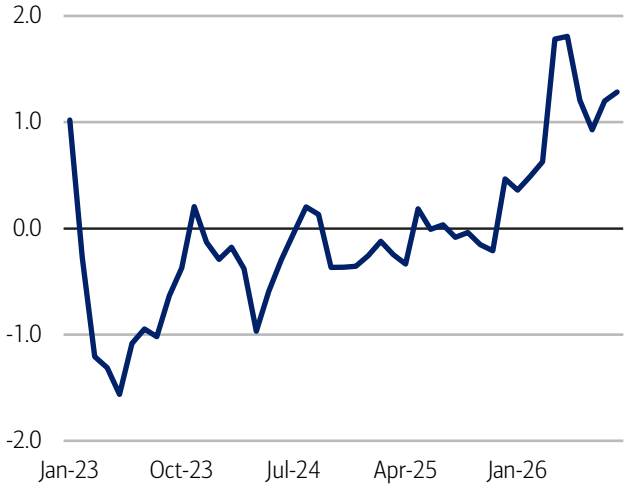
Exhibit 6: Average distribution and inventory transaction growth per business client has increased steadily since the spring of 2025
Payments and transactions to distribution and inventory per client (monthly, YoY%, three-month moving average)



Source: Bank of America internal data

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Exhibit 7: The Global Supply Chain Pressure Index reached its highest reading since the summer of 2022 in May 2026
Global Supply Chain Pressure Index (monthly)



Source: Federal Reserve Bank of New York, Global Supply Chain Pressure Index, https://www.newyorkfed.org/research/gscpi.html.

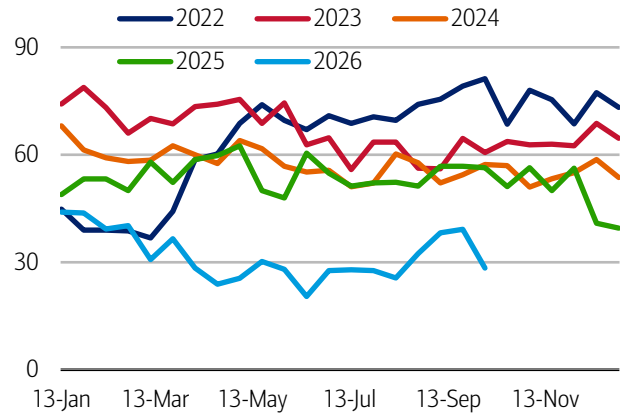
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The trials of trucking

As of October 1, BofA Global Research's proprietary bi-weekly Truckload Demand Indicator (see Methodology) remained below its historical growth signal (Exhibit 9) and the Truck Capacity Indicator, which reflects shippers' views of available capacity, tightened significantly, falling from 39.2 to 28.4 (Exhibit 8). This suggests shippers are seeing fewer trucks available relative to demand than in the previous survey.

According to BofA Global Research, the current trucking upcycle is being driven by constrained supply rather than a significant improvement in freight demand. While any positive movement in demand is an important indicator as the industry moves into peak shipping season between late September and early December, tighter capacity appears to be the dominant force supporting freight rates.

Exhibit 8: Shippers' view of available capacity has fallen significantly as of October 1, 2026
BofA Truckload Capacity Indicator by year (bi-weekly)



Source: BofA Global Research

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Exhibit 9: The Demand Indicator has been sub-60 for the past two months after 4.5 months above
BofA Truckload Demand Indicator (bi-weekly)



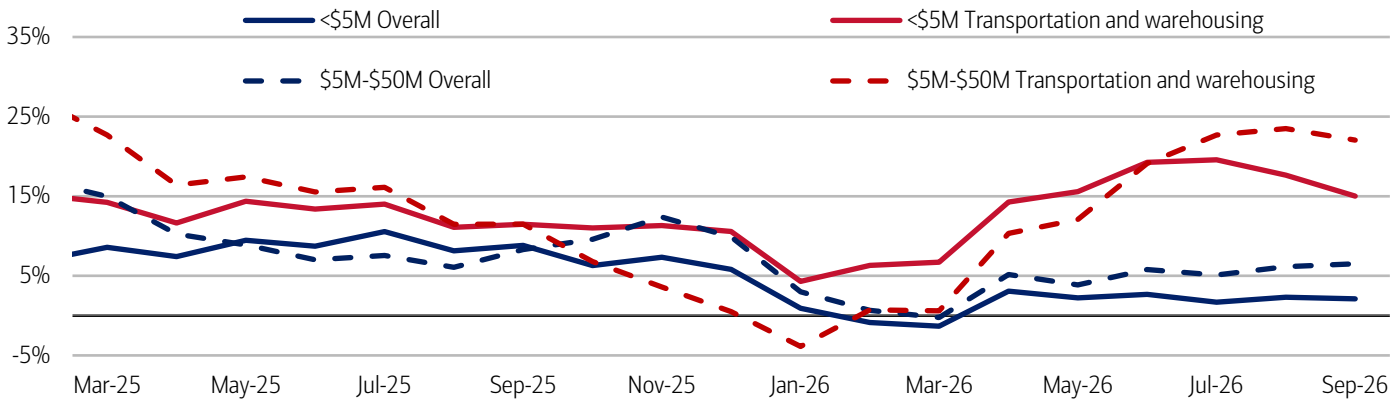
Source: BofA Global Research

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The reduction in capacity reflects several factors, including stricter commercial driver's license (CDL) and training requirements, increased compliance and enforcement measures, and other regulatory changes that have reduced the pool of qualified drivers. This has also likely driven some wage inflation within the industry. According to Bank of America payments data, transportation and warehousing payroll growth per client has surged since the start of the year, especially for mid-sized businesses (Exhibit 10).

Having to pay higher wages suggests labor shortages remain a challenge in the sector. Firms may increasingly need to offer higher compensation to attract and retain workers as trucking capacity tightens. In turn, such capacity puts upward pressure on freight rates, particularly in the spot market where small businesses may be more impacted.

Exhibit 10: Transportation and warehousing payroll growth has surged since the start of the year, especially for mid-sized businesses
Payroll payments per client by industry and business size (monthly, YoY%, three-month moving average)



Source: Bank of America internal data

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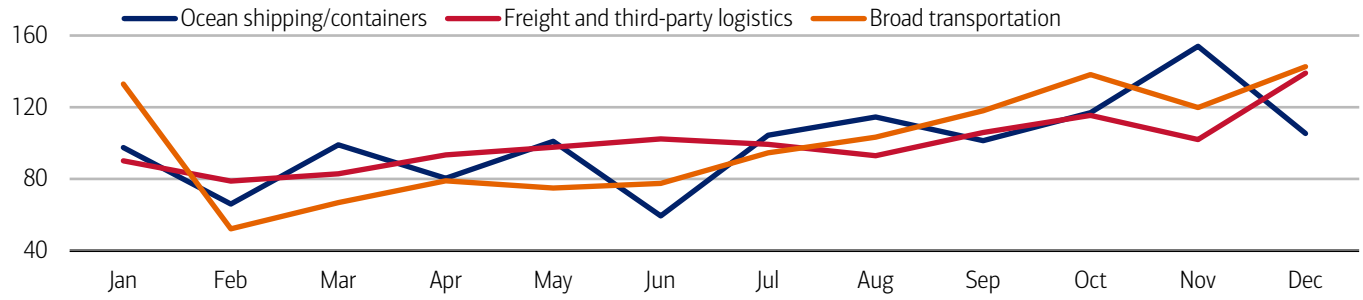
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Logistics payments tend to increase in the second half of the year

September marks a key run-up to the holiday season (read more on signs of holiday spending in the October Consumer Checkpoint), with spending on ocean shipping, containers, and other transportation services steadily accelerating ahead of the peak shopping period.

In fact, over the past two years, the month of November has seen ocean shipping/container spending volumes well above their 2023-2025 monthly average (Exhibit 11). The seasonal pattern likely reflects inventory restocking and holiday-related shipping activity. Ocean shipping volumes typically accelerate during the fall months as businesses prepare for year-end demand.

Exhibit 11: Ocean shipping/container spending volumes were well above their 2023-2025 monthly average in November
Logistics and freight average payments by category (monthly, indexed, 100 = monthly 2023-2025 average)



Source: Bank of America internal data

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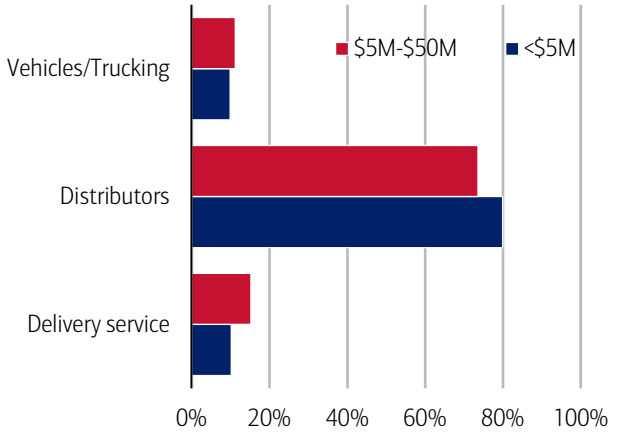
Distributors and inventory payments' growth fell ahead of holiday season

Changes in logistics and freight spending often provide an early signal of inventory trends, as distributors account for a large share of supply chain-related spending (Exhibit 12).

The sharp moderation in distribution and inventory spending growth per client for both small and mid-sized businesses indicates that much of the large-scale inventory accumulation seen earlier in 2025 has already occurred (Exhibit 13). In other words, businesses appear to be replenishing inventories for seasonal demand rather than significantly expanding stockpiles as they did earlier in the year.

Overall, businesses seem to be entering the holiday season in relatively stable financial condition, supported by tariff refunds and positive profitability growth. However, rising logistics costs, tighter trucking capacity and elevated supply chain pressures suggest operating conditions are becoming more challenging.

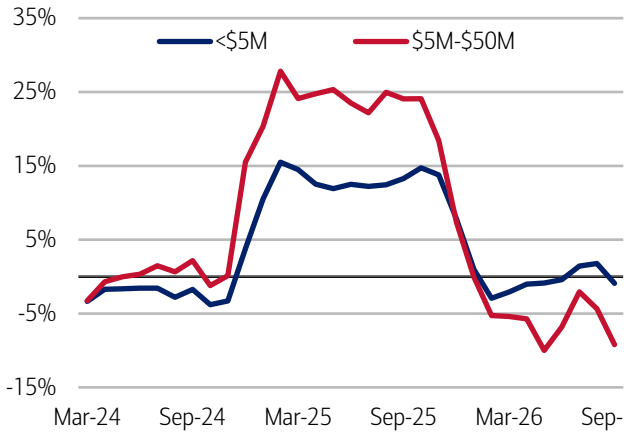
Exhibit 12: Distributors comprise a sizeable share of overall supply chain-related payments
Share of supply chain-related payments by category in 2026 by business size (%)



Source: Bank of America internal data
Note: Supply chain-related payments are those towards companies within logistics, freight, distribution and inventory.

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Exhibit 13: For both small and mid-sized firms, payments growth to distribution and inventory firms accelerated sharply during 2025
Payments to distribution and inventory by business size (monthly, YoY%, three-month moving average)



Source: Bank of America internal data

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Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

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Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

Revenue tiers are determined by the combination of following factors: 1) stated revenue on small businesses credit applications, 2) actual account inflow into Bank of America Deposit Accounts, and 3) third party revenue estimation.

The alternative hiring indicator consists of payments from Bank of America business clients to business-focused hiring firms which include both direct deposits through Automated Clearing House (ACH) and payments via credit and debit cards.

Supply chain-related spending is comprised of payments to logistics, freight, distribution, inventory, delivery service, and trucking companies. These are identified from merchant names and MCC codes.

BofA Global Research's trucking indicators are a bi-weekly diffusion-style survey of shipper demand expectations over the next 0–3 months, built from responses from surveyed shippers.

Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

Additional information about the methodology used to aggregate the data is available upon request.

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Disclosures

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Economy

Consumer Checkpoint: Strength in numbers

09 October 2026

Key takeaways

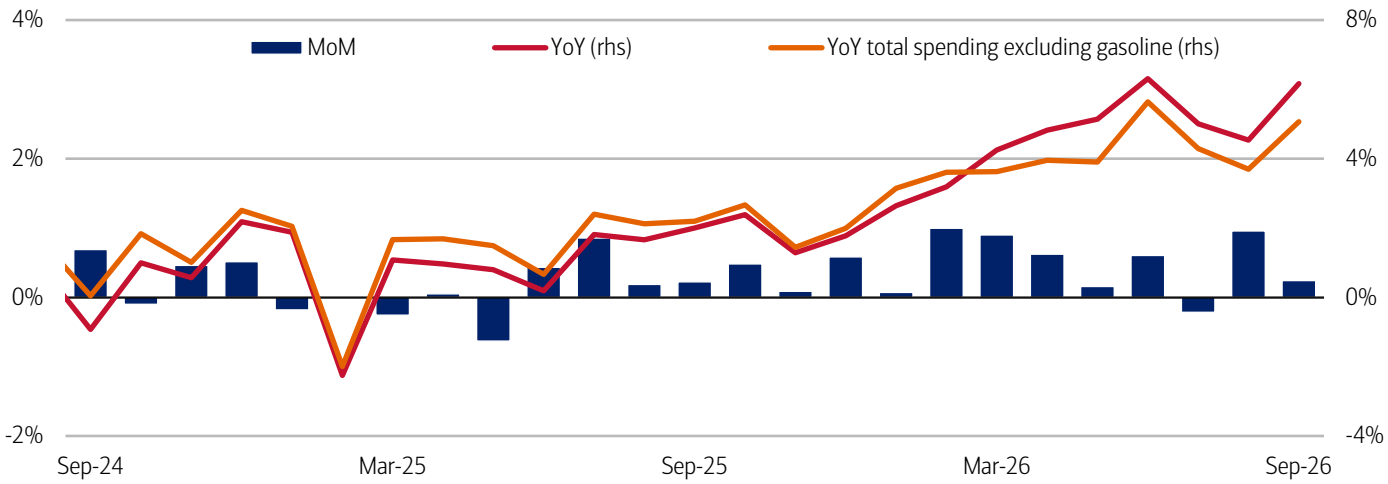
- Consumer spending remained very solid in September. Bank of America aggregated credit and debit card spending per household rose 6.2% year-over-year (YoY), or 5.1% excluding gasoline, both the third strongest growth rates over the past four years. Month-over-month (MoM) spending was more muted, especially without gas.
- Spending shows considerable breadth: the vast majority of discretionary retail and services categories are showing growth. Moreover, this strength extends across income cohorts, with the ongoing recovery of lower- and middle-income households' spending growth.
- With the winter months fast approaching, holiday spending is already increasing at a solid rate, though it's still early. In Bank of America deposit data through September, buy now, pay later (BNPL) usage is rising but not at a rate that is overly concerning. However, nearly one in every five households that use BNPL has a repayment of over $500.

Consumer Checkpoint is a regular publication from Bank of America Institute. It aims to provide a holistic and real-time estimate of US consumers' spending and their financial well-being, leveraging the depth and breadth of Bank of America proprietary data. Such data is not intended to be reflective or indicative of, and should not be relied upon as, the results of operations, financial conditions or performance of Bank of America.

A solid September for spending

Consumer spending continued to look solid in September. Bank of America total aggregated credit and debit card spending per household rose 6.2% year-over-year (YoY), while spending excluding gasoline increased 5.1% YoY. Notably, both are the third strongest growth rates in the past four years. Seasonally-adjusted total card spending rose 0.2% month-over-month (MoM), while spending excluding gasoline was little changed. Overall, spending growth remains well above the 2025 average rate (Exhibit 1).

Exhibit 1: Total card spending per household rose 6.2% YoY and 5.1% YoY excluding gasoline in September
Total card spending growth per household, based on Bank of America aggregated credit and debit card data (monthly, MoM%, seasonally adjusted (SA)) and (monthly, YoY%, non-SA, right-hand side (rhs))



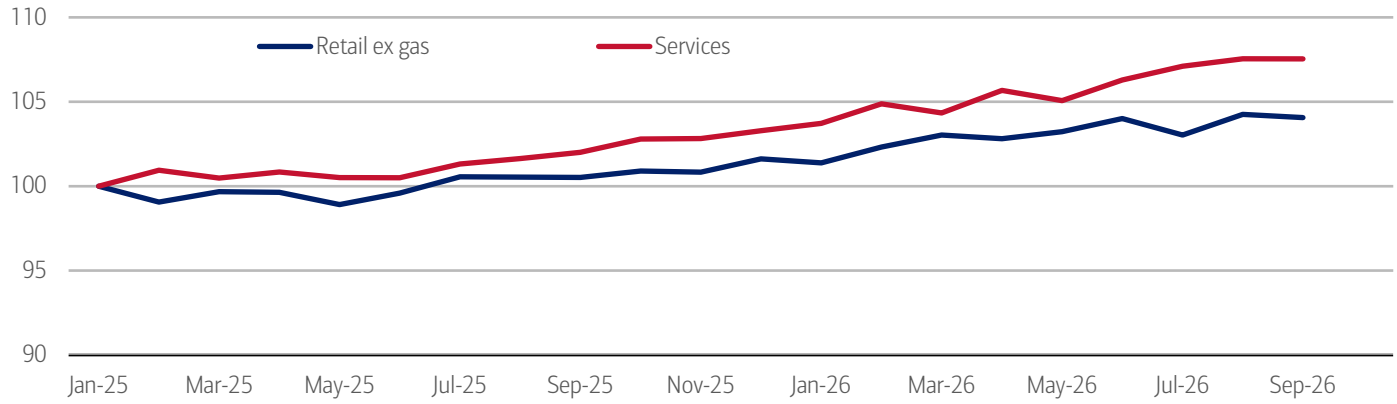
Source: Bank of America internal data

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Most spending categories are growing

One striking thing about recent card spending, in our view, is that it is well balanced across categories. For example, both spending on retail (excluding gasoline) and services grew strongly YoY in September, at 3.5% and 5.4%, respectively. Meanwhile, retail spending is showing notably better momentum than in 2025 (Exhibit 2).

Exhibit 2: Spending is rising across retail and services
Retail (excluding gasoline) and services card spending per household (monthly, seasonally adjusted, index January 2025 = 100)



Source: Bank of America internal data

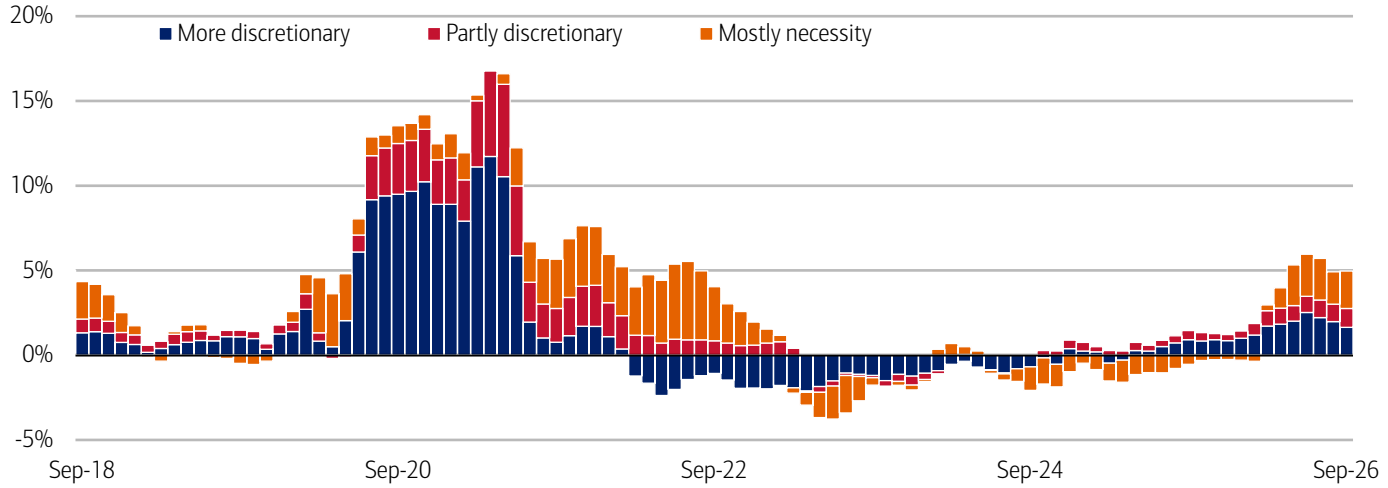
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Well balanced across "nice-to-have" and "need-to-have" retail categories

Is the momentum in retail spending reflective of "nice-to-have" discretionary or "need-to-have" necessity spending? In retail, this is a nuanced question, as many categories sit in a gray area between "necessity" and "discretionary". General merchandise stores, for example, sell both essentials like groceries as well as nice-to-have electronics or even jewelry.

In Exhibit 3, we account for this by grouping retail spending into "more discretionary" (furniture, electronics, building materials, online retailers, clothing retailers and miscellaneous stores), "partly discretionary" (auto, health and personal care retailers, general merchandise) and "mostly necessity" (gas and groceries). Even within these categories the edges will be blurred, but the overall impression is that retail spending growth is widely distributed, with spending rising across all fronts. One caveat: "mostly discretionary" spending has been making a smaller contribution since June.

Exhibit 3: Retail spending is distributed across discretionary and necessity categories
Percentage points contribution to YoY growth by type of retail spending (three-month moving average)



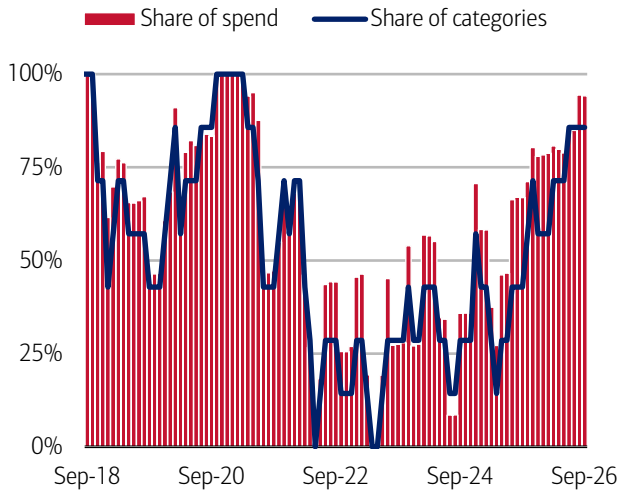
Source: Bank of America internal data. See methodology for category aggregation definitions.

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Looking at the share of "more discretionary" retail spending categories that are increasing YoY underscores that growth is well balanced and broad-based (Exhibit 4). Non-store and online retailers and miscellaneous stores saw some of the fastest growth, while furniture and building materials were among the laggards, though other durable categories showed some strength.

While some of this spending growth likely reflects the impact of higher prices pushing up spending across a swathe of categories, looking just at transactions as a proxy for volumes shows a similar picture, with only slightly less diffusion (Exhibit 5).

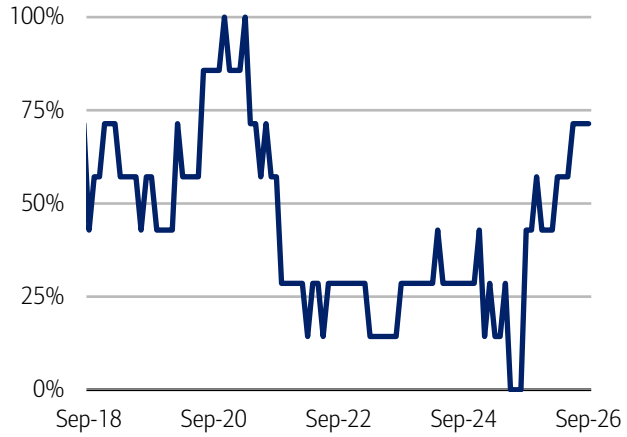
Exhibit 4: Nearly 86% of categories are experiencing positive spending growth, accounting for over 94% of retail spending
Diffusion index across "more discretionary" retail categories by spending (monthly, %)



Source: Bank of America internal data

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Exhibit 5: 71% of retail categories have positive transaction growth, with durables responsible for the declines
Diffusion index across "more discretionary" retail categories by transactions (monthly, %)



Source: Bank of America internal data

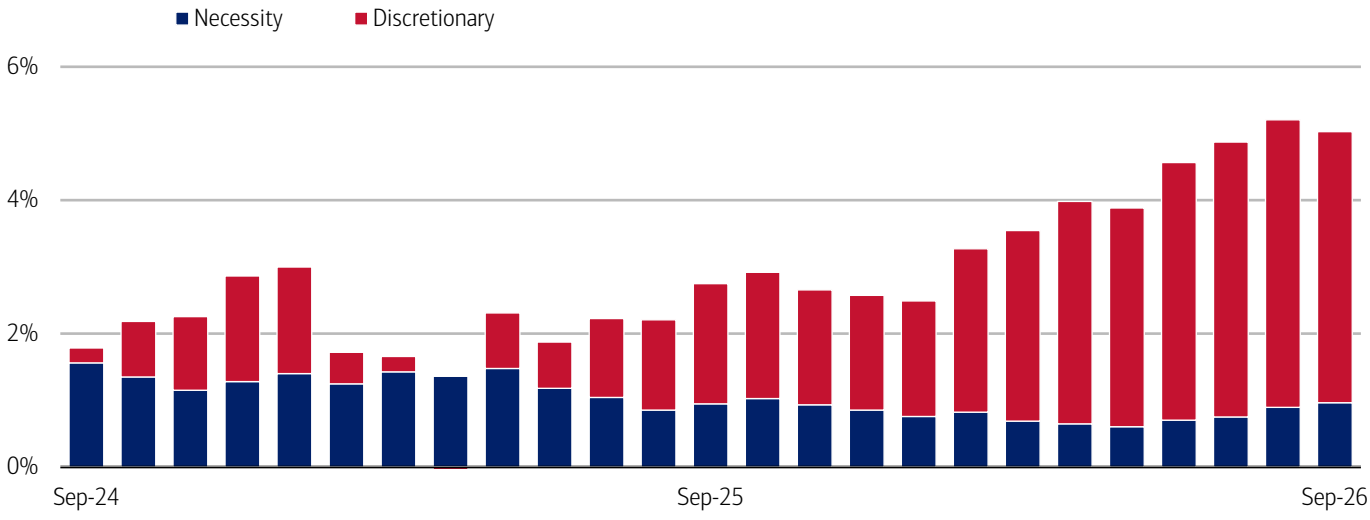
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For services, growth is driven mostly by discretionary spending

In services spending, on the other hand, the boundaries between discretionary and necessity spending are often clearer. Travel spending (e.g., airlines, lodging) can be thought of as, by and large, discretionary. At the same time, utility bills, rent and insurance payments are necessities.

And in services, discretionary spending remains the clear driver, accounting for over three-quarters of YoY spending growth in September (Exhibit 6). Leisure and housing services are major components of growth, though travel and restaurants also made positive contributions. However, as we noted last month strength in airline spending largely reflects price rises rather than transactions growth (read more in Consumer Checkpoint: Still sizzling).

Exhibit 6: Services spending growth has been mostly driven by discretionary spending over the past year
Percentage points contribution to growth by type of services spending (three-month moving average)



Source: Bank of America internal data

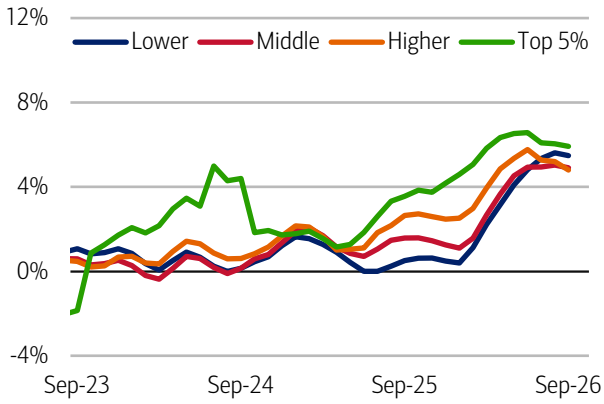
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Lower-income households' spending outpaced all but the top 5%

The breadth in spending growth across categories is also evident across income cohorts. In September, lower-income households' total card spending growth was 5.5% YoY, while the equivalent figures for middle- and higher-income households were 4.9% and 4.8% YoY, respectively. The top 5% of households by income continue to show the fastest spending growth, 5.9% in September, though this has eased somewhat (Exhibit 7).

Overall, the divergences across income cohorts are much smaller than they were in 2025 and the first three months of this year. Just focusing on spending excluding key necessities (gasoline, food and utilities) doesn't meaningfully change the picture of convergence across income cohorts (Exhibit 8).

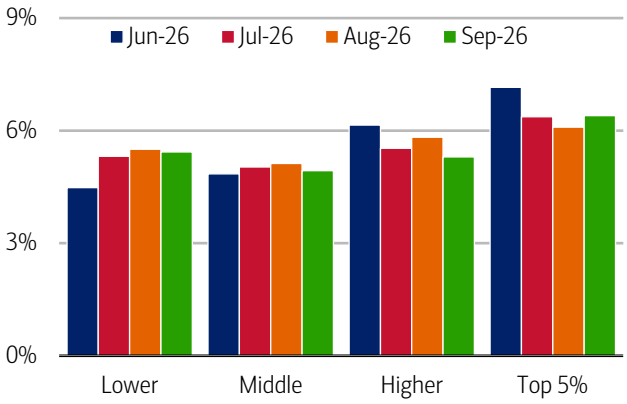
Exhibit 7: The top 5% saw the strongest spending growth of any income cohort…
Total card spending per household by income tercile and the top 5% (three-month moving average, YoY%, SA)



Source: Bank of America internal data

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Exhibit 8: …though discretionary spending for the top 5% has eased some since June
Discretionary card spending per household by income tercile and the top 5% (three-month moving average, YoY%, SA)



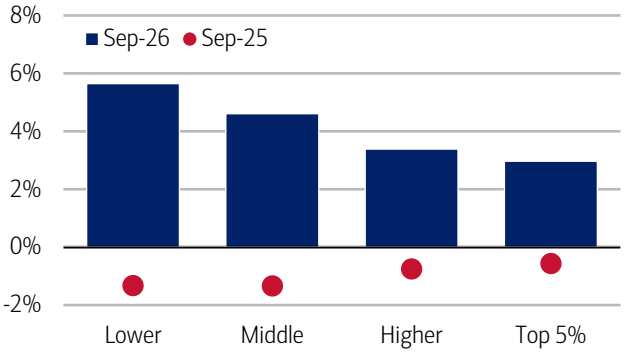
Source: Bank of America internal data

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The strength in lower- and middle-income households' discretionary spending is occurring despite necessity spending shifting from a tailwind to a headwind over the past year amid rising gas prices (Exhibit 9). In our view, a key component of why these households have been able to combine rising necessity spending with strength in discretionary is the labor market and a recovery in their after-tax wage growth (Exhibit 10).

In September, lower-income wage growth eased to 4.5% YoY from 4.7% YoY in August, while growth for middle- and higher-income households rose to 3.8% and 3.7% YoY, respectively. These changes were modest, and lower-income households are continuing to see faster after-tax wage growth than other cohorts, a notable turnaround from the "K-shaped" pattern we observed over much of 2025 and 2026.

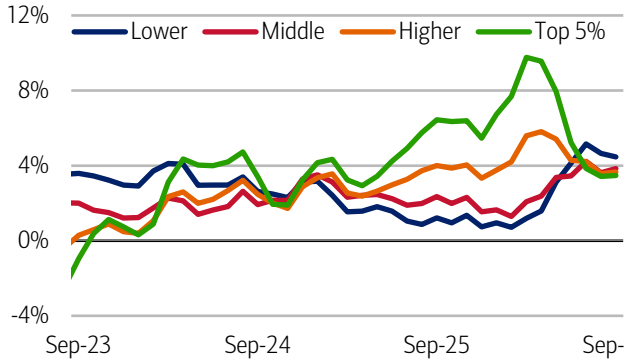
Exhibit 9: Necessity spending has gone from a tailwind to a headwind across all income cohorts
Necessity card spending per household by income tercile and the top 5% (three-month moving average, YoY%, SA)



Source: Bank of America internal data

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Exhibit 10: Lower-income wage growth remains strongest, despite a narrowing gap
After-tax wage and salary growth by household income tercile (three-month moving average, YoY%, SA)

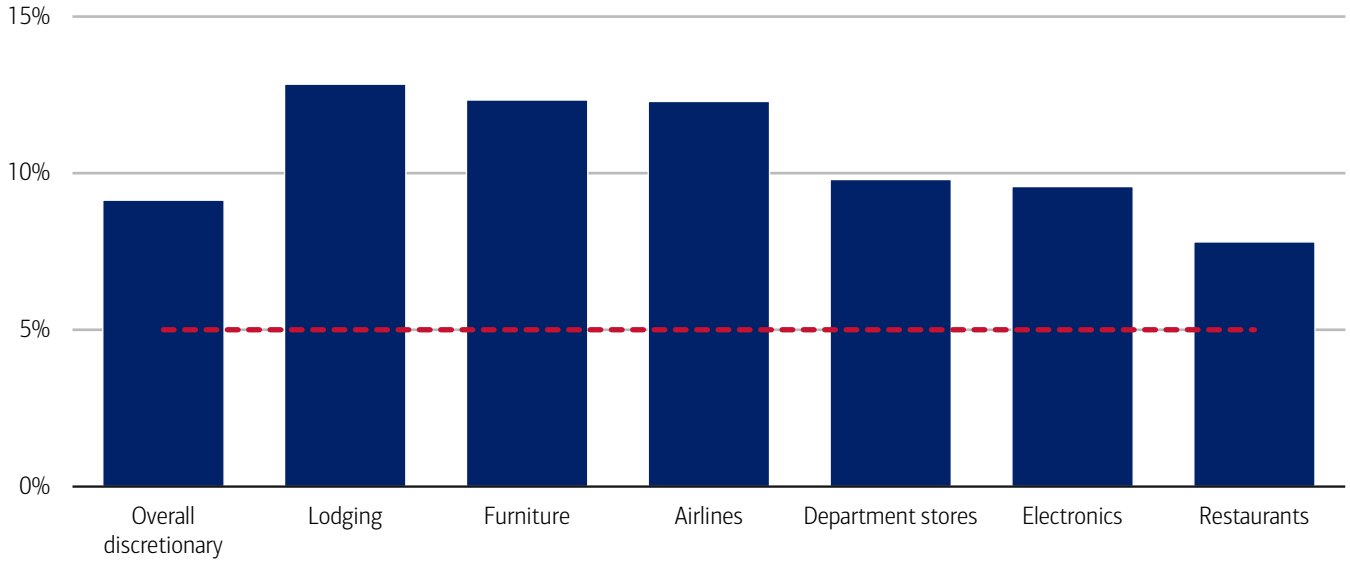


Source: Bank of America internal data

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Should we be concerned about the easing in household spending and after-tax wage growth for those with the top 5% of incomes? For now, probably not, as in our view this spending growth is likely being driven by wealth effects rather than wage gains. Absent a sharp correction in equity markets, there is little reason to expect them to retrench their spending. However, developments here are worth watching as this cohort comprises a significant proportion of overall consumer spending. They also tend to make outsized contributions to several discretionary spending categories (Exhibit 11).

Exhibit 11: The top 5% of households by income make up a disproportionate amount of spending in highly discretionary categories
Share of spending by the top 5% of households by income across select discretionary categories (three-month moving average to September 2026, %)



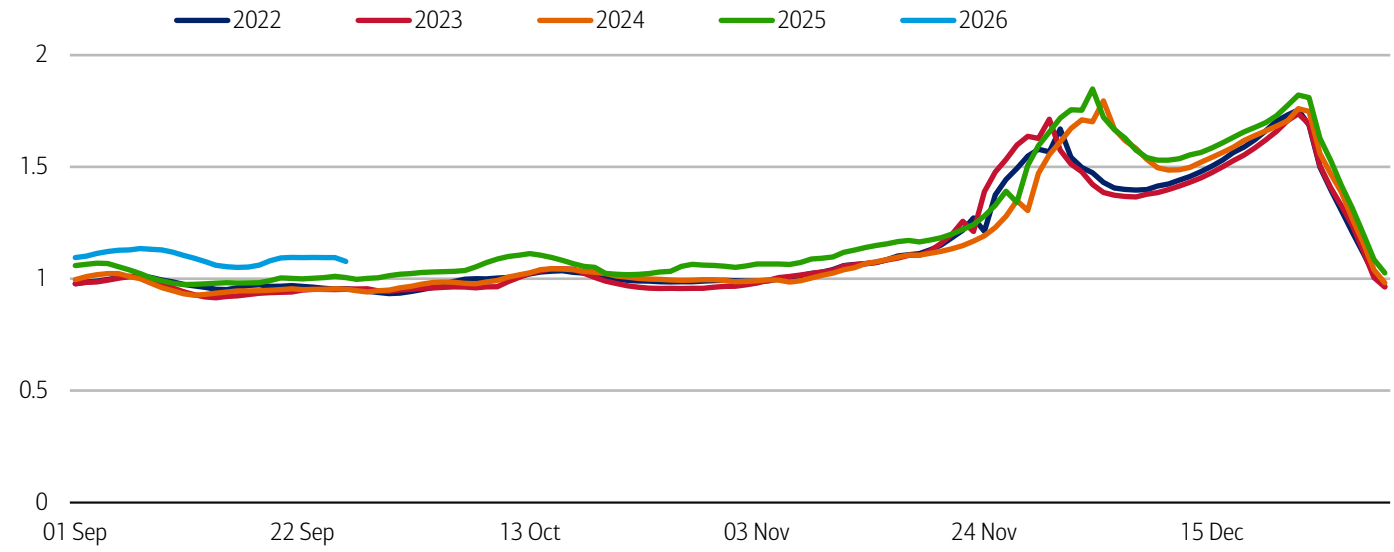
Source: Bank of America internal data

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With summer gone, thoughts turn to the holidays and BNPL...

Now that summer is over, the next big gauge of consumer momentum is how people spend during the run-up to, and over, the holiday season. It is early days, but September spending on holiday items ran well ahead of 2025 (Exhibit 12). Given the backdrop of firmer after-tax wage growth and breadth across retail and services categories, this is perhaps not surprising. However, some of this spending strength could be due to rising prices as opposed to more purchases.

Exhibit 12: It's early days, but holiday spending is strong so far this year
Card spending per household for holiday items by year (seven-day moving average, index September 2025 average = 1)



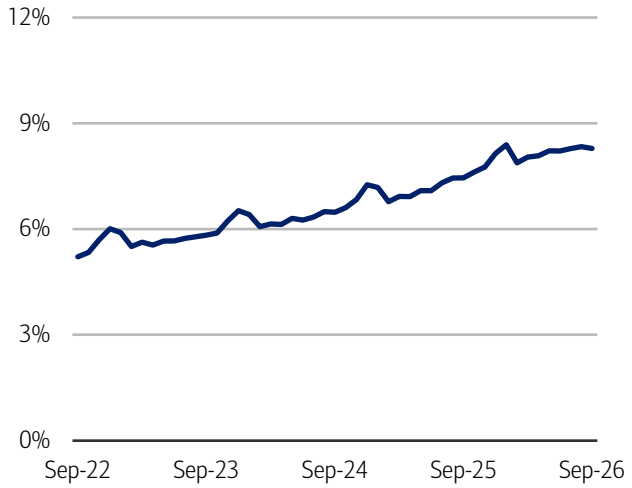
Source: Bank of America internal data
Note: Holiday items include all merchant category codes for which spending in November through December is at least 20% of total annual spending in the category.

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A concern often raised as we move into the holidays is whether consumers will lean on less traditional forms of credit such as buy now, pay later (BNPL) to finance their spending. Frequent BNPL users tend to have higher credit card utilization rates (read more in: Do consumers have wiggle room to absorb higher gas prices?).

Looking at the share of households in Bank of America deposit data with a BNPL payment, we do tend to see a seasonal rise over the holidays (Exhibit 13), though in past years, these increases have quickly reversed. Further, while the underlying trend in BNPL usage is upwards, there are no signs of a sharp acceleration. We do, however, see some evidence that BNPL is being used to finance more expensive purchases, with the share of households making repayments above $500 rising, albeit relatively slowly (Exhibit 14).

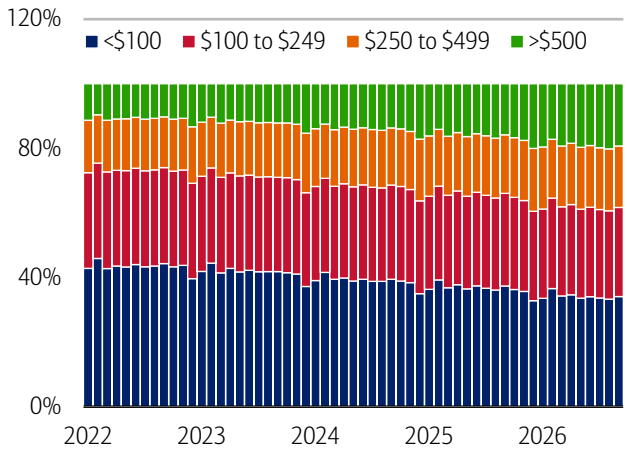
Exhibit 13: 8.3% of Bank of America households had a BNPL payment in September
Share of households with a BNPL payment in the month (monthly, %)



Source: Bank of America internal data

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Exhibit 14: Almost one in five households with BNPL usage has a repayment of over $500
Share of households with BNPL payments by repayment amount (monthly, $ amounts)

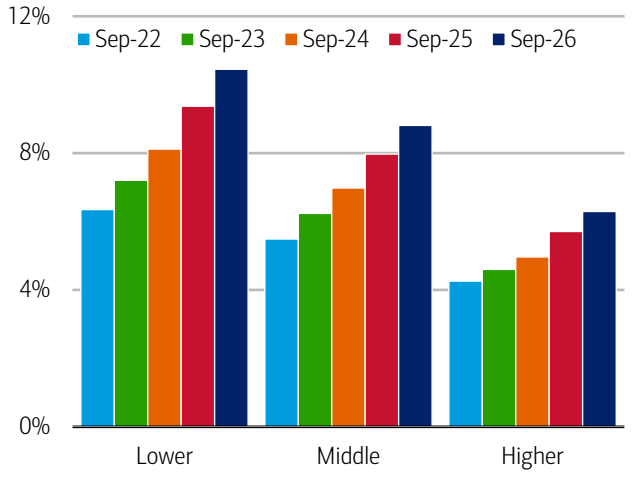


Source: Bank of America internal data

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The tendency towards greater usage is occurring across income cohorts, though is somewhat more pronounced for lower-income households (Exhibit 15). However, higher-income households have the largest share of BNPL payments above $500 (Exhibit 16). Overall, in our view, the increase in the share of BNPL users heading into the season does not appear too concerning, though the rate of usage for some BNPL consumers does bear watching.

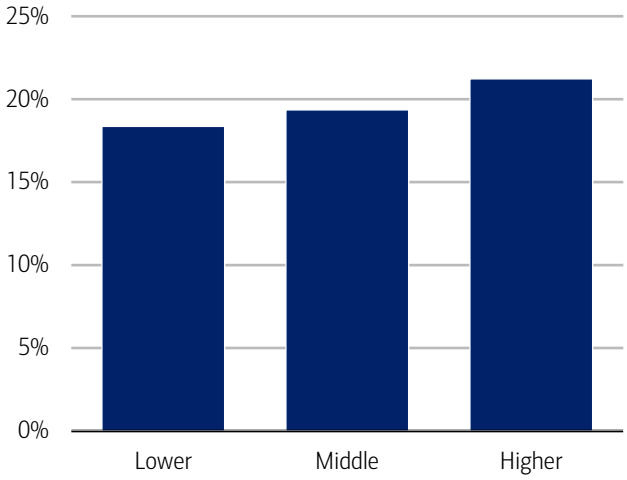
Exhibit 15: Lower-income households have seen larger adoption of BNPL...
Share of households with a BNPL payment in the month by household income tercile (monthly, %)



Source: Bank of America internal data

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Exhibit 16: ....but a larger share of higher-income households have BNPL payments over $500
Share of BNPL payments by spending amount across household income terciles (September 2026, %)



Source: Bank of America internal data

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Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under $5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate (if discussed) can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.
2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.
3. Overall total card spending includes small business card spending while per household card spending does not.
4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).
5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

Store category aggregation for "more discretionary", "partly discretionary", and "mostly necessity" is defined as follows:

More discretionary: furniture, electronics, building materials, hobbies, miscellaneous stores, nonstore retail, clothing.

Partly discretionary: auto, health and personal care, general merchandise.

Mostly necessity: Gasoline, food and beverage.

We consider a measure of services necessity spending that includes but is not limited to groceries, utilities, and gasoline. Discretionary services includes but is not limited to charitable donations, leisure travel, entertainment, and professional/consumer services. Discretionary retail includes but is not limited to general merchandise, miscellaneous, clothing, electronics and furniture. It excludes categories like groceries and gasoline. Holiday spending is defined as items in which spending in the November-December period is usually at least 20% of total annual spending on the category. Value and premium grocers are determined judgmentally on a proprietary analysis of merchants by equity analysts. The "more discretionary" retail category includes furniture, electronics, building materials, online marketplaces, clothing retailers and miscellaneous stores; while "partly discretionary" retail contains auto parts and services, health and personal care retailers, general merchandise and "mostly necessity" retail contains gas and groceries.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation, changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Generations, if discussed, are defined as follows: Gen Z, born after 1995; Younger Millennials: born between 1989-1995; Older Millennials: born between 1978-1988; Gen Xers: born between 1965-1977; Baby Boomer: 1946-1964; Traditionalists: pre-1946.

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Additional information about the methodology used to aggregate the data is available upon request.

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Economy

The Institute Employment Report: September 2026

07 October 2026

Key takeaways

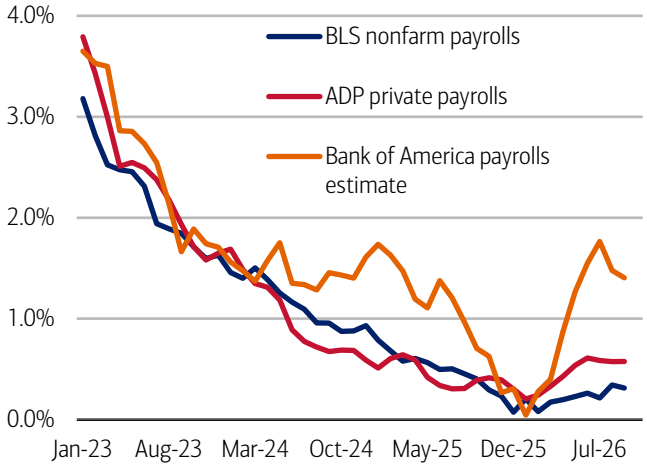
- Job growth softened further in September, according to Bank of America's deposit data, with growth of 1.4% year-over-year (YoY), from 1.5% in August. But falling unemployment payments in September indicate a relatively healthy overall picture.
- Lower-income households' wage growth remains strongest, despite a narrowing gap. After-tax wage growth eased to 4.5% YoY for lower-income households in September, while rising to 3.8% and 3.7% for middle- and higher-income households, respectively.
- Lower tax withholdings and a relative strengthening in the younger/lower-income end of the labor market are driving this story. Additionally, average hours have risen in some lower-paid industries, which is also boosting pay packets.

Job growth slowed in September, but unemployment trends are benign

Our estimate of job growth – which uses Bank of America deposit data to track changes in the number of customer accounts receiving a paycheck (see Methodology) – slowed further in September, to 1.4% year-over-year (YoY) from 1.5% YoY in August. The data can be noisy, partly due to seasonal variation and differences in pay-period timing, but in our view, the latest reading is consistent with the easing in payrolls growth seen in the September Bureau of Labor Statistics (BLS) data (Exhibit 1).

Still, despite slower hiring, there is little sign of a rise in joblessness in Bank of America data. Unemployment payments into customer accounts showed a YoY decline in September (Exhibit 2). So, overall, labor market trends appear resilient.

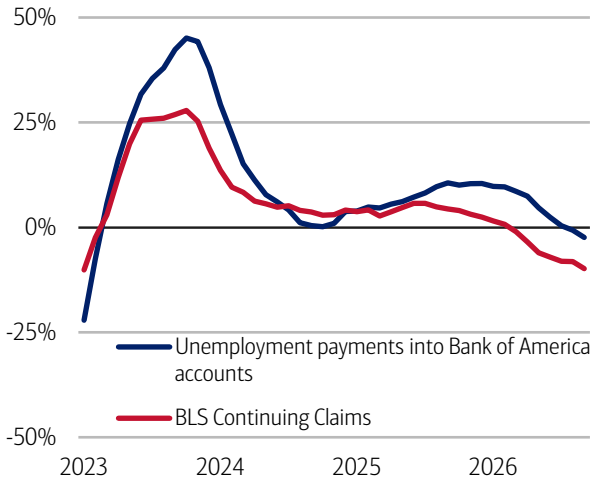
Exhibit 1: September payroll growth showed further slowing in Bank of America deposit account data
Payroll estimates from Bank of America customer deposit account data (three-month moving average, % YoY), the Bureau of Labor Statistics (BLS) and Automatic Data Processing (ADP) (monthly, YoY)



Source: Bank of America internal data, Haver Analytics
Note: BLS and ADP data are seasonally adjusted, Bank of America data is not seasonally adjusted.

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Exhibit 2: Unemployment payments declined YoY in September, according to Bank of America consumer deposit data
Number of households receiving unemployment payments (three-month moving average, YoY%, not seasonally adjusted (NSA)) and BLS Continuing Claims (three-month moving average, YoY%, seasonally adjusted (SA))



Source: Bank of America internal data, Bloomberg

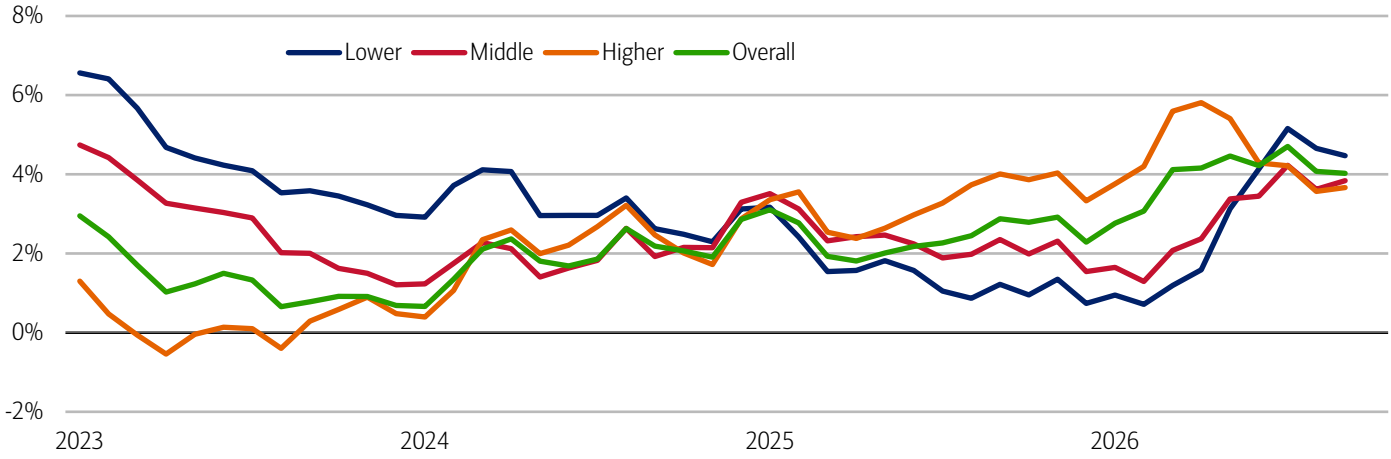
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After-tax wage growth cooled for lower-income, but rose for other cohorts

Lower-income wage growth eased to 4.5% YoY in September from 4.7% YoY in August, while wage growth for middle- and higher-income households rose to 3.8% and 3.7% YoY, respectively. Overall, these changes were modest, and lower-income households are continuing to see faster after-tax wage growth than other cohorts, a notable turnaround from the "K-shaped" pattern we observed over much of 2025 and 2026 (Exhibit 3).

Exhibit 3: Lower-income wage growth remains strongest, despite a narrowing gap

After-tax wage and salary growth by household income terciles, based on Bank of America aggregated consumer deposit account data (three-month moving average, YoY%, SA)



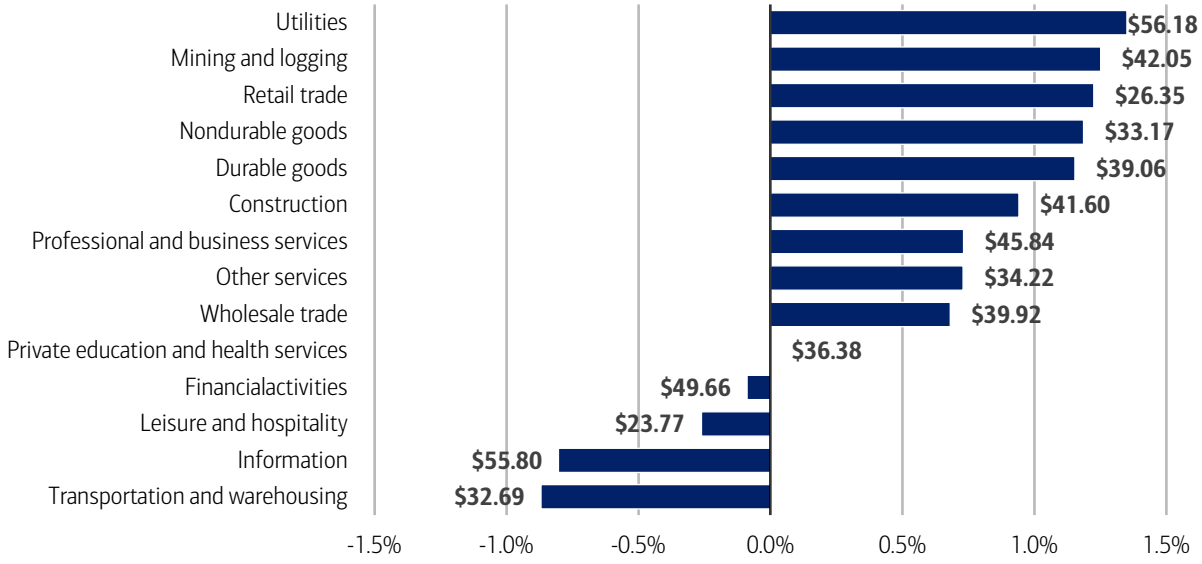
Source: Bank of America internal data

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In our view, the pick-up in lower- and middle-income households' after-tax wage growth continues to reflect the impact of the One Big Beautiful Bill Act (OBBA) in reducing tax withholdings, alongside an underlying rise in job-to-job changes at the younger/lower-income end of the labor market (for more, read: The Institute Employment Report: August 2026). Consistent with that underlying pick up, BLS data shows a solid YoY rise in average hours worked in several relatively lower-paid industries, such as manufacturing, retail and other services. Hourly workers in these industries are likely seeing rising paychecks as a result (Exhibit 4).

Exhibit 4: Some relatively lower-paid industries have seen a rise in hours worked

Average weekly hours by industry, July-September 2026 (% YoY) and average hourly earnings by industry, July-September 2026 ($)



Source: Haver Analytics

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Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under $5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation, changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Generations, if discussed, are defined as follows: Gen Z, born after 1995; Younger Millennials: born between 1989-1995; Older Millennials: born between 1978-1988; Gen Xers: born between 1965-1977; Baby Boomer: 1946-1964; Traditionalists: pre-1946.

Estimate of payroll growth from Bank of America internal data is based on the change in customer accounts receiving a paycheck in the month. An adjustment is made for the difference between overall population growth and customer account growth. Occasionally historic payments can be reclassified, leading to revisions in the series.

An estimate of bonus growth from Bank of America deposit data is calculated by looking at customers who have received an inbound ACH payroll transaction in the last two years. From this sample an estimate of bonuses is derived by looking for payroll transactions which are over 50% higher than the median regular payroll payments received by the customer. Of these payments only those that were received around the same time in each of the last two years are selected.

The job-to-job change rate (j2j rate) is defined as the proportion of customers with an identified change in their employer as a proportion of the total number of customers with employment income. We estimate the median pay rise associated with a j2j change using the pay in the first three months of the new job compared to the same three months a year ago.

Additional information about the methodology used to aggregate the data is available upon request.

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Transformation

Quantum dictionary, part 1: Decoding the fundamentals

21 October 2026

Key takeaways

- What is quantum computing? How does it work? What is a qubit? And when will quantum reach its full potential? Quantum computing challenges our intuition and our understanding of what computers can do. In this two-part dictionary series, we decode key terms behind what could be the next major technological revolution.
- Qubits are the basic building blocks of quantum computers, harnessing properties such as superposition and entanglement to process information. However, qubits are highly sensitive to their surroundings where disturbances can lead to errors in quantum computation, making error correction a key challenge for building quantum computers.
- There is no single approach to building a quantum computer. Researchers and companies are pursuing a range of qubit technologies, from neutral atoms and photons to spin, superconducting, topological and trapped-ion qubits, while quantum computing itself is likely to increasingly work alongside classical computing and AI.

First, let's lay the groundwork...

Quantum computing

Quantum computing uses qubits and quantum properties such as superposition (defined later on) to perform complex calculations and solve certain problems faster than classical computers.[1] From a practical perspective, quantum computing could be used to tackle problems related to optimization, machine learning, simulation and factorization across a range of sectors, including autos and logistics, financial services, energy and climate, health and life sciences and national security.

Current quantum computers can perform 'just' thousands of quantum operations. As the technology advances from thousands to millions, billions and eventually trillions of operations, quantum computers could begin to have commercial applications.

- Millions: Financial services and security could experience some of the first use cases, such as threat detection, fraud prevention, derivatives pricing, bond trading and arbitrage.
- Billions: Quantum could help power AI through advanced data generation capabilities.
- Trillions: Quantum chemistry applications could emerge, including drug discovery, battery simulation and materials discovery.

These are the publicly known applications, according to Oxford Quantum Circuits (OQC). And as quantum computing capabilities improve, some of these use cases could emerge sooner than anticipated.

Now, we'll tackle the core concepts...

Qubits

Qubits (quantum bits) are a quantum computer's basic unit of information. They can exhibit quantum properties such as superposition and entanglement (which we elaborate on below) — properties fundamental to quantum computing.

Quantum gates manipulate the state of qubits, much like logic gates manipulate bits in classical computers. Despite the name, a quantum gate is not a physical object, but rather an operation applied to a quantum system that changes its state.

From physical to logical qubits

Physical qubits are the foundation of a quantum computer's hardware. However, they are highly susceptible to error. To address this, multiple physical qubits can be structured together to encode a logical qubit, allowing errors to be detected and corrected

[1] Schneider, J., & Smalley, I. (2025, June 10). *What is quantum computing?* IBM.

and making the stored quantum information more reliable. In recent years, according to BofA Global Research, logical qubits have become a greater focus for quantum computing companies.

Entanglement

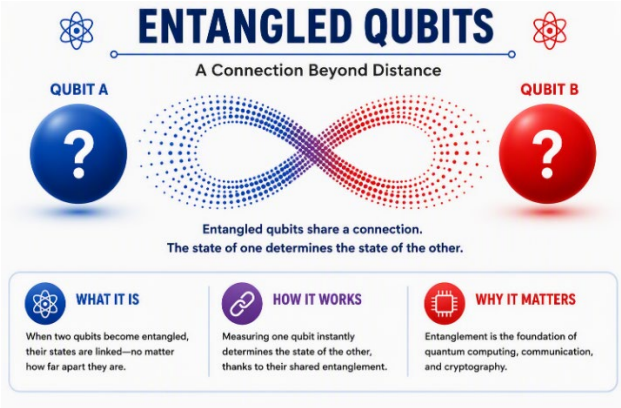
Entanglement is a condition where two or more qubits become linked so that their states are correlated, regardless of distance (Exhibit 1). It allows quantum computers to process information in ways that can be difficult or impossible for classical computers.

So, how does it work? Suppose we have two photons that have opposite polarizations (the orientation of their light waves). Before measurement, neither has a definite polarization. When one is measured as horizontally polarized, the other will be vertically polarized when measured in the same basis, wherever it is. When measured separately, they appear to be random, but when compared, they are correlated (Exhibit 2).

The challenges

Once created, entanglement must be protected from decoherence (where interactions with the environment destroy quantum correlations). Maintaining entanglement over long distances is another challenge. Photons traveling through optical fiber can be absorbed or scattered, making long-distance quantum communication difficult.[2] Quantum repeaters aim to overcome this challenge by extending entanglement across a series of shorter links.

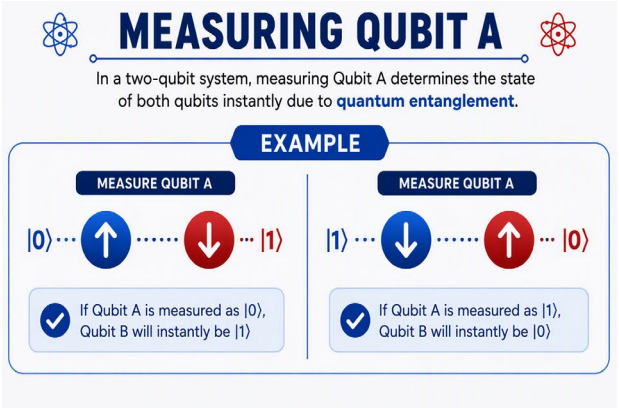
Exhibit 1: Entanglement is a state where at least two qubits become correlated
An infographic of entangled qubits



Source: BofA Global Research

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Exhibit 2: Measuring one qubit in an entangled pair instantly determines the state of the other, no matter the distance
An example of what entanglement means for qubit measurement



Source: BofA Global Research

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Superposition

Unlike a classical bit, which can be one of two states (0 or 1), superposition allows a quantum system to exist in multiple states or configurations at the same time (Exhibit 3). For example, a qubit can exist in the 0 state, the 1 state or any state that is a combination of 0 and 1.[3]

A quantum system can maintain a superposition until it is observed or measured. Once measured, its superposition collapses into a single state. Another way of thinking about superposition is to imagine a sphere with the north pole representing zero and the south pole representing one. Other points on the surface of the sphere represent different superposition states. This allows us to: 1) represent complex forms of information; and 2) simulate atoms in a computer.

Schrödinger's cat

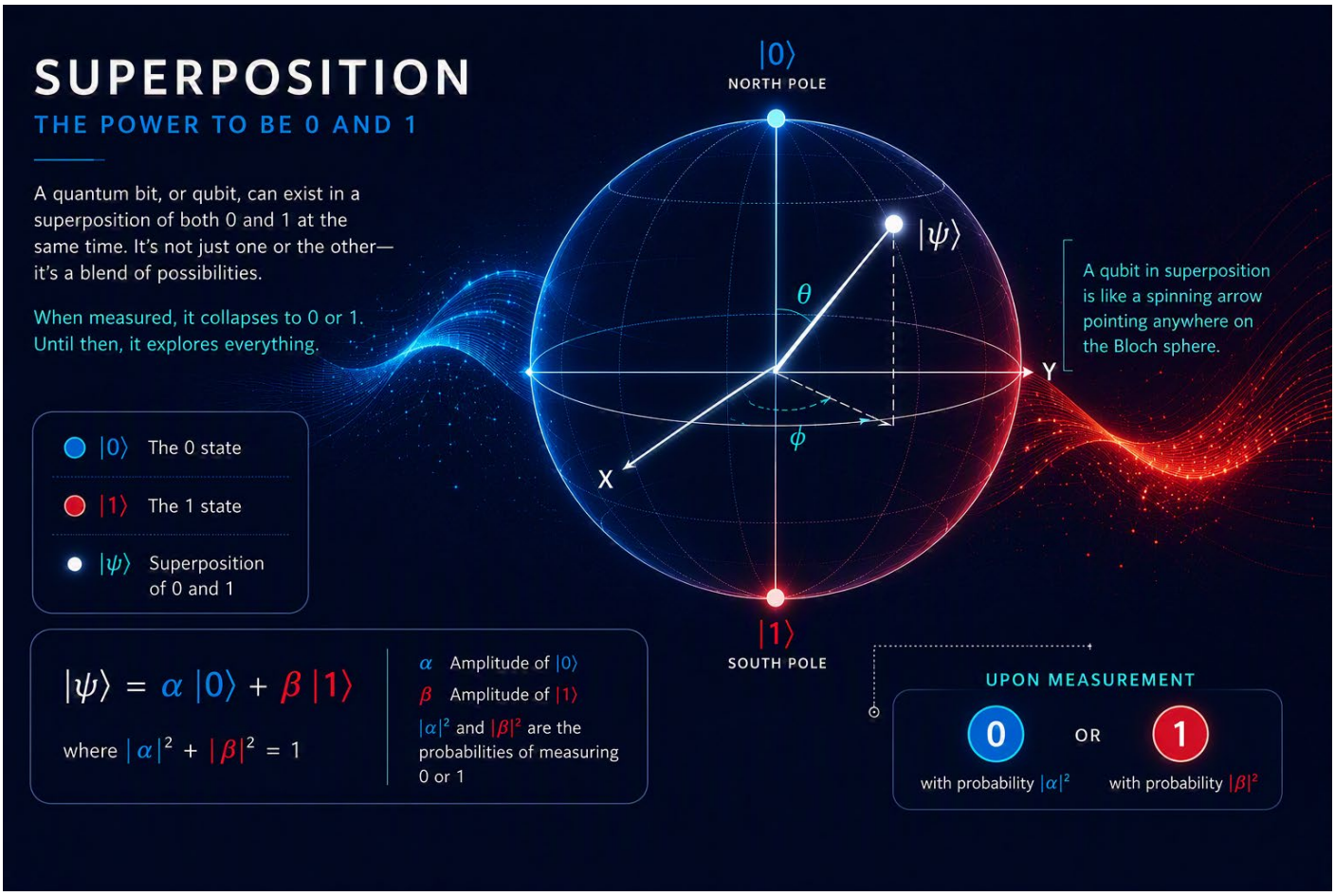
In 1935, physicist Erwin Schrödinger proposed a thought experiment to illustrate the strange implications of quantum mechanics. In this scenario, a cat, a radioactive substance that decays very slowly, and a Geiger counter are put together in a sealed box. According to quantum mechanics, the system can be described as a superposition in which the cat is both alive and dead until the box is opened. Once the box is opened and the system is observed, the superposition collapses into a single outcome.

[2] Rehman, M. (2026, May 11). *What is Quantum Entanglement?* Quantum Insider.
[3] Microsoft. (n.d.). *Superposition.*

Enabling parallel computation versus classical computing

Superposition is a key attribute of quantum computing that allows for quantum parallelism, or the ability to perform multiple computations at the same time. Classical bits can only be in one of two states, 0 or 1. In contrast, a qubit can be in a superposition of 0 and 1, allowing a quantum computer to process multiple possible qubit states at once.[4]

Exhibit 3: Superposition is the condition where a qubit can exist in a superposition of 0 and 1
Illustrating the power of superposition



Source: BofA Global Research
Note: A Bloch sphere is a geometric representation of qubit states.

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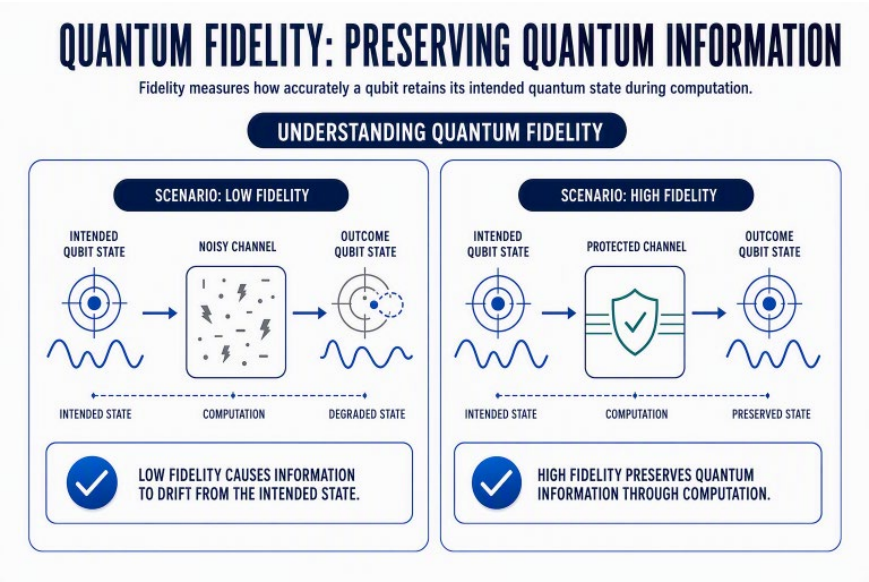
Error rates and fidelity

Fidelity is a measure of how closely a quantum state or operation matches what was intended (Exhibit 4). A higher fidelity generally means fewer errors.

Physical qubits are prone to errors. One way to address this is to combine multiple physical qubits to create a more reliable logical qubit. Quantum error correction (QEC) allows errors in the physical qubits to be detected and corrected, helping protect the information stored in the logical qubit.

[4] Ibid.

Exhibit 4: Fidelity measures how accurately a qubit retains its intended quantum state during computation
Infographic showing the implications of high versus low fidelity



Source: BofA Global Research

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How does quantum error correction work?

At its simplest, QEC relies on redundancy, or storing information across multiple qubits so that errors can be detected and corrected**.**

Consider an example: suppose someone wants to send a message represented by 0. Instead of sending a single 0, the message could be repeated three times as 000, known as a codeword. If one bit were accidentally flipped from 0 to 1 during transmission, the recipient might receive 010. Because two of the three bits are still 0, the recipient could infer that the intended message was 000.

However, if two bits flipped, producing 110 or 011, the recipient could infer the wrong message. If all three flipped to 111, the recipient would have no way to know from the received codeword that errors had occurred.[5] This approach is known as a repetition code and illustrates how redundancy can help protect information from errors.

Repetition codes are a simple form of error correction that can address bit-flip errors, where a 0 changes to a 1 or vice versa. More sophisticated QEC codes are needed to address other quantum errors, such as phase flips.[6] Shor's code, developed by American theoretical computer scientist Peter Shor, uses nine physical qubits to encode one logical qubit and can correct one bit-flip error or one phase flip error, but it can't correct both types of errors at the same time. Meanwhile, Steane's seven-qubit code, developed by Oxford physicist Andrew Steane, can also correct both types of errors. It has the advantage of being fault-tolerant, meaning the error-correction process itself does not introduce additional errors.[7]

At what error level might we start seeing quantum advantage?

Some classical computing systems today operate with extremely high reliability, often described as having "17 nines" of fidelity (99.99999999999999%). By comparison, quantum computers currently achieve fidelity rates of around 99.9%. Gerald Mullally, CEO of OQC, has argued that quantum systems would need to add roughly three more nines of fidelity to achieve commercial relevance, or quantum advantage.

Noise and decoherence

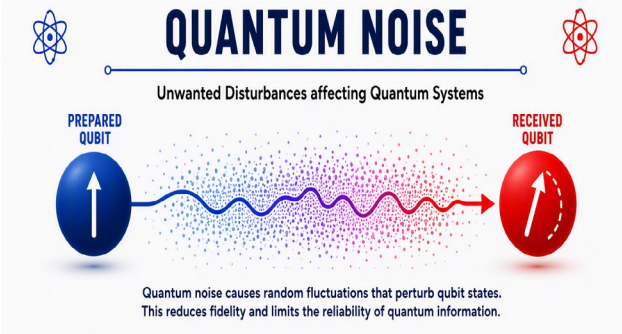
Noise refers to unwanted disturbances that affect quantum systems and can lead to errors in computations (Exhibit 5). It can come from sources such as interactions with the environment, electromagnetic interference and temperature fluctuations. Coherence is a quantum system's ability to maintain its quantum properties, including superposition and entanglement. When coherence is lost through interactions with the environment, the process is known as "decoherence," causing the system to behave more classically (Exhibit 6).[8]

[5] Microsoft Ignite. (n.d.) *Introduction to quantum error correction.*
[6] Ibid.
[7] Ibid.
[8] Quandela. (n.d.) *Quantum Decoherence.*

To fully control qubits for calculations and information transfer, they need to be carefully protected from their surroundings. External factors like noise, energy, changes in Earth's magnetic field, radiation from Wi-Fi or mobile phones or cosmic rays can interfere with the process. Even small disturbances can lead to decoherence, making it difficult to develop large-scale, fault-tolerant quantum computers.[9]

According to QuEra Computing, quantum noise can be mitigated through physical isolation of qubits, the development of more precise control techniques and quantum error correction, which helps detect and correct errors in physical qubits.

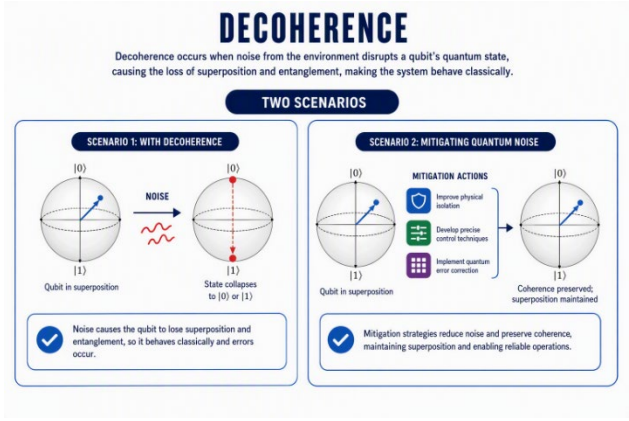
Exhibit 5: Noise refers to unwanted disturbances, e.g., interactions with the environment, that lead to errors in quantum computation
An infographic explaining quantum noise



Source: BofA Global Research

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Exhibit 6: Decoherence is when qubits lose their key quantum properties, superposition and entanglement
An infographic of decoherence



Source: BofA Global Research

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Quantum processing unit (QPU)

Referred to as the brain of the quantum computer, the QPU is the core hardware where qubits reside and where quantum computations take place. QPUs rely on quantum behaviors like superposition and entanglement to tackle problems in areas such as cryptography, simulation and optimization. Just as a CPU (central processing unit) in classical computing contains multiple components, a QPU includes quantum gates (operations that manipulate qubits in a controlled way, similar to logic gates in classical computing) and interconnects, which enable communication between qubits and other quantum processors.[10]

Next, we'll look at broader computing architecture...

Heterogeneous computing

Heterogeneous computing refers to a future computing landscape likely combining classical and quantum computing, potentially unlocking more powerful, faster and cheaper compute and helping solve problems that classical computers cannot. According to BofA Global Research, AI and quantum computing could also converge and complement each other, enabling more transformative use cases.

Quantum modality

A quantum modality refers to the type of physical technology used to create and manipulate qubits. We discuss the modalities in greater depth in the following section.

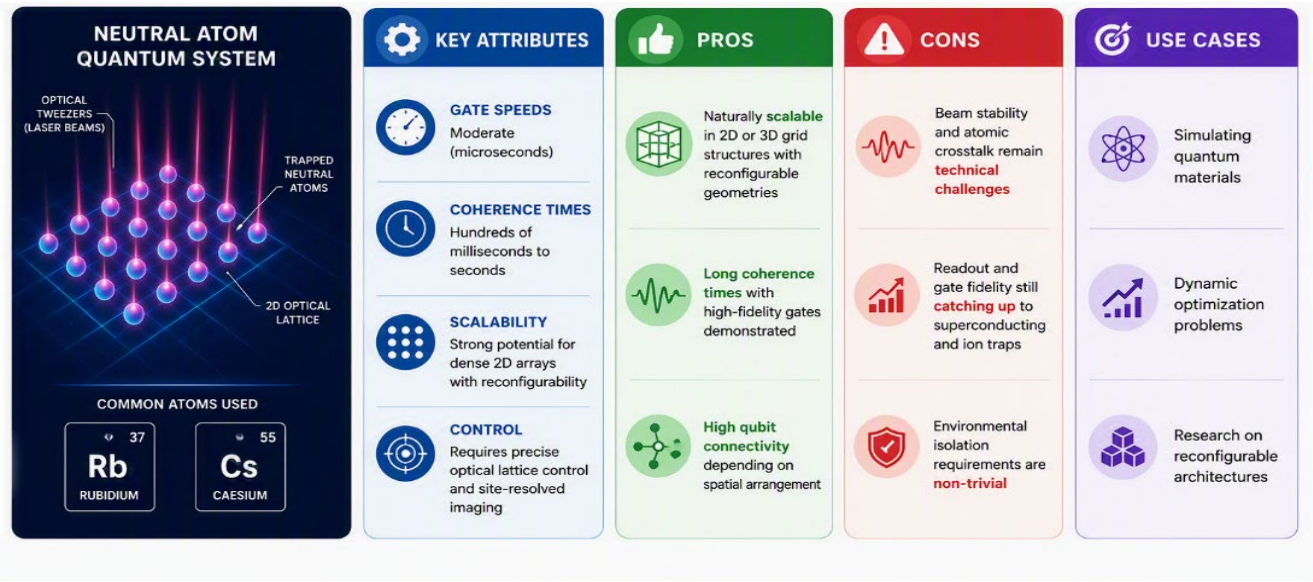
Lastly, we outline qubit technologies...

Neutral atom qubit

A quantum modality that uses individual atoms with no electrical charge as qubits (Exhibit 7). The atoms are held in place using focused lasers known as optical tweezers. Neutral atom qubit systems do not require atoms to be ionized, which means fewer energy constraints and potentially simpler cooling requirements. Atoms such as rubidium or cesium are typically used. The qubits are manipulated by changing the energy levels or positions of the trapped atoms using laser pulses.

[9] QuEra. (n.d.) *Quantum Noise.*
[10] SpinQ. (2025, January 16). *What does a Quantum Computer Look Like? Explain Key Parts.*

Exhibit 7: A quantum modality where atoms with a net electrical charge of zero are the foundation for qubits
Key features of neutral atoms



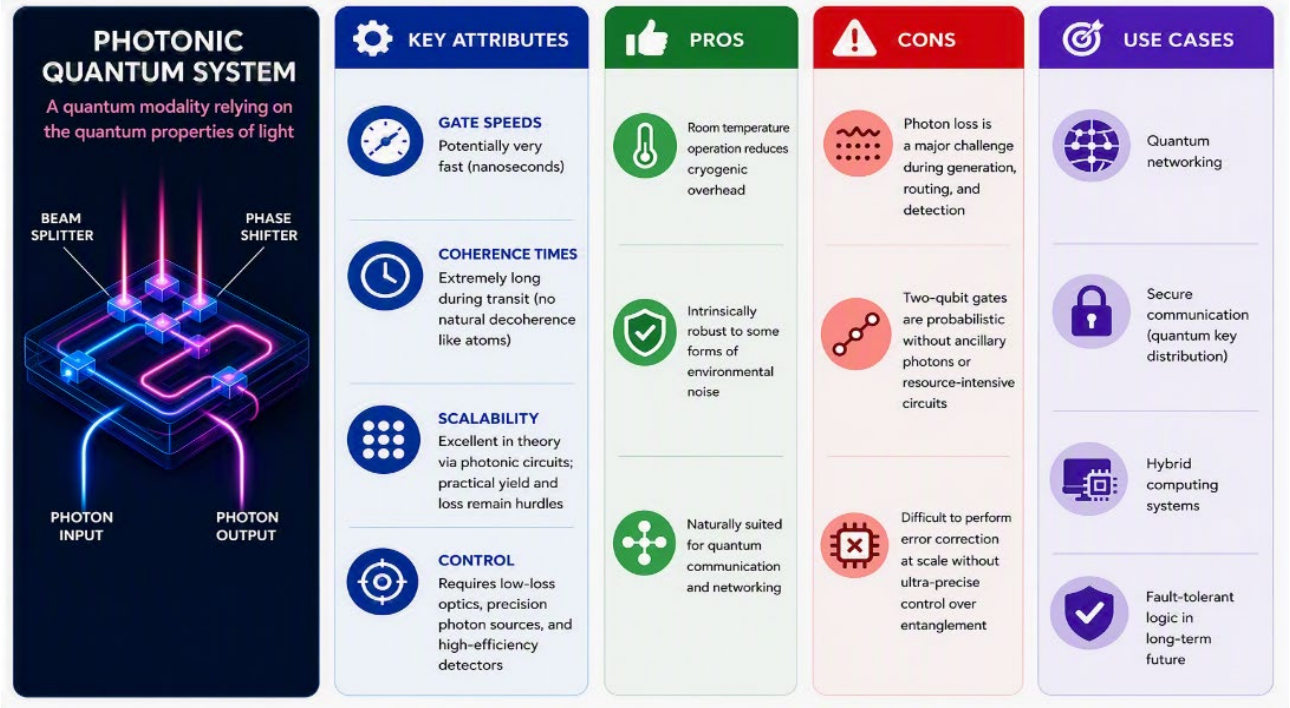
Source: BofA Global Research

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Photonic qubit

A quantum modality that uses particles of light, known as photons, to carry quantum information (Exhibit 8).[11] Optical components such as beam splitters, which divide light along different paths, and phase shifters, which change the phase of light, can be used to perform quantum operations. Photonic quantum computers can use integrated photonic chips to manipulate photons and enable quantum logic.

Exhibit 8: A quantum modality relying on the quantum properties of light
Key features of photonic qubits



Source: BofA Global Research

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[11] Moore, J. (2025, March 28). *The 6 different types of quantum computing technology*. TechTarget.

Spin qubit

A quantum modality that uses the intrinsic spin of electrons or nuclei in solid-state materials, such as silicon or diamond. These spins can be manipulated using magnetic or electric fields, and their states are read using optical or charge-sensing techniques. Spin qubits are particularly appealing because they can leverage existing CMOS (complementary metal-oxide semiconductor; a technology to build processors) manufacturing infrastructure, making it easier to integrate them with conventional processors.

Superconducting qubit

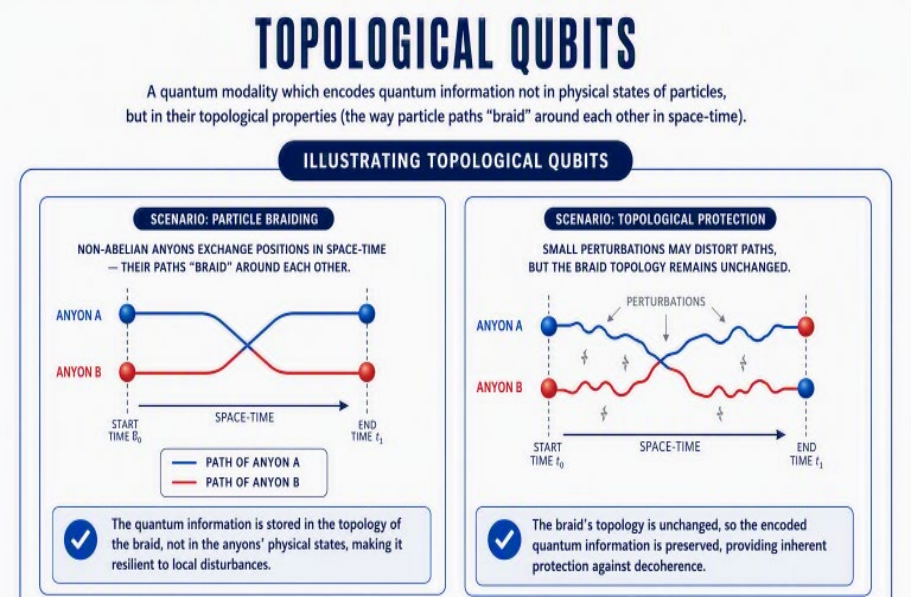
A quantum modality in which electronic circuits create qubits. They are built using Josephson junctions, where two superconductors are separated by a thin insulating layer. Microwave pulses manipulate the qubits' quantum states to run computations.[12]

Specifically, a resistance-free current oscillates back and forth around a quantum loop. An injected microwave signal excites the current into superposition states.[13] To enable this process, superconducting materials are cooled to near absolute zero (around 10-20 millikelvin) using dilution refrigerators, helping reduce thermal noise and maintain coherence.

Topological qubit

A quantum modality that aims to encode quantum information using topological properties rather than relying only on individual particle states (Exhibit 9). One way to think about this is through the paths certain quasiparticles take as they move around one another, creating "braids" in space-time. Because the information can depend on the overall pattern of these braids rather than small details of their paths, the approach is designed to be more resilient to certain disturbances.

Exhibit 9: Resilient to small perturbations but requires special quasiparticles not yet been conclusively demonstrated in nature
An infographic of topological qubits



Source: BofA Global Research

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Trapped-ion qubit

A quantum modality in which ions (electrically charged atoms) are held in place using electromagnetic fields. These ions act as the qubits and are controlled by the system with tightly focused laser beams.[14] The qubits are held inside ultra-high vacuum (UHV) chambers.

[12] Ibid.
[13] Simon Benjamin
[14] Moore, J. (2025, March 28). *The 6 different types of quantum computing technology*. TechTarget.

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